The ABJM $\mathfrak{su}(4)$ alternating chain: complete notes

transfer matrices, ABJM point, localisation, symmetries, separation of variables, and tree-level structure constants in Q-functions

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Abstract

These notes collect everything obtained in the `su4-alternating` thread of the `abjm-c123` repository between 11 and 13 October 2026. They supersede the 9-page notes of 12 October (`notes/abjm-transfer-matrices-symmetries.tex`). The subject is the $\mathfrak{su}(4)$ alternating chain $\mathbf{4}\otimes\bar{\mathbf{4}}\otimes\cdots$ at the ABJM point $\theta_A-\theta_F=2h$, and the tree-level structure constants of the ABJM note in the full $\mathfrak{su}(4)$ sector. The background parts treat one-pass fused transfer matrices, the identification of the ABJM point, localising transfer-matrix words, twisted vacua, highest-weight states and Q-functions, discrete symmetries and selection rules, two-block polarisation forms, and the companion-twist separation-of-variables basis with its dictionary, dual basis and regularisation at the ABJM point. The structure-constant parts give closed odd-bridge formulas at any length, a common pair rotation (the pair κ-family) that produces the even bridges, and a Q-function form of the even-bridge structure constant. The last rests on two facts: the overlaps of the reference bra $\langle W_a^L|$ coincide with those of a companion twisted vacuum, and they factorise over the two middle twist eigenvalues into a polynomial in the F-site single-Q ratios. The resulting formula reproduces the data at $L=2,3$, with an analytic untwisted limit at $L=2$.

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1 Overview

Scope. Research lead: P. Ryan. The question is the computation of tree-level structure constants in ABJM theory in the full $\mathfrak{su}(4)$ sector, using the methods of arXiv:2603.26617 (2603: localising transfer matrices, separation of variables, untwisting limit). Two operators are BPS, of the two-block form of the ABJM note `StructuringABJM.pdf`, and the third is a non-BPS eigenstate of the two-loop dilatation operator. The spin chain is the alternating $\mathfrak{su}(4)$ chain at the ABJM point (§4). Conventions, frames and evidence tags are in §2, and the notebook index is Appendix A.

Timeline.

- 11 Oct: SoV basis and localisation for the alternating chain at two and four sites, companion frame (§5, §13).
- 12 Oct: transfer matrices t_2, t_3; the ABJM point; states, Q-functions and symmetries; selection rules (§3–§11). Twisted vacua; odd bridges at any L; the pair κ-family; the SoV dictionary and the ABJM-point regularisation (§6, §13–§16).
- 13 Oct: the reference-state question, the Q-form of the even bridge, and its untwisted limit (§17, §18).

Headline results.

1. t_2 and t_3 are single-pass transfer matrices with explicit $\wedge^2$ and dual Lax operators (§3).
2. The ABJM chain is the unique point $\theta_A - \theta_F = 2h$ where the Minahan–Zarembo Hamiltonian commutes with the transfer matrices. There $t_3(u_0) = 0$ and $t_2(u_0) = t_1(u_0)t_1(\theta_F)$, with $u_0 = \theta_F + h$ (§4).
3. Three discrete symmetries hold, with Q-function forms. There is no LR selection rule for two-block structure constants, but length-independent parity selection rules exist. The bridge-parity vanishing rule is a property of the note's polarisation, which is blind to all states with $L + p$ odd or $q > 0$ (§8–§12).
4. The companion-twist SoV basis at four sites has an exact Q-dictionary: F sites carry single-Q ratios and A sites 3×3 Casoratians. The dual basis has a triangular, non-diagonal measure. A regularised basis exists at the ABJM point (§13).
5. Odd (one-field) bridges are given in closed form at every length and placement, through an end-growing chain of transfer matrices with proved constants. Their untwisting limit is continuous (§15).

6. At the ABJM point with $z_2 = z_3$, five pair words form a common group rotation of one pair of $\langle W_a^L|$:
$$\langle W_a^L|K_\kappa = (1-\kappa z_1)^2 \langle W_{a'} \otimes W_a^{L-1}|.$$
This gives the even bridges with $M \in \{1, L-1, L\}$ at every L (§16). W_a is the only admissible reference with nonzero cross pairings (§17).

7. The overlaps of $\langle W_a^L|$ are twisted-vacuum overlaps, and they factorise into a polynomial in F-site single-Q ratios:
$$\frac{\langle W_a^L|\Psi\rangle}{\langle 0|\Psi\rangle} = \Big(\frac{a}{\kappa_0}\Big)^n (\lambda_1 u_4)^{-L} B_{z_2}(y) B_{z_3}(y).$$
With this, the even-bridge structure constant is a Q-expression up to the known SoV norm. It reproduces the data at $L = 2, 3$, and at $L = 2$ it has the closed untwisted limit $|C_{M=1}| = K_1 B_1^2/\sqrt{\mathbb{B}_0}$ (§18).

How the pieces fit.

2603 factor	ABJM analogue here	status
$\mathbb{A}_{\ell,\kappa}$ (rotated block)	pair word $P_\kappa(\tau)$, limit K_1	one pair; Q-form known (§16)
$\mathbb{W}_\omega$ (reference overlap)	$B_{z_2}(y)B_{z_3}(y)$, limit B_1^2	closed form; exact at $L \le 3$ (§18)
$\sqrt{\mathbb{B}}$ (norm)	SoV norm $\langle 0|\Psi\rangle/\|\Psi\|$	known (Paul); cross-checked (§13)

2 Conventions, frames and reference states

This section fixes notation for all other sections: evidence tags and standard points (§2.1); the chain, Lax operator, twists and dictionaries (§2.2); the physical and companion frames (§2.3); the four reference bras (§2.4); and the $GL(4)$ covariance of the transfer matrices (§2.5).

2.1 Evidence levels, standard points, data

Evidence levels. [exact]: rational or symbolic arithmetic in a wolfbook notebook. [hp]: arbitrary-precision numerics (40–120 digits) in a notebook. [num, 10^{-n}]: machine- or fixed-precision numerics with residual 10^{-n}. [python]: python only (`checks/`), not (yet) re-done in a notebook. All new computations are done in wolfbook notebooks (`Paul/Mathematica/su4-alternating/*.wb`); the notebook that contains a result is given in square brackets. Part of the work of 12–13 October was first done in python while the Wolfram kernel was unavailable (licence expired on the evening of 12 October; a shared kernel faulted on the morning of 13 October) and ported afterwards. The state of every notebook (stored outputs, partial reruns) is recorded in App. A.

Standard points. $h = \frac{1}{2}$ throughout.

point	values
generic	$\theta = (\frac{2}{7}, -\frac{3}{5}, \frac{8}{11}, -\frac{11}{8}[, \frac{5}{13}, -\frac{7}{9}])$ on sites $1, \dots, 4\,[, 5, 6]$; $z = (\frac{3}{2}, \frac{5}{4}, \frac{2}{3}, \frac{4}{7})$; twisted vacuum $w = (\frac{7}{5}, \frac{9}{8}, \frac{3}{7}, \frac{5}{9})$
ABJM, pair by pair	$\theta_{A,n} = \theta_{F,n} + 2h$ with distinct $\theta_{F,n}$, e.g. $\theta_F = (-\frac{3}{5}, -\frac{11}{8}, -\frac{7}{9}, \frac{5}{6})$; twist with $z_2 = z_3$: $z = (\frac{3}{2}, \frac{5}{4}, \frac{5}{4}, \frac{4}{7})$; $a = \frac{1}{3}$
ABJM, homogeneous	$\theta_A = 2h = 1$, $\theta_F = 0$ (spectra and untwisted data; $u_0 = \theta_F + h = \frac{1}{2}$)
ε-chain (untwisting)	$z = 1+\varepsilon\varphi$, $\varphi = (\frac{9}{10}, \frac{7}{20}, \varphi_3, -\frac{11}{10})$ with $\varphi_3 = \frac{7}{20}$ ($z_2 = z_3$: K_κ, ε-scans) or $\varphi_3 = \frac{3}{5}$ ($z_2 \neq z_3$: Q-form assembly, companion frame regular); $\kappa = 1 - c\varepsilon$, $c = \frac{3}{10}$, $a = (\varphi_1 - c)/(\varphi_1 - \varphi_4)$; $\theta_{F,n} = \varepsilon\theta_n^0$, $\theta^0 = (0, \frac{37}{100}, -\frac{13}{25})$, $\theta_{A,n} = \theta_{F,n} + 2h$; $\varepsilon = 10^{-2}, \dots, 10^{-5}$ ($1/100, \dots, 1/1600$ in the [hp] runs)

Data (details in App. A). The folder `data/` holds `abjm-hw-states.wxf` (all highest-weight cyclic eigenstates at $L = 2, 3, 4$ pairs with energies, LR data and C_ℓ), `abjm-tau.wxf` and `abjm-tau-L4.wxf` (eigenvalue polynomials $\bar{t}_{1,2,3}(u)$), `abjm-Qfunctions.wxf` (Q_1, Q_{12}, Q_{123}, roots, degrees) and `abjm-twoblock-forms.json` (polarisation forms). Each `.wxf` file has a `"description"` key documenting its conventions; load with `Import[file,"WXF"]`. The $L = 4$ key of `abjm-hw-states.wxf` is about 3 GB after decompression and must not be imported; the $L = 2, 3$ states are also stored as `hw_L2.json` and `hw_L3.json` in the folder `epsexp/` of `checks/session-2026-10-13/`.

2.2 The chain, Lax operator, twists and dictionaries

Chain. $2L$ sites (L pairs), A F A F …, periodic. An A site carries the $\mathbf{4}$, $\rho_A(E_{ij}) = e_{ij}$, and the group acts by g. An F site carries the $\bar{\mathbf{4}}$, $\rho_F(E_{ij}) = -e_{ji}$, and the group acts by g^{-T}. ABJM: $\mathrm{tr}(Y^{A_1} Y^\dagger_{B_1} \cdots)$; $L = 2, 3, 4$ pairs $= 4, 6, 8$ sites, dimensions $256, 4096, 65536$. $P_{ab} = \sum e_{ij} \otimes e_{ji}$ (same-type sites), $K_{ab} = \sum e_{ij} \otimes e_{ij}$ (A–F pair; $K = |S\rangle\langle S|$, $S = \sum_i e_i \otimes e_i$). Bras are row vectors and operators act from the right. On a pair, $\rho(g) = g \otimes g^{-T}$, so $\langle v \otimes u|\rho(g) = \langle vg \otimes ug^{-T}|$ and the bilinear pairing vu^T between an A vector and an F vector is invariant.[1] A product bra $(v \otimes u)^{\otimes L}$ is called BPS if $vu^T = 0$.

Naming (Paul). A "fundamental" site is the one whose SoV layer is built with t_1^n: this is the F site ($\rho_F(E_{ij}) = -e_{ji}$, highest-weight vector e_4, group acts by g^{-T}). An "anti-fundamental" site is built with t_n, $n = 0, \ldots, 3$: this is the A site ($\rho_A(E_{ij}) = e_{ij}$, highest-weight vector e_1, group acts by g). The $L = 2$ notebook `L2-ffbar-SoV-localisation.wb` uses the opposite names ("f" $= \rho\, e_{ij}$, "$\bar{\text{f}}$" $= \rho\, {-e_{ji}}$); the $L = 4$ notebook follows Paul. In terms of Ryan–Volin's labels this is the swap $\lambda \leftrightarrow \lambda^*$: $u + h \sum e_{ij} \otimes \rho(E_{ij})$ is RV's $u - \hbar \sum E_{ij} \otimes \pi(E_{ji})$ with $\pi = \rho^*$.

Lax, monodromy, fusion.

$$\mathcal{L}_s(u) = \sum_{ij} e_{ij} \otimes \big((u - \theta_s)\delta_{ij} + h\rho_s(E_{ij})\big), \qquad \mathcal{L}_A = u - \theta + hK, \quad \mathcal{L}_F = u - \theta - hP, \tag{1}$$

$T = \mathcal{L}_1 \mathcal{L}_2 \cdots \mathcal{L}_{2L}$ (site 1 leftmost), twist on the right, $T^G = TG$. Quantum minors: rows antisymmetrised, column k at $u - (k-1)h$. The transfer matrices are

$$t_a(u) = \sum_{|I|=a} \mathrm{qmin}_{I,I}[T^G](u) = \sum_{I,J} \mathrm{qmin}_{I,J}[T](u)\, \det G_{J,I}, \qquad a = 0, \ldots, 4, \tag{2}$$

with $t_1^G = \sum_{ij} T_{ij} G_{ji}$ and $t_4 = \mathrm{qdet}$. Reduced transfer matrices $\bar{t}_a$ have the scalar prefactors removed (§3); t^G without index means t_1^G.

Twists. Companion twist

$$G_c(z) = \begin{pmatrix} c_1 & -c_2 & c_3 & -c_4 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad c_k = e_k(z), \tag{3}$$

diagonal (physical) twist $\Lambda = \mathrm{diag}(z)$. With V the Vandermonde matrix with columns $v(z_j) = (z_j^3, z_j^2, z_j, 1)^T$ one has $G_c V = V\Lambda$, i.e. $v(z_j)$ is the right eigenvector of G_c for eigenvalue z_j; the left eigenvectors are the rows $\tilde{u}_j$ of V^{-1}. We write $\lambda_j = 1/\prod_{m \neq j}(z_j - z_m)$ for the leading coefficient of the j-th Lagrange polynomial, so $(V^{-1})_{j1} = \lambda_j$, and $\gamma := G^{-T}$. The frame map $t^{G_c} R = R t^\Lambda$ is given in §2.3.

[1] The earlier note wrote the same action on column components, $v \to g^T v$ (A), $w \to g^{-1} w$ (F).

Vacua. On the pair vacuum $e_1 \otimes e_4$ (per pair) one has, as bras, $\langle \mathrm{vac}|T_{ij}(u) = 0$ for $i > j$ (lower triangular) and $\langle \mathrm{vac}|T_{ii} = V_i\langle \mathrm{vac}|$, and as kets $T_{ij}|\mathrm{vac}\rangle = 0$ for $i < j$ with the same diagonal eigenvalues. For pair-periodic θ,

$$V_1 = (u-\theta_A+h)^L(u-\theta_F)^L, \quad V_2 = V_3 = (u-\theta_A)^L(u-\theta_F)^L, \quad V_4 = (u-\theta_A)^L(u-\theta_F-h)^L, \tag{4}$$

and in general V_i is the product of the site factors. The explicit one-pair statement is displayed in §4. The vacuum eigenvalue of t^Λ is $\tau_{\mathrm{vac}}(u) = \sum_i z_i V_i(u)$. The SoV vacuum $\langle 0| = e_1 \otimes e_4 \otimes e_1 \otimes e_4 \cdots$ in the companion frame is the Ryan–Volin Gelfand–Tsetlin vacuum: it is the unique bra killed by every strictly lower-triangular $T_{ij}(u)$ of the untwisted monodromy and eigen for the T_{ii}, generic and ABJM θ [exact] [`sov-companion-L4.wb`]. The two meanings of "vacuum" (diagonal-frame $\langle \mathrm{vac}^L|$ and companion-frame $\langle 0|$) are different bras; see §2.4.

Dictionary to Ryan–Volin (arXiv:2002.12341). RV's Lax is $u - \hbar P^\nu$, $P^\nu = \sum E_{ij} \otimes \pi^\nu(E_{ji})$, with $T = \mathcal{L}_L \cdots \mathcal{L}_1$ and quantum minors and B as in our conventions.

this note	Ryan–Volin
F site: $\mathcal{L}_F = u-\theta-hP$, $\rho_F(E_{ij}) = -e_{ji}$	$\pi(E_{ij}) = e_{ij}$, fundamental, $\nu = (1,0,0,0)$, $\theta^{\mathrm{RV}} = \theta_F$
A site: $\mathcal{L}_A = u-\theta+hK$, $\rho_A(E_{ij}) = e_{ij}$	$\pi(E_{ij}) = -e_{ji} \simeq \nu = (1,1,1,0)$ with $\pi^{(1,1,1,0)}(E_{ij}) = \delta_{ij} - e_{ji}$, hence $\theta_A^{\mathrm{RV}} = \theta_A - h$
$T = \mathcal{L}_1 \cdots \mathcal{L}_{2L}$, $T^G = TG$, $G = G_c(z)$	the same operator; our site s is RV's site $2L+1-s$, $\hbar = h$; G_c is RV's (2.19)
$t_a(u)$, $a = 0, \ldots, 4$	$\mathbb{T}_{a,1}(u)$
V_1; $V_2 = V_3$; V_4	ν_4; $\nu_2 = \nu_3$; ν_1, with $\nu_j(u) = \prod_\alpha (u - \theta_\alpha - \hbar\nu_j^\alpha)$
$\langle 0| = (e_1 \otimes e_4)^L$ (companion)	GT vacuum (4.12), $\langle 0|T_{j1}(u) = \delta_{j1}\nu_n(u)\langle 0|$ (3.19)
ABJM point $\theta_A - \theta_F = 2h$	$\theta_A^{\mathrm{RV}} - \theta_F^{\mathrm{RV}} = h \in \hbar\mathbb{Z}$: violates the genericity assumption (2.9)

Quantum minors and $B(u)$ (2.25), and the basis (4.12), agree verbatim with RV; the other equation numbers are counted from the arXiv source.[2] Hence $\pi = -\rho^T$: our A site is RV's dual irrep, our F site RV's fundamental. RV1 (arXiv:1810.10996) covers only one rectangular representation on every site and does not apply to the alternating chain; RV2 gives the left SoV basis for any irrep per site.

Dictionary to the ABJM note and 2603. Note (`StructuringABJM.pdf`): $O_2 = \mathrm{tr}(Y_1^\dagger Y^4)^L$, $O_3 = \mathrm{tr}(\tilde{Y}_1^\dagger \tilde{Y}^4)^L$, $\tilde{Y}_1^\dagger = Y_1^\dagger - Y_4^\dagger$, $\tilde{Y}^4 = Y^1 + Y^4$; (3.6): $C = \mathcal{N}\big(\langle Y_1^\dagger Y^4|^{L-M} \otimes \langle \tilde{Y}_1^\dagger \tilde{Y}^4|^M\big)|\Psi\rangle$. On one pair,

$$W_a := (e_1 + ae_4) \otimes (e_4 - ae_1) = \langle \mathrm{vac}|\rho(1 + aE_{14}), \tag{5}$$

so W_0 is the O_2 pair and W_{-1} the O_3 pair. 2603 (Bargheer, Bercini, Lefundes, Ryan, arXiv:2603.26617): $\mathfrak{so}(6)$ chain of $\mathbf{6} = \wedge^2\mathbf{4}$ with vacuum $Z = e_1 \wedge e_2$, $\hbar = i$ and centred shifts (up to gauge $Q_I^{\mathrm{BBLR}}(u) \leftrightarrow Q_I^{\mathrm{RV}}\big(u + \frac{\hbar}{2}(|I| - 1)\big)$), site 1 leftmost as here; on a site in the $\mathbf{4}$ its Lax is $u - iP$, our F site with $h = i$. Their LR symmetry $\tau_+ = \tau_-$ gives $C = 0$ unless the state is LR symmetric (Sec. IV, (18)–(20)); O_2 is O_3 with the first ℓ sites rotated by $1 - \kappa$ twist, and $C = \mathcal{N}\,\mathbb{W}\,\mathbb{A}/\sqrt{\mathbb{B}}$ (their (11)–(14)).

2.3 Frames

Physical (diagonal) frame. The twist is $\Lambda = \mathrm{diag}(z)$. *The states Ψ in every structure-constant formula of these notes are eigenstates of $t^\Lambda(u)$ in the diagonal frame.* The pair word K_κ is a word in t^Λ transfer matrices; the identity $\langle W_a^L|K_\kappa = (1-\kappa z_1)^2\langle W_{a'} \otimes W_a^{L-1}|$ and the untwisting limit $z \to 1$ are diagonal-frame statements (§16).

[2] They may differ by one from the published numbering; only (2.25), (4.12) and (1.2) were matched against the published text.

Companion frame (SoV only). The SoV basis is built with the companion twist $G_c(z)$. The two frames are related by

$$t^{G_c(z)}(u)\,R = R\,t^{\Lambda}(u), \qquad R = \bigotimes_{\text{sites}} \left(V^{-T} \text{ on A},\ V \text{ on F}\right) \tag{6}$$

[exact] for $t_{1,2,3}$ on one and two pairs [`even-bridge-Qform.wb` Part 0, `Wa-Fonly-L6.wb`; first found [python] in `checks/chk_frame.py`, 10^{-15}]. States and bras are transported as $|\Psi_c\rangle = R|\Psi\rangle$ and $\langle B_c| = \langle B|R^{-1}$, so $\langle B_c|\Psi_c\rangle = \langle B|\Psi\rangle$; equivalently $\langle x_{\text{diag}}| = \langle x_{\text{comp}}|R$. *Every bra is transported with the R of the twist z of* Ψ, including the twisted vacuum $\langle 0_w|$, whose own twist is w. Eigenvalues and Q-functions are frame independent, so the Q-expressions of §18 (for example $P_\kappa(\tau)$) are the same in both frames. The companion image of a pair of diagonal-frame vectors is $\langle v\otimes u|R^{-1} = \langle vV^T \otimes uV^{-1}|$.

Two cautions. (i) At coincident twist eigenvalues the Vandermonde matrix $V(z)$ is singular. The pair word K_κ needs $z_2 = z_3$; there the companion frame does not exist, the diagonal-twist chain has a $\mathfrak{gl}(2)$ symmetry and there is no SoV basis. The two are combined at $z_2 - z_3 = O(\varepsilon)$ in the untwisting scaling, which suffices for the limit (§16). (ii) The literal conjugate $V^{-1}G_aV$ is not the companion-frame image of G_a (§2.5).

Weight selection. For a bra $\langle W_a^L| = \langle \text{vac}^L|e^{aJ}$ with $J = \rho(E_{14})$ summed over sites,

$$\langle W_a^L|\Psi\rangle = \frac{a^n}{n!}\langle \text{vac}^L|J^n|\Psi\rangle, \qquad n = L - \mu_1, \tag{7}$$

for Ψ of weight $(\mu_1, 0, 0, -\mu_1)$, and $\langle W_a^L|\Psi\rangle = 0$ for every other weight; the torus gives $\langle W_a^L|\Psi\rangle = a^n\langle W_1^L|\Psi\rangle$ (weight counting; confirmed in the ε-scans [python] [`scan_eps.py` in `checks/session-2026-10-13/epsexp/`]).

2.4 The four reference bras

The reference bras appearing in these notes are

- $\langle \text{vac}^L|$, the Bethe vacuum of t^Λ;
- $\langle 0|_{\text{comp}}$, the SoV (Gelfand–Tsetlin) vacuum;
- $\langle 0_w|$, the twisted vacuum, i.e. the vacuum eigenbra of $t^{G_c(w)}$;
- $\langle W_a^L|$, the Borel rotation of the Bethe vacuum used for the ABJM structure constants.

All four are BPS product states $(v \otimes u)^{\otimes L}$ with $vu^T = 0$. Table 1 gives the per-pair vectors in both frames, Table 2 the properties. Here $v(x) = (x^3, x^2, x, 1)$, $\tilde{u}_j$ is row j of V^{-1} and $\lambda_j = 1/\prod_{m\neq j}(z_j - z_m)$. A BPS reference $(v \otimes u)^{\otimes L}$ is *good* if $u\Lambda v^T = 0$ (the README criterion; in the companion frame it reads $u_cG_cv_c^T = 0$, and it equals the obstruction $q(w, z)$ of the twisted-vacuum notebooks, 232551/1558000 at the standard point). Good references localise like $\langle 0|$.

The overlap identity. For every Ψ of weight $(\mu, 0, 0, -\mu)$,

$$\langle 0_w|\Psi\rangle = (v_1u_4)^L\langle W_a^L|\Psi\rangle, \qquad a = \frac{v_4}{v_1} = -\frac{(z_1 - z_2)(z_1 - z_3)(w_1 - z_1)}{(z_4 - z_2)(z_4 - z_3)(w_1 - z_4)}, \tag{8}$$

and $\langle W_a^L|\Psi\rangle = 0$ on all other weights [exact] for $L = 2$, all eigenstates of the sectors $(2, 0, 0, -2)$, $(1, 0, 0, -1)$, $(0, 0, 0, 0)$, to 40 digits [hp] [`even-bridge-Qform.wb` Part 1(b), `Wa-sov-expansion-L4.wb`]; the argument is weight counting and holds at any L. Take $w_2 = z_2$, $w_3 = z_3$: then $u_2 = u_3 = 0$, the F sites carry only e_1 and e_4, an A site in e_2 or e_3 would give a positive 2- or 3-weight that no F site cancels, so on these weights only the A components v_1, v_4 contribute and each pair acts as $v_1u_4W_a$, with $vu^T = 0$ giving $a = v_4/v_1 = -u_1/u_4$. The A sites of

bra	diagonal frame (A vector $\otimes$ F vector, per pair)	companion frame
$\langle \mathrm{vac}^L\|$	$e_1 \otimes e_4$	$v(z_1) \otimes \tilde{u}_4$
$\langle 0\|_{\mathrm{comp}}$	$(\lambda_1,\lambda_2,\lambda_3,\lambda_4) \otimes (1,1,1,1)$	$e_1 \otimes e_4$
$\langle 0_w\|$	$v_j = \lambda_j \prod_{m\neq j}(w_1 - z_m)$, $u_j = \dfrac{\prod_{m\leq 3}(z_j - w_m)}{\prod_{m\leq 3}(w_4 - w_m)}$	$v(w_1) \otimes u_w$, u_w = row 4 of $V(w)^{-1}$
$\langle W_a^L\|$	$(e_1 + a e_4) \otimes (e_4 - a e_1)$	$\big(v(z_1) + a\, v(z_4)\big) \otimes (\tilde{u}_4 - a\tilde{u}_1)$

Table 1: Reference bras per pair in the two frames. The diagonal-frame image of $\langle 0|_{\mathrm{comp}}$ is exact; $\langle 0|_{\mathrm{comp}}$ lies in the Borel orbit of $\langle \mathrm{vac}^L|$ and is finite as $z \to 1$ (§13). The diagonal image of $\langle 0_w|$ exists only for $z_2 \neq z_3$.

bra	what it is	$u\Lambda v^T$	overlap with an eigenstate Ψ of t^Λ
$\langle \mathrm{vac}^L\|$	Bethe vacuum of t^Λ, eigenvalue τ_{vac}	0	$\langle \mathrm{vac}^L\|\Psi\rangle = 0$ for $\Psi \neq$ vac
$\langle 0\|_{\mathrm{comp}}$	RV Gelfand–Tsetlin vacuum; the word basis on it is z-independent	0 (good)	$\langle 0\|\Psi\rangle$ is the SoV normalisation (known); does not admit the five-word K_κ (§17)
$\langle 0_w\|$	vacuum eigenbra of $t^{G_c(w)}$, eigenvalue $\sum_i w_i V_i$; shares the SoV basis with a state of twist z	$-\dfrac{\prod_i (w_1 - z_i)}{\prod_{m\leq 3}(w_4 - w_m)}$	finite Q-sum $\mathbb{W}_w$ (§18); admits no κ-family at $z_2 \neq z_3$ (§17)
$\langle W_a^L\|$	vacuum eigenbra of t^{G_a} (§2.5); never of a companion twist; Q-functions those of $\langle \mathrm{vac}^L\|$	$a(z_4 - z_1)$	$= (v_1 u_4)^{-L} \langle 0_w\|\Psi\rangle$ for $w = (w_1, z_2, z_3, w_4)$ on weights $(\mu, 0, 0, -\mu)$, zero otherwise; admits K_κ (§16)

Table 2: Properties of the four reference bras. v, u are the A and F vectors of the diagonal frame.

$\langle 0_w|$ still have e_2, e_3 components; this is why the two bras differ although their relevant overlaps agree. The Mobius map $a(w_1)$ sends $w_1 = z_1$ to $a = 0$ (the vacuum) and $w_1 = z_4$ to $a = \infty$ (the lowest-weight vacuum $e_4 \otimes e_1$). In the limit $w_1 \to \infty$ one finds $a \to \kappa_0$ with

$$\kappa_0 = -\frac{(z_1 - z_2)(z_1 - z_3)}{(z_4 - z_2)(z_4 - z_3)}, \tag{9}$$

and the twisted vacuum becomes, in the companion frame, $(e_1 \otimes \tilde{u})^{\otimes L}$ with $\tilde{u} = e_4^T (G_c - z_2)(G_c - z_3)$ (§18).

Why $\langle W_a^L|$ is not a twisted vacuum. Its eigenvalue is τ_{vac}, since conjugating the twist does not change eigenvalues, so its Q-functions are those of $\langle \mathrm{vac}^L|$. If a twist sharing the companion SoV basis had $\langle W_a^L|$ as an eigenbra, its dual-basis wave functions would be those of $\langle \mathrm{vac}^L|$ and completeness would force $\langle W_a^L| \propto \langle \mathrm{vac}^L|$. Consistently, the SoV coefficients of $\langle W_a^L|$ are dense (all 256 words at $L = 2$, every power of a) [exact] [`Wa-sov-expansion-L4.wb`], and the t_1-twist space of $\langle W_a|$ at $L = 2$ is 11-dimensional and contains no companion matrix $G_c(w)$ for $a \neq 0$ [exact] [`twisted-vacuum-companion-check.wb`]. Its companion-frame image is $\langle W_a^L|R^{-1} = [(v(z_1) + a\, v(z_4)) \otimes (\tilde{u}_4 - a\tilde{u}_1)]^{\otimes L}$ [exact]; for $a = 0$ this is the companion Bethe vacuum $\langle 0_{w=z}|$, not the SoV vacuum.

2.5 $GL(4)$ covariance and the twist of $\langle W_a^L|$

Covariance. For $g \in GL(4)$ and $k = 1, 2, 3$,

$$\rho(g)\, t_k^\Lambda(u)\, \rho(g)^{-1} = t_k^{G'}(u), \qquad G' = g^{-T} \Lambda g^T = (g \Lambda g^{-1})^T, \tag{10}$$

and more generally $\rho(g)t^G\rho(g)^{-1} = t^{g^{-T}Gg^T}$ with $\rho(g) = g \otimes g^{-T}$ on A–F sites. It follows from the invariance of the Lax operator, $\rho(g)\mathcal{L}_s(u)\rho(g)^{-1} = (g^T \otimes 1)\mathcal{L}_s(u)(g^{-T} \otimes 1)$, and was checked exactly on one pair (two sites) for a generic upper-triangular g, $k = 1, 2, 3$, against the three other candidate twists $g^{-1}\Lambda g$, $g\Lambda g^{-1}$, $g^T\Lambda g^{-T}$, which fail [exact] [`reference-admissibility-Kkappa.wb` §6; python `checks/session-2026-10-13/refmap/covariance.py`]. Hence for every word X,

$$\langle \mathrm{vac}|\rho(g)\, X[t^\Lambda] = \langle \mathrm{vac}|\, X[t^{G'}]\, \rho(g), \tag{11}$$

verified exactly for the whole pair operator K_κ and seven group elements (W_a, torus$\times W_a$, two block-BPS classes, a broken class, a generic Borel element, a product) [exact]. For $v = e_1 g$, $u = e_4 g^{-T}$ one has $uf(\Lambda)v^T = [f(G')]_{41}$, so the goodness $u\Lambda v^T$ is G'_{41} and the admissibility criterion of §17 is a statement about the conjugated twist G'.

The twist G_a. Put

$$G_a = \Lambda + a(z_1 - z_4)E_{41} = e^{aE_{41}}\Lambda e^{-aE_{41}} = (g\Lambda g^{-1})^T\big|_{g=e^{-aE_{14}}} \text{ [exact].} \tag{12}$$

Applying the covariance with $g = e^{aE_{14}}$ to the twist G_a gives $g^{-T}G_a g^T = \Lambda$, so

$$\rho(e^{aE_{14}})\, t^{G_a}\, \rho(e^{aE_{14}})^{-1} = t^\Lambda \qquad \text{and} \tag{13}$$

$$\langle W_a^L|\, t_k^{G_a}(u) = \tau_{\mathrm{vac},k}(u)\, \langle W_a^L|, \qquad k = 1, 2, 3, \tag{14}$$

[exact] for $L = 2$, generic θ, z, $a = \frac{1}{3}$, three values of u [`even-bridge-epsilon-expansion.wb`, step 0a]. The twist seen by the *words* acting on $\langle W_a| = \langle \mathrm{vac}|\rho(e^{aE_{14}})$ is $G' = \Lambda + a(z_4 - z_1)E_{41}$, the opposite sign. The other sign $\Lambda + a(z_4 - z_1)E_{41}$ is not an eigenbra twist, Λ itself is not, and $\langle \mathrm{vac}|$ is not an eigenbra of t^{G_a}; the twist $\Lambda + ((z_4 - z_1)/a)E_{14}$ is an eigenbra twist, but with an eigenvalue different from τ_{vac} (that of the $e_4 \otimes e_1$ vacuum) [exact].

The T_{14} identity. With $T_{14} = \mathrm{tr}(TE_{41})$ the $(1, 4)$ element of the monodromy,

$$t^{G_a}(u) = t^\Lambda(u) + a(z_1 - z_4)\, T_{14}(u) \qquad (t = t_1,\ 256 \times 256) \text{ [exact],} \tag{15}$$

and therefore, for every eigenstate Ψ of t^Λ and every u,

$$\big(\tau_{\mathrm{vac}}(u) - \tau_\Psi(u)\big)\langle W_a^L|\Psi\rangle = a(z_1 - z_4)\, \langle W_a^L|T_{14}(u)|\Psi\rangle, \tag{16}$$

verified on all 41 eigenstates of the sectors $(2, 0, 0, -2)$, $(1, 0, 0, -1)$, $(0, 0, 0, 0)$, minimum eigenvalue gap $2 \cdot 10^{-3}$, deviation $< 10^{-29}$ at 30 digits [hp] [`even-bridge-epsilon-expansion.wb`]. The infinitesimal form is $[J, t^\Lambda(u)] = -(z_1 - z_4)T_{14}(u)$ and $[J, T_{14}(u)] = 0$, checked at finite ε on random vectors to 10^{-16} [python] [`checks/session-2026-10-13/epsexp/pt.py`]. It makes the whole twist dependence of $\langle W_a^L|\Psi\rangle$ explicit: $a^n(\varphi_1 - \varphi_4)^n$ at order ε^n (§18).

Companion-frame image. $VG_aV^{-1} = G_c(z) + a(z_1 - z_4)\, v(z_4) \otimes \tilde{u}_1$ [exact]: a rank-one perturbation of G_c, not a companion matrix for $a \neq 0$, in agreement with the negative statement above. The literal $V^{-1}G_aV$ is not the companion-frame map ($V\Lambda V^{-1} = G_c$ but $V^{-1}\Lambda V \neq G_c$).

3 Transfer matrices

[`dual-lax-t3.wb`] The higher transfer matrices are single-pass. From the one-site quantum minors (symbolic in u, θ, h), checked identically against the quantum-minor t_a on the chain at 2 and 4 sites

for companion, diagonal, generic and trivial twist [exact]:

$$t_3(u) = \prod_s g_s^{(3)}(u)\ \mathrm{tr}\big[\bar{\mathcal{L}}_1 \cdots \bar{\mathcal{L}}_{2L}\ \det G\, G^{-T}\big], \tag{17}$$

$$\bar{\mathcal{L}}_A = (u-\theta+h) - hP, \qquad \bar{\mathcal{L}}_F = (u-\theta-3h) + hK,$$

$$g_A^{(3)} = (u-\theta-h)(u-\theta-2h), \qquad g_F^{(3)} = (u-\theta)(u-\theta-h),$$

$$t_2(u) = \prod_s g_s^{(2)}(u)\ \mathrm{tr}\big[\mathcal{L}_1^{(2)} \cdots \mathcal{L}_{2L}^{(2)}\ \wedge^2 G\big], \tag{18}$$

$$\mathcal{L}_s^{(2)} = u - \tilde{\theta}_s + h\sum_{ab} D(e_{ab}) \otimes \rho_s(E_{ab}), \qquad g_A^{(2)} = u-\theta-h, \quad g_F^{(2)} = u-\theta,$$

with the sign gauge $(-1)^k$ on the dual index of t_3, D the derivation action on $\wedge^2\mathbb{C}^4$, $\tilde{\theta}_A = \theta$, $\tilde{\theta}_F = \theta + h$, and $(\wedge^2 G)_{JI} = \det G_{J,I}$. The relative A/F shift of the evaluation point is 0 for t_1, h for t_2 and $4h$ for t_3; the earlier naive dual Lax missed this shift and does not commute with t_1, t_3 (`checks/abjm_dual_lax_try.py`).

All k in one form. Define for $k = 1, \ldots, 4$ the fused Lax operator on $\wedge^k\mathbb{C}^4$,

$$t_k(u) = \prod_s g_s^{(k)}(u)\ \mathrm{tr}\big[\mathcal{L}_1^{(k)} \cdots \mathcal{L}_{2L}^{(k)}\ \wedge^k G\big], \qquad \mathcal{L}_s^{(k)} = u - \tilde{\theta}_s + h\sum_{ab} D_k(e_{ab}) \otimes \rho_s(E_{ab}), \tag{19}$$

with $\tilde{\theta}_A = \theta$, $\tilde{\theta}_F = \theta + (k-1)h$, $g_A^{(k)} = \prod_{j=1}^{k-1}(u-\theta-jh)$, $g_F^{(k)} = \prod_{j=0}^{k-2}(u-\theta-jh)$, D_k the derivation action on $\wedge^k\mathbb{C}^4$. For $k = 2, 3$ this reproduces the forms above, and for $k = 4$ it gives qdet (for an A site $(u-\theta+h)(u-\theta-h)(u-\theta-2h)(u-\theta-3h)$ times $\det G$). It holds for any twist matrix G, including the non-diagonal twists G_a and G' of §2.5. It was calibrated against the quantum-minor definition for $k = 1, \ldots, 4$ on chains A F, A F A F and F A with a random twist, relative error 10^{-16} to $3 \cdot 10^{-14}$ [num, 10^{-14}] [`pair-group-rotation.wb` §1; python `checks/session-2026-10-12/pairgroup/py/`].

Reduced and regularised transfer matrices. $\bar{t}_a$ denotes the reduced transfer matrix without the scalar prefactor $\prod_s g_s^{(a)}$, so that $t_a = \prod_s g_s^{(a)}\, \bar{t}_a$ and $\tau_a = \prod_s g_s^{(a)}\, \bar{t}_a$ on states. $\hat{t}_k$ denotes t_k with exactly vanishing scalar factors $(u-\theta-jh) = 0$ dropped (needed at the ABJM point, §4); for a double zero from two coinciding F inhomogeneities, $\hat{t}_3(u_0) = \frac{1}{2}t_3''(u_0)$.

Vector form. One pass per vector: t_3 takes 0.16 s at 6 sites and a few seconds at 8 sites with the sparse site action; $[t_a, t_b] = 0$ to 10^{-16} at 6 sites [num, 10^{-16}]. For $L = 3$ (4096-dimensional) the monodromy is never built: $t_k^G(u)\psi = \prod g \sum (\wedge^k G)_{JI} T_{IJ}\psi$ and $T_{IJ}(u)\psi$ are propagated site by site on the auxiliary-block array, and bras are acted on by the transposed site operators. These vector builders were validated against the explicit one- and two-pair matrices to 10^{-16} [exact] [`even-bridge-epsilon-expansion.wb` §10, `Wa-Fonly-L6.wb`]. $T_{14} = t_1^{E_{41}}$.

Q-side. The eigenvalues τ_a satisfy the Baxter equation of §7; $\tau_2 = \prod g^{(2)}\bar{t}_2$, $\tau_3 = \prod g^{(3)}\bar{t}_3$.

4 The ABJM point

4.1 Minahan–Zarembo Hamiltonian and uniqueness of $\theta_A - \theta_F = 2h$

[`abjm-chain.wb`] The two-loop Hamiltonian of Minahan–Zarembo (arXiv:0806.3951; Bak–Rey arXiv:0807.2063; the 't Hooft prefactor $\lambda^2/2$ dropped; form quoted from memory and confirmed by the commutation below) is

$$H_{\mathrm{MZ}} = \sum_{l=1}^{2L} \big(2 - 2P_{l,l+2} + P_{l,l+2}K_{l,l+1} + K_{l,l+1}P_{l,l+2}\big). \tag{20}$$

It is Hermitian, $\mathfrak{su}(4)$-invariant, and commutes with the translation S by one pair, $[H_{\rm MZ}, S] = 0$ [exact]. The $L = 2$ spectrum is $0\,(\times 84)$, 4, $8\,(\times 120)$, $12\,(\times 30)$, $16\,(\times 20)$, 20.

- [exact] ($L = 2$, u symbolic): $[H_{\rm MZ}, \bar{t}_a(u)] = 0$ for $a = 1, 2, 3$ at $\theta_A - \theta_F = 2h$, and not at $\theta_A = \theta_F$. With $d = \theta_A - \theta_F$ symbolic the gcd of all commutator coefficients is proportional to $d - 2h$, so the point is unique. [num, 10^{-15}] at $L = 3$.
- $U(u) = \bar{t}_1(\theta_F + u)\,\bar{t}_3(\theta_A - h + u)$. At $u = 0$, $\bar{t}_1$ has P on the F sites, $\bar{t}_3$ has P on the A sites, and the A–F dressings are $(x + hK)(-x - 4h + hK) = x(-x - 4h)$ with $x = \theta_F - \theta_A$. Hence $U(0) = c\,S$ with $c = h^4[x(-x - 4h)]^2$ at $L = 2$ [num, 10^{-16}], vanishing at $d \in \{0, 4h\}$. At the ABJM point $x = -2h$, $c = 16h^8$, $U(0) = S/16$, and

$$H_{\rm MZ} = U(0)^{-1}U'(0) + 12 \qquad (L = 2)\ \text{[exact]}. \tag{21}$$

The shift-operator statement $U(0) \propto S$ is the translation-operator identity that reappears in §4.3 for the regularised words.

Superseded: computations at $\theta_A = \theta_F$. There $U(0) = 0$, S drops out of the t-algebra, $H_{\rm MZ}$ does not commute with the t's, and an odd-momentum highest-weight partner is degenerate with a cyclic state in t_1, t_2, t_3 (only S separates them); matching it through the untwisting limit required pair-periodic θ and a twisted shift $\hat{S} = S \circ \rho_{3..L}(\Lambda)$. All of this is an artefact of the singular point [`abjm-bridges.wb` §1–2].

4.2 Exact identities at the ABJM point

In this subsection $\theta_A - \theta_F = 2h$ for every pair, and

$$u_0 := \theta_F + h = \theta_A - h, \qquad \tau_F := \tau_1(\theta_F), \tag{22}$$

with θ_F, θ_A those of pair 1; the other pairs have arbitrary $\theta_{F,n}$ with $\theta_{A,n} = \theta_{F,n} + 2h$. t_a are the full transfer matrices of §3.

Operator identities.

$$t_3(u_0) = 0, \qquad t_2(u_0) = t_1(u_0)\,t_1(\theta_F), \tag{23}$$

and the mirror identities (used on the ket side),

$$t_2(\theta_A) = t_1(\theta_A)\,t_1(u_0), \qquad t_3(\theta_A + h) = 0. \tag{24}$$

The two vanishing statements are immediate from the prefactors of §3: $g_F^{(3)}(u) = (u - \theta)(u - \theta - h)$ vanishes at $u = \theta_F + h$ and $g_A^{(3)}(u) = (u - \theta - h)(u - \theta - 2h)$ at $u = \theta_A + h$. The content lies in the two t_2 identities, which involve $u_0 = \theta_F + h = \theta_A - h$, and in the regularised $\hat{t}_3(u_0)$ of §4.3. The first pair is verified as operator identities at $L = 2, 3, 4$ for the diagonal twists used in these notebooks [exact] [`even-bridge-epsilon-expansion.wb` §10 (one pair), `even-bridge-lpairs.wb`, `sov-abjm-point-limit.wb`]. The mirror pair was found in the SoV norm construction [python] [`checks/session-2026-10-12/sovformula/`, report §9d] and is used by the regularised ket basis of `abjm-sov-norm-L4.wb`.

On states. $\tau_3(u_0) = 0$ and $\tau_2(u_0) = \tau_1(u_0)\,\tau_F$ for every eigenstate. These are the collisions of the eigenvalues of Ryan–Volin's B at the ABJM point: 13 distinct eigenvalues per pair instead of 16, i.e. $169 = 13^2$ at two pairs. Q-reading, in the Baxter convention $\sum_{a=0}^{4}(-1)^a t_a(u)Q(u + h - ah) = 0$: at $u = \theta_F$ the prefactors kill $t_{a\geq 2}$, so $Q_j(\theta_F + h) = \tau_F Q_j(\theta_F)$ for all four solutions. With $D_a = \det_{i\leq 3;\ b\in\{0,\ldots,3\}\setminus\{a\}} Q_i\big(\theta_A - h + h(1 - b)\big)$, which gives $\tau_a(\theta_A - h) = D_a/D_0$ at generic θ, the identity $Q_j(u_0) = \tau_F Q_j(u_0 - h)$ makes the columns at $\theta_A - h$ and $\theta_A - 2h$ of the A-site Casoratian proportional, so

$$D_0 = D_3 = 0, \qquad D_2 = \tau_F\,D_1. \tag{25}$$

Since $D_0 = Q_{123}(\theta_A - h)$, the generic dictionary degenerates (0/0 for $a = 3$); this is the Q-side of the B collisions and is resolved by the regularised basis of §4.4.[3] The LR map of §8, $\mathcal{P}\,\bar{t}_1(u)\mathcal{P}^{-1} = \bar{t}_3(u+h)$, sends $\bar{t}_1(\theta_F)$ to $\bar{t}_3(\theta_F + h) = \bar{t}_3(\theta_A - h)$ exactly at $d = 2h$.

Pair-vacuum zero pattern. [`pair-group-rotation.wb` §2] On one pair, with $(\mathcal{L}_A\mathcal{L}_F)_{jk}$ the (j,k) element in the auxiliary space,

$$\langle e_1e_4|(\mathcal{L}_A\mathcal{L}_F)_{jk}(u) = (u-\theta_A)(u-\theta_F)\,\delta_{jk}\langle e_1e_4| - h(u-\theta_A)\,\delta_{k4}\langle e_1e_j| + h(u-\theta_F)\,\delta_{j1}\langle e_ke_4| - h^2\delta_{j1}\delta_{k4}\langle S|. \tag{26}$$

Seven entries vanish identically: the six lower-triangular ones and $(2,3)$. There are bonus zeros at $u = \theta_A$ for $(j,4)$, $j = 2,3$ (and for the diagonal entries $(2,2),(3,3),(4,4)$), and at $u = \theta_F$ for $(1,k)$, $k = 2,3$ (and for $(1,1)$). Only $(1,4)$ never vanishes; it carries the singlet $-h^2\langle S|$. [exact] (sympy exact; the formula holds for all (j,k)).

Localisation on a pair, and why only the ABJM point. In the $\wedge^k$ auxiliary space let $D_k = \{I : 1 \notin I,\ 4 \in I\}$. Row I of $\langle e_1|\mathcal{L}_A^{(k)}$ is off-diagonal only if $1 \in I$, and row I of $\langle e_4|\mathcal{L}_F^{(k)}$ is off-diagonal only if $4 \notin I$. So on D_k every vacuum site is diagonal, with I-independent eigenvalues $(u - \tilde{\theta}_A)$ and $(u - \tilde{\theta}_F - h)$. If the Lax product of pair 1 at the evaluation point u_* has column support inside D_k, the index stays in D_k through pairs $2,\dots,L$ and the twist only closes the trace:

$$\langle \mathrm{vac}^L|t_k(u_*) = \lambda_k(u_*)\,\langle \mathrm{vac}|t_k^{(1\ \mathrm{pair})}(u_*) \otimes \langle \mathrm{vac}^{L-1}|, \tag{27}$$

$$\lambda_k(u) = \prod_{n\geq 2}\ \prod_{j=0}^{k-1}(u - \theta_{A,n} - jh)\ (u - \theta_{F,n} - kh)\ \prod_{j=0}^{k-2}(u - \theta_{F,n} - jh), \tag{28}$$

for every twist, and by the covariance of §2.5 for every global rotation of the vacuum, with no goodness condition. The F site at $u_* = \tilde{\theta}_F$ removes all columns without a 4; the A site at $u_* = \theta_A - h$ removes all columns containing a 1. Hence:

- $k = 1$: $t_1(\theta_F)$ localises for any geometry;
- $k = 3$: $\hat{t}_3(\theta_A - h)$ localises for any geometry;
- $k = 2$: both conditions are needed at $u_* = \theta_F + h = \theta_A - h$, i.e. $t_2(\theta_F + h)$ localises on a pair *only at the ABJM point* $\theta_A - \theta_F = 2h$.

[exact], $L = 2,3$ pairs [`pair-group-rotation.wb` §§2–3]. A scan of all single $t_k(\theta_F + mh)$ on four sites with a random twist and a random BPS reference, $k = 1,2,3$, $m = -6,\dots,7$, $d = (\theta_A - \theta_F)/h = -5,\dots,5$ (binary-exact floats, [python] [`checks/session-2026-10-12/pairgroup/`]; $d = -2,\dots,4$ in the notebook):

d	single words localising on pair 1
$-5,\dots,1,3,5$	$t_1(\theta_F)$, $\hat{t}_3(\theta_A - h)$
2 (ABJM)	$t_1(\theta_F)$, $t_2(\theta_F + h)$, $\hat{t}_3(\theta_F + h) = \hat{t}_3(\theta_A - h)$
4	$t_1(\theta_F)$ ($t_3(\theta_A - h)$ vanishes identically)

The localised algebra on pair 1 is therefore 5-dimensional at the ABJM point, $\{1,\ X_1 = t_1(\theta_F),\ X_2 = t_2(\theta_F + h),\ X_3 = \hat{t}_3(\theta_F + h),\ X_3X_1\}$, and 4-dimensional off it. This is why all generic-θ searches for a pair rotation failed, and it is the mechanism behind the pair κ-family of §16.[4]

[3] In the python convention of the 2026-10-12 reports, $\sum_a (-1)^a \tau_a Q(u - ah) = 0$ with $Q^{\mathrm{RV}}(u) = Q(u-h)$, the same columns sit at $\theta_A - 3h$.

[4] The python check `pairword_check.py` and the text of the notebook of 12-10 state that every localising word, single or product, equals the localisation constant $\prod \lambda_k$ times the same word on the one-pair chain. The rerun of 13-10 shows this for the five K_κ words (the only product word there, X_3X_1, has constant $\lambda_1\lambda_3$), but four other two-factor words, $t_1(\theta_F)t_1(\theta_F + h)$, $t_1(\theta_F)\hat{t}_3(\theta_F + 3h)$, $t_1(\theta_F + 2h)\hat{t}_3(\theta_F + h)$ and $\hat{t}_3(\theta_F + h)\hat{t}_3(\theta_F + 2h)$, localise with constants 51/31 or 71/31 times $\prod \lambda_k$ [`pair-group-rotation.wb` cell 17; `notes/rerun-2026-10-13.md`]. K_κ is unaffected.

4.3 The shift operator in the regularised words

Write $X_1 = t_1(\theta_F)$ and $\hat{X}_3 = \hat{t}_3(u_0)$. At $L = 2$, untwisted, homogeneous ($\theta_A = 1$, $\theta_F = 0$, $h = \frac{1}{2}$) one has the exact operator identity

$$\hat{X}_3 X_1 = 24\, h^7 \lambda_1 \lambda_3\, S, \qquad \lambda_1 = \tfrac{1}{2},\ \lambda_3 = \tfrac{3}{8}, \quad 24h^7\lambda_1\lambda_3 = \tfrac{9}{256}, \tag{29}$$

[exact] [`even-bridge-epsilon-expansion.wb` §10, `Wa-Fonly-L6.wb` (ket and bra port test)]. Here $\hat{X}_3 = \frac{1}{2}t_3''(u_0)$ because the prefactor of t_3 has a double zero at u_0, one from each F site, and λ_1, λ_3 are the localisation constants of the other pair with the vanishing factor dropped (the λ_k of §4.2, here $\lambda_1(\theta_F)$ and $\lambda_3(u_0)$); S is the translation by one pair, which at $L = 2$ swaps the two pairs (the ε-notes call it P). The full operator $\hat{X}_3 X_1$ is not a scalar. On the eigenvalue side, with $A = \tau_F/(h^2\lambda_1)$ and $C = \hat{\tau}_3(u_0)/(6h^5\lambda_3)$ (twist $y = 1$) the identity gives

$$A\,C = 4 \qquad \text{on translation-invariant states } (S = +1), \tag{30}$$

i.e. $\tau_F \hat{\tau}_3(u_0) = 24h^7\lambda_1\lambda_3$ is state independent. It holds on all 6 cyclic highest-weight states at $L = 2$ (exact operator identity) and on all 26 at $L = 3$ [python] [`an3.py` in `checks/session-2026-10-13/epsexp/`]; the Q-side use is in §16.

This is the same statement as the shift test $U(0) \propto S$ of §4.1, in the regularised normalisation: $\hat{X}_3 = \prod_{\text{rest}} g^{(3)}(u_0)\, \bar{t}_3(u_0)$ where the remaining prefactors at $L = 2$ are $(6h^2)^2 h^2 = 36h^6 = \frac{9}{16}$, and $\frac{9}{16} \cdot \frac{1}{16} = \frac{9}{256}$ reproduces the constant above from $U(0) = S/16$ (an arithmetic cross-check of the two statements, made here).

4.4 Consequences for the SoV basis

[`sov-abjm-point-limit.wb`, `checks/abjmlimit/`; not yet rerun, [python] with mpmath at 10^{-45}] Ryan–Volin's genericity assumption (2.9) fails and B is degenerate (§13 has the full account). In short:

- With pair-by-pair $\theta = (\theta_{F,1} + d, \theta_{F,1}, \theta_{F,2} + d, \theta_{F,2})$ the generic words have rank 256; at $d = 2h$ the rank drops to $81 = 9^2$ (nine per resonant pair), and for one pair $\det M(d) \propto d^{23}(d - h)^{11}(d - 2h)^7(d - 3h)^3(d - 4h)^4$.
- B has Jordan blocks at $d = 2h$: 10 simple and 3 of size two per pair, i.e. 13 distinct eigenvalues.
- The regularised basis is the $d \to 2h$ limit of the RV basis. Per resonant A site $O(a, k) \in \{1,\ t_1(u_0),\ W,\ Z\}$ with $W = t_2'(u_0) - t_1'(u_0)t_1(\theta_F)$ for $k \le 2$ ($t_2(u_0)$ at $k = 3$) and $Z = t_3'(u_0) = g_3'\,\hat{t}_3(u_0)$; the words $\langle 0|O_1 t_1(\theta_2)^{k_2} O_3 t_1(\theta_4)^{k_4}$ have rank 256, and on states $\langle x|\Psi\rangle = \langle 0|\Psi\rangle \prod_{\text{pairs}} \lambda(a, k)\tau_F^k$ with $\lambda \in \{1,\ x = \tau_1(u_0),\ w = \tau_2'(u_0) - \tau_1'(u_0)\tau_F$ ($k \le 2$) or $x\tau_F$ ($k = 3$), $y = \tau_3'(u_0)\}$ [python]. The data x, w, y come from the u-derivative of the twisted Baxter equation.
- Fixed-bra coefficients diverge like $\delta^{-\#\text{resonant pairs}}$ but overlaps converge (10^{-45}). The K_κ word of §16 is built from exactly these regularised pair words: $X_2 = t_1(u_0)X_1$, $X_3 \propto Z$.
- Fully pair-periodic θ (coincident inhomogeneities) collapses the generic words to rank 70 and needs shift derivatives up to order 7: the homogeneous limit is a separate open problem. Use pair-by-pair θ.

5 Localising words and free-parameter rotations

[`L2-ffbar-SoV-localisation.wb` (2 sites), `L4-ffbar-free-rotations.wb` (4 sites), `checks/chk_common13.py`, `checks/chk_rotref.py`]

Setting and notation. Chain A F A F (four sites = two pairs) unless stated, companion twist $G = G_c(z)$ at the standard point ($h = \frac{1}{2}$, $\theta = (2/7, -3/5, 8/11, -11/8)$, $z = (3/2, 5/4, 2/3, 4/7)$), $\theta_{ab} = \theta_a - \theta_b$, $\sigma_k = e_k(z)$ (so $\det G = \sigma_4$), $\gamma := G^{-T}$. Bras are row vectors and operators act from the right. Two kinds of site operators are used. $\rho_s(g)$ is the representation matrix of $g \in GL(4)$ on site s, i.e. g on an A site and g^{-T} on an F site, so $\rho_A(\gamma^p) = G^{-pT}$ and $\rho_F(\gamma^p) = G^p$. $M_{[s]}$ is the plain 4×4 matrix M acting on the row vector of site s ($v \mapsto vM$); thus $\rho_F(g) = (g^{-T})_{[s]}$. A word W *localises* on a set of sites S of a reference Ω if $\langle\Omega|W = c \bigotimes_{s\in S} x_s \otimes (\Omega \text{ on the other sites})$ with c a scalar; overlaps with an eigenstate Ψ then reduce to eigenvalue polynomials times $\langle\Omega|\Psi\rangle$. Words are built from $t_1(\theta_s + mh)^{\pm 1}$ and $t_3(\theta_s + mh)$ with twist $G_c(z)$ (§3). The reference $\langle 0| = e_1 \otimes e_4 \otimes e_1 \otimes e_4$ is the SoV vacuum (§13); a "generic rotated vacuum" is $\Omega = \langle 0|\rho(g)$ with g a generic element of $GL(4)$ (the cross-check element of the notebook has $\det g = -58$).

qKZ form. $\mathcal{L}_A(\theta) = hK$ has rank one and $\mathcal{L}_F(\theta) = -hP$, so t_1 at an inhomogeneity transports the twist onto one site and leaves R-matrices. At two sites, for any twist matrix G (symbolic) [exact]

$$\begin{aligned} t_1(\theta_1) &= h\,(G^T)_{[1]}(\theta_{12} - hK), & t_1(\theta_2) &= h\,G_{[2]}(\theta_{12} - hK), \\ t_1(\theta_1)\,t_1(\theta_2)^{-1} &= (G^T)_{[1]}(G^{-1})_{[2]} = \rho(G^T), \end{aligned} \tag{31}$$

the global twist rotation. At four sites [exact]

$$\begin{aligned} t_1(\theta_1) &= h\,(G^T)_{[1]} R_{14} R_{13} R_{12}, & t_1(\theta_2) &= -h\,R_{23} R_{24} G_{[2]} R_{21}, \\ t_1(\theta_3) &= h\,R_{32} R_{31} (G^T)_{[3]} R_{34}, & t_1(\theta_4) &= -h\,G_{[4]} R_{41} R_{42} R_{43}, \end{aligned} \tag{32}$$

with $R_{ab} = \theta_{ab} \pm hK_{ab}$ for an A–F pair and $\theta_{ab} \pm hP_{ab}$ for a same-type pair. The mechanism behind everything below: the R-factors on either side of the twist matrix are scalars on the reference as long as the singlet overlap of the site vectors of each A–F pair vanishes (and same-type pairs are symmetric); then the word is a pure site rotation.

Bare vacuum: every site localises [exact]. Let $X = t_1(\theta_1)^{-1}$, $Y = t_1(\theta_2)$, $Z = t_1(\theta_3)$, $W = t_1(\theta_4)^{-1}$. Then

$$\begin{aligned} \langle 0|X &= c_1 \langle 0|\rho_1(\gamma), & c_1 &= \frac{1}{h\,\theta_{12}(\theta_{13}+h)\theta_{14}}, \\ \langle 0|Y &= l_2 \langle 0|\rho_2(\gamma), & l_2 &= h\,\theta_{12}\theta_{23}(\theta_{24}-h), \\ \langle 0|Z &= l_3 \langle 0|\rho_3(\gamma^{-1}), & l_3 &= h\,\theta_{32}(\theta_{31}+h)\theta_{34}, \\ \langle 0|W &= c_4 \langle 0|\rho_4(\gamma^{-1}), & c_4 &= \frac{-1}{h\,\theta_{43}(\theta_{42}-h)\theta_{41}}. \end{aligned} \tag{33}$$

Write $(O_1, O_2, O_3, O_4) = (X, Y, Z, W)$, $(\lambda_1, \ldots, \lambda_4) = (c_1, l_2, l_3, c_4)$ and $o_s = O_s/\lambda_s$. Then each site has one free parameter, $\langle 0|(1 - \kappa\, o_s) = v_s(\kappa) \otimes \mathrm{vac}_{\bar{s}}$ (the vacuum on the other three sites), with

$$\begin{aligned} v_1 &= e_1^T(1 - \kappa G^{-T}) = (1, 0, 0, \kappa/\sigma_4), & v_2 &= e_4^T(1 - \kappa G) = (0, 0, -\kappa, 1), \\ v_3 &= e_1^T(1 - \kappa G^T) = (1 - \kappa\sigma_1, -\kappa, 0, 0), & v_4 &= e_4^T(1 - \kappa G^{-1}) = \big(\kappa, -\kappa\sigma_1, \kappa\sigma_2, \sigma_4 - \kappa\sigma_3\big)/\sigma_4. \end{aligned} \tag{34}$$

The corresponding group elements are $h_1 = 1 - \kappa G^{-T}$, $h_2 = (1 - \kappa G)^{-T}$, $h_3 = 1 - \kappa G^T$, $h_4 = (1 - \kappa G^{-1})^{-T}$, with $\rho_A(h) = h$, $\rho_F(h) = h^{-T}$. The two end identities (X and W) also hold for a generic global rotation $\langle 0|\rho(g)$; the two interior ones do not (see below). Also $t_1(\theta_s)\,t_3(\theta_s + mh)$ is an exact scalar at $(s, m) = (1, -1), (2, 3), (3, -1), (4, 3)$, so these t_3 operators are not new: $t_3(\theta_1 - h) = q_1 t_1(\theta_1)^{-1}$ with $q_1 = -6h^4 \det G\, \theta_{12}(\theta_{12}-h)(\theta_{12}-2h)(\theta_{12}-4h)\,(\theta_{13}+h)(\theta_{13}-h)(\theta_{13}-2h)(\theta_{13}-3h)\,\theta_{14}(\theta_{14}-h)(\theta_{14}-2h)(\theta_{14}-4h)$, and the one-site rotation of site 1 is also a t_3-word, as used in the SoV basis (§13) [exact] [`L4-ffbar-free-rotations.wb`, cells 10–25].

On eigenstates, $\langle 0|(1 - \kappa o_s)|\Psi\rangle = (1 - \kappa\hat{o}_s)\langle 0|\Psi\rangle$ with $\hat{o}_s$ the eigenvalue of o_s:

$$\hat{o}_1 = \frac{h\theta_{12}(\theta_{13}+h)\theta_{14}}{\tau_1(\theta_1)}, \quad \hat{o}_2 = \frac{\tau_1(\theta_2)}{h\theta_{12}\theta_{23}(\theta_{24}-h)}, \quad \hat{o}_3 = \frac{\tau_1(\theta_3)}{h\theta_{32}(\theta_{31}+h)\theta_{34}}, \quad \hat{o}_4 = \frac{-h\theta_{43}(\theta_{42}-h)\theta_{41}}{\tau_1(\theta_4)} \tag{35}$$

[num, 10^{-51}] on all 256 eigenstates (60-digit eigenvectors from exact weight blocks); $\langle 0|\Psi\rangle \neq 0$ for all 256 eigenstates (smallest $6.2 \cdot 10^{-4}$), i.e. $\langle 0|$ is cyclic for the Bethe algebra. Of the sixteen products $X^{e_1}Y^{e_2}Z^{e_3}W^{e_4}$ exactly those with $e \in \{0000$, one entry, 1100, 0110, 0011$\}$ (single sites and adjacent pairs) are pure site rotations, so there

$$\langle 0| \prod_{s\in S}(1-\kappa_s o_s)|\Psi\rangle = \prod_{s\in S}(1-\kappa_s \hat{o}_s)\,\langle 0|\Psi\rangle \qquad (S=\{s\} \text{ or an adjacent pair}). \tag{36}$$

Same-type pairs pick up an exchange term from $R = \theta \pm hP$: the pure product rotations $\Pi_{13} = \langle 0|\rho_1(\gamma)\rho_3(\gamma^{-1})$, Π_{24}, Π_{123}, Π_{234} (rotations γ, γ^{-1} alternating along the listed sites) are, as words on $\langle 0|$, $\Pi_{13} = [(\theta_{13}-h)o_1o_3 + h]/\theta_{13}$, $\Pi_{24} = [(\theta_{24}-h)o_2o_4+h]/\theta_{24}$, $\Pi_{123} = [(\theta_{13}-h)o_1o_2o_3 + ho_2]/\theta_{13}$, $\Pi_{234} = [(\theta_{24}-h)o_2o_3o_4 + ho_3]/\theta_{24}$, e.g. on $\{1,3\}$

$$\Big[(1-\kappa_1\hat{o}_1)(1-\kappa_3\hat{o}_3) - \frac{h}{\theta_{13}}\kappa_1\kappa_3(\hat{o}_1\hat{o}_3 - 1)\Big]\langle 0|\Psi\rangle, \tag{37}$$

and for three consecutive sites $\langle v_1(\kappa_1)\otimes v_2 \otimes v_3 \otimes \mathrm{vac}_4|\Psi\rangle = [ABC - \frac{h}{\theta_{13}}\kappa_1\kappa_3(\hat{o}_1\hat{o}_3 - 1)B]\langle 0|\Psi\rangle$, $A = 1-\kappa_1\hat{o}_1$, etc. [num, 10^{-51}]; $\{1,2,3\}$ and $\{2,3,4\}$ are the maximal site sets on which the product deformation is polynomial in the o_s. The exchange dressings of the two end operators that are not site rotations are

$$\begin{aligned}
\langle 0|t_1(\theta_1) &= h\theta_{12}\theta_{14}\big[\theta_{13}\langle 0|\rho_1(\gamma^{-1}) + h\langle 0|\rho_3(\gamma^{-1})\big], \\
\langle 0|t_1(\theta_4) &= -h\theta_{41}\theta_{43}\big[\theta_{42}\langle 0|\rho_4(\gamma) - h\langle 0|\rho_2(\gamma)\big] \ \text{[exact]}
\end{aligned} \tag{38}$$

(block $\{1,3\}$, A–A exchange; block $\{2,4\}$, F–F exchange). *Both ends* $\{1,4\}$ do not factorise: $\langle 0|o_1o_4 = \Pi_{14} + \mu\, Q_{14}$ with $\Pi_{14} = \langle 0|\rho_1(\gamma)\rho_4(\gamma^{-1})$, $\mu = -h/((\theta_{14}-4h)\det G)$, with $Q_{14} = \langle S_{14}|\rho_4(\gamma^{-1})\otimes e_4^T \otimes e_1^T$ a singlet-channel covector on sites 1,4, so longer words are needed (the four failing products $e = 1001, 1011, 1101, 1111$ lie in the span of the words plus four singlet covectors $\langle S_{14}|\rho_4(\gamma^n)\otimes\{\mathrm{vac}_2, \mathrm{vac}_2\gamma\}\otimes\{\mathrm{vac}_3, \mathrm{vac}_3\gamma^{-1}\}$ exactly for $n=-1$) [exact]. Why the interior sites localise only on the bare vacuum: for $g=1$ the singlet term of $t_1(\theta_2)$ vanishes because the singlet overlap is $e_1\cdot(e_4^TG)^T = e_1\cdot e_3 = 0$; for a generic rotation $\langle 0|\rho(g)$ it does not.

Global symmetries and counting. The global torus $\rho(X)$, $X \in \mathbb{C}[G^T]$, commutes with all t_a (eigenvalue $\prod_i f(z_i)^{\mu_i}$ for $X = f(G^T)$ on a state of weight μ); the four κ_s together with the torus give a Jacobian of rank 7 [exact]. Reach on the 256 states [exact]: global rotations see only the 84-dimensional vacuum multiplet $[2,0,2]$; the 16 products $\prod_s(1-\kappa_s o_s)$ see all six irrep types; κ's plus torus span 193 of 256. *No higher twist powers* [num, 10^{-30}]: of 2288 fused transfer matrices $T_\lambda(\theta_\alpha + mh)^{\pm 1}$, $|\lambda| \le 6$ (at most 4 rows), acting on $\langle 0|$, 680 vanish, 86 localise on at most three sites (56 are quantum determinants), and the 71 that localise on at most two sites reduce to the seven operators $t_1(\theta_s)^{\pm 1}$ and the scalar; none gives γ^p, $|p| \ge 2$, on a site: a single word moves a site only within span$\{\mathrm{vac}, \mathrm{vac}\,\gamma^{\pm 1}\}$ [`L4-ffbar-free-rotations.wb`, cells 14–18].

Generic globally rotated vacuum Ω: only the ends localise [exact]. For $\Omega = \langle 0|\rho(g)$, g generic, only $t_1(\theta_1)^{-1}$ and $t_1(\theta_4)^{-1}$ are single-site rotations, with the constants c_1, c_4 above, and the end blocks are

$$\langle\Omega|t_1(\theta_1)^{-1}t_1(\theta_2) = c_{12}\langle\Omega|\rho_1(\gamma)\rho_2(\gamma), \qquad c_{12} = \frac{\theta_{23}(\theta_{24}-h)}{(\theta_{13}+h)\theta_{14}} = c_1 l_2, \tag{39}$$

and the mirror block $\{3,4\}$. Of the sixteen two-site SoV words $\langle 0|t_a(\theta_1 - h)t_1(\theta_2)^k$ only $\{1, t_3(\theta_1 - h)\}\times\{1, t_1(\theta_2)\}$ localise on the block $\{1,2\}$, with $\langle\Omega|W = c_{ak}\langle\Omega_2|W^{(2)}\otimes\langle\Omega_2|$ ($W^{(2)}$ the same word on the two-site chain with the same twist), $c_{01} = \theta_{23}(\theta_{24}-h)$, $c_{30} = (\theta_{13}-h)(\theta_{13}-2h)(\theta_{13}-3h)(\theta_{14}-h)(\theta_{14}-2h)(\theta_{14}-4h)$, $c_{31} = c_{30}c_{01}$, independent of g (the analogue of 2603 (G2), one operator per site) [exact] [cell 12]. Interior sites do not localise because of the singlet term just described.

Common rotations and good references [python] (10^{-14}). On $\langle 0|$ write $R_s = \langle 0|\rho_s(G^T)$ for $s = 1, 3$, $R_{13} = \langle 0|\rho_1(G^T)\rho_3(G^T)$. Then $\langle 0|t_1(\theta_3) = l_3 R_3$, $\langle 0|t_1(\theta_1) = \alpha(\theta_{13}R_1 + hR_3)$ with $\alpha = h\theta_{12}\theta_{14}$, and $\langle 0|t_1(\theta_1)t_1(\theta_3) = m\,R_{13}$ exactly ($m = 0.056461$ at the point, closed form not fitted). Hence, with $g = 1 - \kappa G^T$,

$$\langle 0|\,\Xi_{13}(\kappa) = \langle 0|\rho_1(g)\rho_3(g), \qquad \Xi_{13}(\kappa) := 1 + a\,t_1(\theta_1) + b\,t_1(\theta_3) + c\,t_1(\theta_1)t_1(\theta_3),$$
$$a = -\frac{\kappa}{\alpha\theta_{13}}, \qquad b = -\frac{\kappa(\theta_{13} - h)}{\theta_{13}l_3}, \qquad c = \frac{\kappa^2}{m}, \tag{40}$$

a common one-parameter rotation of sites 1 and 3 with overlap $\Xi_{13}(\kappa)|_\tau\langle 0|\Psi\rangle$, where $\Xi_{13}|_\tau$ is the same polynomial in $\tau_1(\theta_1), \tau_1(\theta_3)$ (2603 resolvent structure; 10^{-14}) [`checks/chk_common13.py`]. It is a *global* rotation: $\langle 0|\rho_1(g)\rho_3(g)$ sees only the first row of g, so $g \to g' = 1 - \kappa(\sigma_1 E_{11} + E_{12}) \propto \exp(-\kappa' E_{12})$ fixes e_4^T and $\langle 0|\rho_1(g)\rho_3(g) = \langle 0|\rho(g')$ on all four sites (not with g itself, since G_c^T contains E_{34}). The $\{2, 4\}$ analogue is generated by E_{34}. Earlier "no common rotation" statements, built from $t_1(\theta_1)^{-1}$, which gives G^{-T} on site 1 against G^T on site 3, are superseded.

Good references [python] [`checks/chk_rotref.py`, 10^{-14}]: a global rotation $v \otimes w \otimes v \otimes w$ with $v \cdot w = 0$ is as good a reference as $\langle 0|$ (all four sites localise, both common rotations exist with the same a, b, c) iff $w\,G\,v^T = 0$, i.e. iff the conjugated twist keeps $(G)_{41} = 0$. For $v = e_1^T p(G^T)$, $w = e_4^T q(G)$, since $(G^k)_{41} = 0$ for $k \le 2$ and $(G^3)_{41} = 1$: the rotation is global iff $\deg p + \deg q \le 2$, good iff $\deg p + \deg q \le 1$. So $\langle 0|\,\Xi_{13}(\kappa_1)\Xi_{13}(\kappa_2) = (e_1^T(1 - \kappa_1 G^T)(1 - \kappa_2 G^T) \otimes e_4)^{\otimes 2}$ ($(G^T)^2$ on a pair of sites) and $\langle 0|\,\Xi_{13}(\kappa_1)\Xi_{24}(\kappa_2)$, with Ξ_{24} the $\{2, 4\}$ analogue, is one global rotation (A: $e_1^T(1 - \kappa_1 G^T)$, F: $e_4^T(1 - \kappa_2 G)$) with overlap $\Xi_{13}(\kappa_1)|_\tau\,\Xi_{24}(\kappa_2)|_\tau\langle 0|\Psi\rangle$, polynomial and without resolvent. References with $\deg p + \deg q = 2$ are global but not good, and a third step leaves the global orbit. The nine words $\langle 0|X_1^a X_3^b$ ($X_1 = t_1(\theta_1)$, $X_3 = t_1(\theta_3)$, $a, b \le 2$) are independent and span exactly $\{e_i \otimes e_4 \otimes e_j \otimes e_4 : i, j \le 3\}$; the global rotations inside are $v \otimes e_4 \otimes v \otimes e_4$ with $v \in \mathrm{span}(e_1, e_2, e_3)$ (generated by E_{12}, E_{13}, E_{11} of the $\mathfrak{gl}(3)$ on $\{1, 2, 3\}$), and $(G^T)^2$ on one site alone is also in the span (combinations of words beat the single-T_λ scan above); $v_4 \neq 0$ is not. On the two-parameter state Ω_2 (not a good reference, $wGv^T \neq 0$) only end-growing words localise, as fixed twist rotations: $t_1(\theta_1)^{-1} \to \rho_1(\gamma)$, $\cdot\,t_1(\theta_2) \to \rho_{12}(\gamma)$, $\cdot\,t_1(\theta_3)^{-1} \to \rho_{123}(\gamma)$, $\cdot\,t_1(\theta_4) \to \rho(\gamma)$ (the torus), mirror from site 4 with γ^{-1}; free parameters only on one end site [`checks/scan_*.py`, `ABJM-3pt-transfer-matrices.md` §4].

Where this goes next. The companion-frame results above are the template. The physical references of the ABJM three-point function are diagonal-frame global rotations, not $\langle 0|$ (§14); the free parameters they need on a pair come from the pair κ-family at the ABJM point (§16), whose admissible references are classified in §17. The twisted vacua of the next section replace $\langle 0|$ by an eigenbra of a different companion twist and ask which of the words above survive.

6 Twisted vacua as references

[`twisted-vacuum-localisation-L2.wb`, `twisted-vacuum-localisation-L4.wb`, `twisted-vacuum-blocks-L3.wb`, `twisted-vacuum-wordspan-L2.wb`, `twisted-vacuum-untwisting.wb`, `twisted-vacuum-companion-check.wb`]

Question, scope, status. A reference that is an eigenbra of the transfer matrices of a *different* companion twist $G_c(w)$ has overlaps that are finite SoV Q-sums (§18). The question of this section is which words of the transfer matrices of the twist z of the state Ψ act on such a reference as free-parameter rotations, in the sense of §5. Sizes are given in sites (2 sites = one pair); the suffixes of the notebook names are inconsistent (sites in two of them, pairs in the others). The notebooks were run exactly (rational arithmetic, or 50–60 digits where marked) on the shared kernel before the Wolfram licence expired on 2026-10-12 and have no stored outputs; results are pasted in markdown cells and a rerun is pending. The tags below refer to those runs. Standard point: $h = \frac{1}{2}$, $\theta = (2/7, -3/5, 8/11, -11/8)$, $z = (3/2, 5/4, 2/3, 4/7)$, $w = (7/5, 9/8, 3/7, 5/9)$; $G := G_c(z)$.

6.1 The vacuum of the twist $G_c(w)$

Let $V(x)$ be the Vandermonde matrix with columns $(x_j^3, x_j^2, x_j, 1)^T$. Put

$$v_w = (w_1^3, w_1^2, w_1, 1), \qquad u_w = \text{row 4 of } V(w)^{-1},$$
$$(u_w)_k = (-1)^{k-1}\frac{\sigma_{k-1}(w_1,w_2,w_3)}{\prod_{m\le 3}(w_4 - w_m)}, \quad k = 1,\dots,4, \tag{41}$$

with σ_j the j-th elementary symmetric polynomial ($\sigma_0 = 1$), so that $G_c(w)v_w^T = w_1 v_w^T$ (right eigenvector), $u_w G_c(w) = w_4 u_w$ (left eigenvector) and $u_w \cdot v_w = 0$; u_w depends on w_4 only through its normalisation. The twisted vacuum is

$$\langle 0_w| := (v_w \otimes u_w)^{\otimes L} \qquad (\text{A sites } v_w,\ \text{F sites } u_w). \tag{42}$$

It is an exact eigenbra of t_1, t_2, t_3 with twist $G_c(w)$ and the vacuum eigenvalue [exact] (two sites: all a; four sites: $a = 1, 3$; §3 of the first two notebooks listed above). At two sites

$$\tau_a(u) = \sum_{i_1<\cdots<i_a}\prod_{k=1}^{a} w_{i_k}\lambda_{i_k}(u-(k-1)h), \qquad \lambda_i(u) = (u - \theta_1 + h\delta_{i1})(u - \theta_2 - h\delta_{i4}), \tag{43}$$

and at four sites (A F A F)

$$\tau_1(u) = w_1(u-\theta_1+h)(u-\theta_2)(u-\theta_3+h)(u-\theta_4) + (w_2+w_3)\prod_s(u-\theta_s)$$
$$+ w_4(u-\theta_1)(u-\theta_2-h)(u-\theta_3)(u-\theta_4-h). \tag{44}$$

Among product states with A site a right and F site a left eigenvector of $G_c(w)$, the eigenbras are exactly those with different eigenvalue indices ($A = C_i$, $F = R_j$, $i \neq j$: 12 states, four sites) [exact]. At $w = z$ the bra is the Bethe vacuum of $t^{G_c(z)}$ (in the diagonal frame of §6.2 it is $\langle \text{vac}^L| = (e_1 \otimes e_4)^{\otimes L}$). The bare vacuum $e_1 \otimes e_4$ is *not* an eigenbra of $t^{G_c(w)}$ (rank-two image, two sites) [exact]. The sections below take $w \neq z$ and let words of $t^{G_c(z)}$ act.

6.2 The obstruction $q(w, z)$

Define

$$q(w,z) := u_w\, G_c(z)\, v_w^T = -\frac{\prod_{i=1}^4(w_1 - z_i)}{\prod_{m\le 3}(w_4 - w_m)} = -(u_w)_1\prod_i(w_1 - z_i) \text{ [exact]}, \tag{45}$$

which follows from $G_c(z)v(x)^T = x\,v(x)^T - \chi_z(x)e_1^T$ with $v(x) = (x^3, x^2, x, 1)$ and $\chi_z(x) = \prod_i(x - z_i)$, together with $u_w \cdot v_w = 0$ (verified symbolically). At the standard point $q = 232551/1558000$. It vanishes iff $w_1 \in \{z_1, \dots, z_4\}$; moving w_4 into $\{z\}$ changes nothing (same word patterns). In the diagonal frame, with $\langle x_{\text{diag}}| = \langle x_{\text{comp}}|R$, $R = V(z)^{-T}$ on A and $V(z)$ on F,

$$(v_w)_{\text{diag}} = \big(l_j(w_1)\big)_j, \quad l_j(x) = \prod_{m\neq j}\frac{x - z_m}{z_j - z_m}, \qquad (u_w)_{\text{diag}} = \Big(\prod_{m\le 3}\frac{z_j - w_m}{w_4 - w_m}\Big)_j, \tag{46}$$

so $(v_w)_{\text{diag}} = e_i$ when $w_1 = z_i$, and $q = u_{\text{diag}}\Lambda v_{\text{diag}}^T$ is exactly the goodness quantity $w\,G\,v^T$ of §5 transported to the diagonal frame (the README criterion $u\Lambda v^T = 0$) [exact]. So $\langle 0_w|$ is a "good reference" in the sense of §5 iff $w_1 \in \{z\}$.

Mechanism (two sites, twist $G = G_c(z)$) [exact]: $t_1^z(\theta_1) = h(G^T)_{[1]}(\theta_{12} - hK)$ and $t_1^z(\theta_2) = hG_{[2]}(\theta_{12} - hK)$ with the rotation standing to the *left* of K, so on a bra the rotation acts first and K then sees the rotated pair; $\langle 0_w|K = 0$ but $(\langle 0_w|\rho(\text{rot}))K = q\,\langle S|$, $S = \sum_i e_i \otimes e_i$. Hence

$$\langle 0_w|t_1^z(\theta_2) = h\theta_{12}\,(v_w \otimes u_w G) - h^2 q\,\langle S|, \qquad \langle 0_w|t_1^z(\theta_1) = h\theta_{12}\,(v_w G^T \otimes u_w) - h^2 q\,\langle S|. \tag{47}$$

Inverse words always localise, because $\langle 0_w|K = 0$ makes the resolvent $(\theta_{12} - hK)^{-1}$ trivial on it:

$$t_1(\theta_1)^{-1} \to v_w G^{-T}, \quad t_1(\theta_2)^{-1} \to u_w G^{-1}, \quad t_3(\theta_1 - h) \to v_w G^{-T}, \quad t_3(\theta_2 + 3h) \to u_w G^{-1},$$
$$t_1(\theta_1)t_1(\theta_2)^{-1} = G^T \otimes G^{-1} \text{ [exact]}. \tag{48}$$

6.3 Generic w: only the ends and end-growing chains

Table of the two-site action ($v = v_w$, $u = u_w$, $G = G_c(z)$; words acting on $\langle 0_w|$; all [exact], single point $w = (7/5, 9/8, 3/7, 5/9)$, and $w = (3/2, 9/8, 3/7, 5/9)$ for the locus $w_1 = z_1$ of §6.5):

word	generic w ($q \neq 0$)	locus $w_1 = z_i$ ($q = 0$)
$t_1(\theta_1)^{-1}$, $t_3(\theta_1 - h)$	site 1: vG^{-T}	trivial (scalar)
$t_1(\theta_2)^{-1}$, $t_3(\theta_2 + 3h)$	site 2: uG^{-1}	site 2: uG^{-1}
$t_1(\theta_1)$	rank 4, entangled	trivial (scalar)
$t_1(\theta_2)$	rank 4, entangled	site 2: uG
$t_1(\theta_1)t_1(\theta_2)^{-1}$	A: vG^{T}, F: uG^{-1}	site 2: uG^{-1}
$t_1(\theta_1)^{-1}t_1(\theta_2)$	A: vG^{-T}, F: uG	site 2: uG
$t_1(\theta_1)t_1(\theta_2)$, $t_1(\theta_1)^{-1}t_1(\theta_2)^{-1}$	rank 4, entangled	site 2: uG, uG^{-1}
$t_1(\theta_s + mh)$, $m \neq 0$	rank 3–4, not a product	site 2 only: $u(1 + \kappa G)$, κ fixed by m
t_2 at tested points	rank 3–4 or zero	site 2 only, not in span$\{1, G^{\pm 1}\}$ (needs G^2)

At four sites (A F A F) with generic w ($q \neq 0$; w = standard, and two more points) [exact] [`twisted-vacuum-localisation-L4.wb` §§5–8]: only the end inverses move a single site, $t_1(\theta_1)^{-1}$ by G^{-T} and $t_1(\theta_4)^{-1}$ by G^{-1}, and only the end-growing chains $t_1(\theta_1)^{-1}t_1(\theta_2)\,[t_1(\theta_3)^{-1}[t_1(\theta_4)]]$ give product states, with sites $1, 3 \to \cdot G^{-T}$ and sites $2, 4 \to \cdot G$ (the torus element truncated at site k); the mirror chains from site 4 give site 4: G^{-1}, 3: G^{T}, 2: G^{-1}, 1: G^{T}. All are *fixed* rotations with no free parameter. Interior single words entangle the end block: $t_1(\theta_2)$ entangles $\{1, 2\}$ and leaves 3,4 alone, $t_1(\theta_3)$ entangles $\{3, 4\}$, and the forward end words $t_1(\theta_1)$, $t_1(\theta_4)$ are fully entangled. At six sites the chain $t_1(\theta_1)^{-1}t_1(\theta_2)t_1(\theta_3)^{-1}\cdots$ through all six letters is an exact fixed rotation of sites $1, \ldots, k$ after k letters (A sites by G^{-T}, F sites by G) [`twisted-vacuum-blocks-L3.wb`, Task G: 60 digits and exact rational at four sites]; the difference $t_1(\theta_2) - t_1(\theta_1)$ is not block-localised (Schmidt ratios 10^{-4}, rank 4 across $\{1, 2\}$). For the rotated pair the pairings with the unrotated reference are $B_1 = u_w G^{-1} v_w^T = -0.10075$ and $B_2 = q = 0.14926$, both non-zero (six sites, 60 digits).

6.4 Word span on pair 1

[`twisted-vacuum-wordspan-L2.wb`, four sites, exact] Let $\mathcal{S}$ be the span of $\langle 0_w|W$ over all words W with at most two factors from $\{t_1(\theta_s + mh)\ (m = -2..2),\ t_3(\theta_s + mh)\ (m = -1..3),\ t_1(\theta_s)^{-1}\}$ (703 words, zero matrices dropped), and $\mathcal{S}_{12} \subset \mathcal{S}$ the subspace with sites 3,4 unchanged ($= v_w \otimes u_w$). For generic w [exact]: $\dim \mathcal{S} = 37$ and $\dim \mathcal{S}_{12} = 4$,

$$\mathcal{S}_{12} = \mathrm{span}\{\, v \otimes u,\ vG^{-T} \otimes u,\ vG^{-T} \otimes uG,\ S - \kappa\, v \otimes uG \,\}, \qquad \kappa = \frac{\theta_{12}}{hq}, \tag{49}$$

the last element being $\langle 0_w|t_1(\theta_2)$ up to a scalar (entangled, rank 4). The vectors $v \otimes uG$, $vG^T \otimes u$, S, $vG^T \otimes uG$, $v \otimes uG^{-1}$ are *not* in $\mathcal{S}_{12}$. Adding 400 random three-factor words raises $\dim \mathcal{S}$ to 93 but leaves $\dim \mathcal{S}_{12} = 4$. The rank-one elements of $\mathcal{S}_{12}$ form only two one-parameter lines,

$$\begin{aligned} &\text{(a)}\ \ vG^{-T} \otimes u(\alpha + \beta G): \quad B_1 = u_w G^{-1} v_w^T,\ \ B_2 = \beta q; \\ &\text{(b)}\ \ v(\alpha + \beta G^{-T}) \otimes u: \quad B_1 = \beta\, u_w G^{-1} v_w^T,\ \ B_2 = 0, \end{aligned} \tag{50}$$

where $B_1 = v' \cdot u_w$ and $B_2 = v_w \cdot u'$ are the cross pairings of the rotated pair (v', u') with the unrotated reference pair (the invariants that decide whether the rotated block is extremal with respect to the rest, §10). In each line only one of the two sites carries a free parameter, the other a fixed rotation (G^{-T} on the A site in (a)) or none (the F site in (b)). At the locus $w_1 = z_1$: $\dim \mathcal{S} = 9$, $\dim \mathcal{S}_{12} = 3$, every element has rank one with $v' = v_w$ frozen and u' free in a three-dimensional space, $B_1 = 0$ [exact].

6.5 The locus $w_1 \in \{z\}$: A sites frozen, F sites free

For $w_1 = z_i$ the A vector v_w is an eigenvector of $G_c(z)^T$ (in the sense $v_w G^T = z_i v_w$), so every A rotation acts as a scalar and all words reduce to rotations of the F sites alone; the F vector u_w is not an eigenvector of G and is free. The pairings of rotated and unrotated pairs vanish ($B_1 = v \cdot u = 0$, $B_2 = -\kappa q = 0$: the rotated pair is extremal) [exact]. The words that rotate F sites are the *forward* words in $t_1(\theta_{2k})$ (the "wrong" powers $t_1(\theta_2)^{-1}$, $t_1(\theta_4)$ do not localise; $t_1(\theta_1)^{\pm 1}$, $t_1(\theta_3)^{\pm 1}$ are scalars). With $\lambda = h\theta_{12}\prod_{k\geq 1}\theta_{2,2k+1}(\theta_{2,2k+2} - h)$ (product over the further pairs)

$$\langle 0_w|\big(1 + a\, t_1(\theta_2)\big) = \langle 0_w|(1 - \kappa G)_{[2]}, \qquad a = -\frac{\kappa}{\lambda}, \qquad \lambda = \begin{cases} h\theta_{12}\theta_{23}(\theta_{24} - h) & \text{4 sites} \\ h\theta_{12}\theta_{23}(\theta_{24} - h)\theta_{25}(\theta_{26} - h) & \text{6 sites} \end{cases} \tag{51}$$

[exact] at 2 and 4 sites, [num, 10^{-15}] at 6 sites.[5] This is a free-parameter rotation $u_w \to u_w(1 - \kappa G)$ of the first F site.

Two F sites in common (4 and 6 sites): the words $\{1, t_1(\theta_2), t_1(\theta_4), t_1(\theta_2)t_1(\theta_4)\}$ give

$$\langle 0_w|\big(1 + a\, t_1(\theta_2) + b\, t_1(\theta_4) + c\, t_1(\theta_2)t_1(\theta_4)\big) = \langle 0_w|(1 - \kappa G)_{[2]}(1 - \kappa G)_{[4]}, \tag{52}$$

$$a = -\frac{\kappa(\theta_{24} - h)}{\theta_{24}\lambda}, \qquad b = -\frac{\kappa}{\alpha_4}, \qquad c = \frac{\kappa^2}{\gamma}, \tag{53}$$

with the exchange dressing $\langle 0_w|t_1(\theta_4) = \alpha_4\big(G_{[4]} + \frac{h}{\theta_{24}}G_{[2]}\big)\langle 0_w|$ and $\langle 0_w|t_1(\theta_2)t_1(\theta_4) = \gamma\, G_{[2]}G_{[4]}\langle 0_w|$, where at six sites $\alpha_4 = -h\theta_{41}\theta_{42}\theta_{43}\theta_{45}(\theta_{46} - h)$ (the factors $\theta_{45}(\theta_{46} - h)$ are absent at four sites) and $\gamma = \lambda\alpha_4(\theta_{24} + h)/\theta_{24}$ (residual 10^{-15} of the least-squares fit; the closed forms of α_4, γ were identified by matching at four θ-points, relative tolerance 10^{-6}) [num, 10^{-6}]. At three pairs (six sites) the eight words $\prod_k t_1(\theta_{2k})^{e_k}$ reach $(1 - \kappa G)$ on all three F sites with coefficients $d_S\kappa^{|S|}$ ($S \subseteq \{2, 4, 6\}$, residual $3 \cdot 10^{-15}$), using $\langle 0_w|t_1(\theta_6) = \alpha_6\big(G_{[6]} + \frac{h}{\theta_{46}}G_{[4]} + r_2 G_{[2]}\big)\langle 0_w|$, $\alpha_6 = -h\theta_{61}\theta_{62}\theta_{63}\theta_{64}\theta_{65}$, $r_2 = h(\theta_{46} + h)/(\theta_{26}\theta_{46})$ (same fit status). The three-pair target is not an eigenbra of t_1^z (relative defect $2.6 \cdot 10^{-4}$). On the locus the single words $t_1(\theta_4 + h)$ and $t_3(\theta_2 + 2h)$ also give $(1 - \kappa G)$ on site 4 and $(1 - \kappa G^{-1})$ on site 2, but with κ fixed by the word.

6.6 The diagonal frame and untwisting: neither gives the note's W_{-1}

[`twisted-vacuum-untwisting.wb`] The F-site rotation $u_w \to u_w(1 - \kappa G)$ of §6.5 becomes $f \to f(1 - \kappa\Lambda)$ in the diagonal frame, $f = (u_w)_{\text{diag}}$ (frame map of §6.2). The frame map is [exact]; the overlaps with the saved highest-weight states are [num, 10^{-15}]. Untwisting: $z = 1 + \varepsilon\varphi$, $\kappa = 1 - c\varepsilon$, $w_{2,3,4}$ fixed.

- *Locus $w_1 = z_1$.* The pair becomes $e_1 \otimes (e_4 + \alpha_2 e_2 + \alpha_3 e_3)$ with $\alpha_j = -f_j(\varphi_j - \varphi_4)/(c - \varphi_4)$, $f_j \to (\varphi_1 - \varphi_j)/(\varphi_1 - \varphi_4)$ (F vector normalised to $f_4 = 1$); the A vector is frozen at e_1 and both pairings with the reference vanish identically. The overlaps with the saved states see only the BPS vacuum (sector with all A sites $= Y^1$).

- *Generic w, family (a) of §6.4.* The finite-twist pairings $B_1 = u_w G^{-1} v_w^T$, $B_2 = \beta q$ are non-zero. With $\alpha = -\beta + c\varepsilon$ and $w_{2,3}$ generic the F vector tends to $(c + \beta\varphi_4)e_4 + \beta\sum_{j=2,3}(\varphi_j - \varphi_4)e_j$ (its e_1 component is $O(\varepsilon^2)$ relative) and A tends to $e_1 + O(\varepsilon)$. With $w_2 = z_2$, $w_3 = z_3$ the F vector tends to $e_4 + \tilde{\alpha}e_1$, $\tilde{\alpha} = -\beta\,\dfrac{(\varphi_1 - \varphi_2)(\varphi_1 - \varphi_3)(\varphi_1 - \varphi_4)}{(\varphi_2 - \varphi_4)(\varphi_4 - \varphi_3)(c + \beta\varphi_4)}$, so $B_2 = \tilde{\alpha} \neq 0$ and $B_1 = O(\varepsilon) \to 0$; this bra sees only the states $[L - 1, 0, L - 1]$.

- For comparison, the note's bra $\text{vac}^{L-\ell} \otimes W_{-1}^{\ell}$ with ℓ rotated pairs sees the vacuum and $[L - 2, 0, L - 2]$ and vanishes on $[L - 1, 0, L - 1]$ (§9(ii): $L + p$ odd).

Neither limit is the note's $W_{-1} = (e_1 - e_4) \otimes (e_4 + e_1)$: that pair needs an $O(1)$ rotation $e_1 \to e_1 - e_4$ of the A site, while all A rotations available from words on $\langle 0_w|$ are fixed powers of the twist.

[5] The session report of 2026-10-12 (`session-2026-10-12-agent-reports.md` §1) quotes the six-site case as exact; the notebook run (`twisted-vacuum-blocks-L3.wb`) was at machine precision, which is what is tagged here.

6.7 Relation to $\langle W_a^L|$ and to 2603

$\langle W_a|$ is not a companion eigenbra [`twisted-vacuum-companion-check.wb`, one pair, exact, t_1 only]. The set of twist matrices g for which $\langle W_a|$ is an eigenbra of t_1^g is 11-dimensional: $g_{24} = -g_{21}/a$, $g_{34} = -g_{31}/a$, $g_{42} = ag_{12}$, $g_{43} = ag_{13}$, $g_{44} = g_{11} + ag_{14} - g_{41}/a$. It contains $\Lambda + a(z_1 - z_4)E_{41}$ and $\Lambda + ((z_4 - z_1)/a)E_{14}$ but no companion matrix: in the diagonal frame for no a, in the companion frame ($\langle W_a|R^{-1}$) only for $a = 0$. Hence the twisted vacua are not references for W_a itself; they enter through the *overlap* identity $\langle 0_w|\Psi\rangle = (v_1u_4)^L\langle W_a^L|\Psi\rangle$ for $w = (w_1, z_2, z_3, w_4)$ and Ψ of weight $(\mu, 0, 0, -\mu)$ (§18), while the free parameters on W_a^L come from the pair κ-family (§16).

The reading of 2603 [reading of arXiv:2603.26617, `notes/rv-sov-dictionary.md` §5; no computation]. The site of 2603 is the $\mathbf{6} = \wedge^2\mathbf{4}$ with vacuum $Z = e_1 \wedge e_2$, and its κ-family is the one-point word

$$1 - \kappa\, t_+(\theta_1) + \kappa^2\, t_0(\theta_1) \qquad (t_0 = \wedge^2 \text{ transfer matrix}), \tag{54}$$

which equals the *group element* $\wedge^2(1 - \kappa g)$ acting on the site. Their localisation (App. K) needs only $\langle Z|\mathcal{L}_{jk} = 0$ for $j < k$ together with the bonus zeros $\mathcal{L}_{21} = \mathcal{L}_{43} = 0$, so it holds for *any global rotation* of $\langle Z|$ and *no goodness condition* enters. The rotated vector stays null, and its pairing with the unrotated one is quadratic in $D = 1 - \kappa g$, hence finite and non-extremal; only one κ per block is available. Untwisting takes $z, \kappa, \omega \to 1$ together, the rows of the SoV matrices vanish and are compensated by the normalisation $\mathcal{N}_\ell$, and the final structure constant is $\mathcal{N}\,\mathbb{W}_\omega\mathbb{A}_{\ell,\kappa}/\sqrt{\mathbb{B}}$. The ABJM analogue is a *common pair rotation* (A by $D = 1 - \kappa\Lambda$, F by $D^{-1} \propto \operatorname{adj} D$, cubic): it is automatically BPS, its pairings with the reference are $-\kappa\, u \cdot \Lambda v$ and $\sum_j v_j u_j/(1 - \kappa z_j)$ ($O(1)$ for non-good references) and on W_a^L it maps pair 1 to $W_{a'}$ with a' free. This is what §16 constructs at the ABJM point; the contrast with the generic-w and locus results above (frozen A sites, fixed powers) is the reason the twisted vacua are used only for overlaps.

7 Highest-weight eigenstates and Q-functions

States [`abjm-LR-symmetry.wb`] [num, 10^{-13}]. All highest-weight (HW) cyclic ($S = 1$, pair shift) joint eigenstates of $\bar{t}_1, \bar{t}_3$ at the ABJM point: $6, 26, 152$ at $L = 2, 3, 4$ pairs, 184 in all. All are H_{MZ} eigenstates ($\leq 3.5 \cdot 10^{-13}$); the joint spectrum is non-degenerate (smallest gap 0.0065 at $L = 4$). (Pitfall found and fixed: degenerate kernel vectors returned by `Eigensystem` need not be orthonormal.) Each state has a weight (w_1, w_2, w_3, w_4), $\sum w_i = 0$, and Dynkin labels $[p, q, r] = [w_1 - w_2, w_2 - w_3, w_3 - w_4]$. They are saved with energy, weight, Dynkin labels, LR data and the bridge numbers C_ℓ in `data/abjm-hw-states.wxf` (load, do not recompute). An independent check on all 184 states: $\max|E_{ij}\Psi| \leq 5.5 \cdot 10^{-15}$ for the six E_{ij}, $i < j$, and every non-zero component has weight w [num, 10^{-14}] [`twoblock-polarisation-forms.wb` §1, run].

Q-functions [`abjm-Qfunctions.wb`] [num, 10^{-11}]. On the vacuum $\sum_a (-1)^a\tau_a D^a = (1 - V_1D)(1 - V_2D)(1 - V_3D)(1 - V_4D)$, $D = e^{-h\partial_u}$ (checked numerically), so the polynomial gauge is $Q \cdot \Phi$ with $\Phi(u) = V_4(u)\Phi(u - h)$ and the Baxter equation reads

$$\sum_{a=0}^{4}(-1)^a\tau_a(u)\, Q(u + h - ah) \prod_{b=a}^{3} V_4(u - bh) = 0, \tag{55}$$

with $\tau_2 = \prod g^{(2)}\bar{t}_2$, $\tau_3 = \prod g^{(3)}\bar{t}_3$, $\tau_4 = \mathrm{qdet}$ (on an A site $(u-\theta+h)(u-\theta-h)(u-\theta-2h)(u-\theta-3h)$).[6] Vacuum solutions have degrees $0, L + 1, L + 2$. Q_1, Q_2, Q_3 are the polynomial solutions ordered by

[6] This is the polynomial-gauge convention of `abjm-Qfunctions.wb`. The SoV sections (§13 onwards) use $\sum_a (-1)^a\tau_a(u)Q(u - ah) = 0$ with $Q_i = z_i^{u/h}\Phi(u)q_i$, which differs from the one here by a shift of the argument and the gauge factor.

degree, $Q_{12} = \mathrm{Cas}(Q_1, Q_2)/\mathrm{Cas}_{\mathrm{vac}}$, $Q_{123} = \mathrm{Cas}(Q_1, Q_2, Q_3)/\mathrm{Cas}_{\mathrm{vac}}$ (shifts $0, -h, -2h$). For all 184 states $\deg(Q_1, Q_{12}, Q_{123}) = (K_1, K_2, K_3)$ with

$$K_2 = L - \tfrac{p+r}{2} - q, \qquad K_1 = \tfrac{L-r+K_2}{2}, \qquad K_3 = \tfrac{L-p+K_2}{2} \qquad (\text{Dynkin } [p,q,r]), \tag{56}$$

Casoratian remainders $\leq 8 \cdot 10^{-12}$. Eigenvalue polynomials by interpolation at $2L+1$ points, 8 parallel subkernels at $L = 4$. *Data:* `data/abjm-tau.wxf`, `data/abjm-tau-L4.wxf`, `data/abjm-Qfunctions.wxf`.

8 Discrete symmetries

[`abjm-LR-symmetry.wb`, `abjm-other-symmetries.wb`] [`num`, 10^{-16}]. Operators commuting with H_{MZ}: $\mathcal{P}$ = (Weyl reversal $e_a \mapsto e_{5-a}$)∘(one-site shift), $\mathcal{Q}$ = (signed Weyl $e_1 \to e_4$, $e_2 \to e_3$, $e_3 \to e_2$, $e_4 \to -e_1$)∘(reflection $s \mapsto 2L+1-s$):

$$\mathcal{P}\bar{t}_1(u)\mathcal{P}^{-1} = \bar{t}_3(u+h), \quad \mathcal{Q}\bar{t}_1(u)\mathcal{Q}^{-1} = \bar{t}_1(2h-u), \quad \mathcal{P}^2 = S, \quad \mathcal{Q}S\mathcal{Q}^{-1} = S^{-1}. \tag{57}$$

The mechanism for $\mathcal{P}$: at the ABJM point $\theta_A - \theta_F = 2h$ the dual Lax of $\bar{t}_3$ at $u+h$ is the Lax of $\bar{t}_1$ with the roles of A and F exchanged, $\bar{\mathcal{L}}_A(u+h) = (u-\theta_F) - hP$, $\bar{\mathcal{L}}_F(u+h) = (u-\theta_A) + hK$ against $\mathcal{L}_A(u) = (u-\theta_A) + hK$, $\mathcal{L}_F(u) = (u-\theta_F) - hP$ (§3). $\mathcal{P}$ maps HW states of $[p,q,r]$ to HW states of $[r,q,p]$ (it exchanges the $\mathbf{4}$ and $\bar{\mathbf{4}}$ slots). Symmetric states (eigenvalue polynomials exact to 10^{-8} relative; Q-function forms checked with fixed shifts on all 184 states):

symmetry	eigenvalues	Q-functions	also
LR ($\mathcal{P}$)	$\tau_1(u) = \tau_3(u+h)$	$Q_1(u) \propto Q_{123}(u+h)$	
P ($\mathcal{Q}$)	$\tau_1(u) = \tau_1(2h-u)$	$Q_1(u) \propto Q_{123}(2h-u)$	$\tau_2(u) = \tau_2(c-u)$
LP ($\mathcal{PQ}$)	$\tau_1(u) = \tau_3(3h-u)$	$Q_1(u) \propto Q_1(h-u)$	$\tau_2(u) = \tau_2(c-u)$

Reflection turns the Baxter equation into its dual, hence P exchanges $Q_1 \leftrightarrow Q_{123}$ and LP maps Q_1 to itself. Every non-LR state has an LR partner $\mathcal{P}\Psi$ with $Q_1^{\mathrm{partner}} = Q_{123}(u+h)$; the spectrum is LR singlets plus LR doublets, as in 2603.

L	states	LR	P	LP	all three	LR only	P only	LP only	none
2	6	4	6	4	4	0	2	0	0
3	26	8	18	12	8	0	10	4	4
4	152	24	60	40	20	4	40	20	68

$\mathcal{Q}$-eigenstates have eigenvalue $(-1)^p$ in this sign convention; every four-pair state is either a $\mathcal{Q}$-eigenstate or part of a $\mathcal{Q}$-pair.

9 Structure constants of the note's configuration

[`abjm-bridges.wb`, `abjm-LR-symmetry.wb`] Bra for bridge ℓ (in sites): the first ℓ sites carry O_2 fields (A e_1, F e_4), the rest O_3 fields (A $e_1 - e_4$, F $e_4 + e_1$); $C_\ell = \langle B_\ell|\Psi\rangle/\|\Psi\|$ with a bilinear bra and the phase of Ψ fixed by its largest component, normalised so that the vacuum has $C_\ell = 1$. Even bridges $\ell = 2(L-M)$ are whole pairs (M pairs of O_3 fields); odd bridges change a single field.

$L=2$ state	E_{MZ}	nonzero C_ℓ
$[2,0,2]$ vacuum	0	all 1
$[0,2,0]$	16	—
$[1,0,1]$ parity pair	12, 12	$C_1 = C_3 = (1 \mp i\sqrt{3})/\sqrt{48}$, $\lvert C_1\rvert = 1/\sqrt{12}$
singlet	20	$C_2 = -1/\sqrt{40}$
singlet	4	$C_2 = 1/\sqrt{24}$

The closed forms are numerical recognitions of the data. The odd-bridge value $|C_1| = 1/\sqrt{12}$ is reproduced by the one-field bridge formula through the untwisting limit at the ABJM point (10^{-11}–10^{-14}, §15). The two even-bridge values $C_2 = -1/\sqrt{40}, 1/\sqrt{24}$ are now reproduced by two independent routes: (i) the pair κ-family K_κ of §16, untwisted with $\kappa = 1/z_4$, $a = -1$ (rate $O(\varepsilon)$) [num, $O(\varepsilon)$] [`pair-group-rotation.wb`], and (ii) the Q-function expression of §18, $|C| = \lim \mathcal{N}_{\text{pre}}\, \mathbb{A}\,\mathbb{W}/\sqrt{\mathbb{B}}$ with $(\mathbb{A}, \mathbb{W}, \mathbb{B}) \to (10, 25, 2.5 \cdot 10^6)$ and $(2, 9, 7776)$, i.e. $250/\sqrt{2.5 \cdot 10^6} = 1/\sqrt{40}$ and $18/\sqrt{7776} = 1/\sqrt{24}$ ($4 \cdot 10^{-16}$ extrapolated) [`even-bridge-Qform.wb`, `abjm-sov-norm-L4.wb`] [hp]. Their closed forms as polynomials in the polarisation are in §12.[7]

Vanishing at $L = 2, 3, 4$:

(i) *R-charge:* $C_\ell = 0$ unless $\Psi \in [p, 0, p]$, and then only for $L - p \le \ell \le L + p$ (the bra has only e_1, e_4 components, so $w_2 = w_3 = 0$).

(ii) *Bridge parity:* if $L + p$ is odd, $C_\ell = 0$ for even ℓ, i.e. $C_\ell \neq 0$ only for $\ell \equiv L + p \pmod 2$; if $L + p$ is even both parities occur. It holds on the whole HW cyclic $[p, 0, p]$ sector, for LR and non-LR, $\mathcal{Q}$-symmetric and paired states alike, so the note's bra has no component along these subspaces. It is "kinematic" only in this sense:[8] it is a property of the *polarisation* of the bra, not of the states. For the even bridges, F/F_{vac} of §12 is divisible by $x + y$ when $L + p$ is odd, and $x + y = 0$ at the note's polarisation; at another polarisation the same states have non-zero even-bridge overlaps (e.g. the $L = 2$ $[1, 0, 1]$ pair: $(x + y)/\sqrt{12} = 0.144338$ at P^* against $C_2 = 0$) [python].

(iii) *No LR selection rule:* unlike 2603 ($C = 0$ unless $\tau_+ = \tau_-$), non-LR states have nonzero C_ℓ, e.g. the $L = 2$ adjoint pair with $|C_1| = 1/\sqrt{12}$, and the $L = 3$ $[1, 0, 1]$ pair at $E = 6.456$ ($\ell = 2, 3, 4$). The $\mathcal{Q}$-pairs also have non-zero C. The bra B_ℓ is not $\mathcal{P}$-covariant ($\mathcal{P}$ moves the bridge to start on an F site), so the 2603 mechanism $\langle W|(t_+ - t_-) = 0$ is not available.

(iv) *Isolated dynamical zeros at $L = 4$:* the LR $[1, 0, 1]$ state at $E = 32$ has $C_\ell = 0$ for all ℓ (its even-bridge component is one of the accidental zeros of §12, which vanish for every polarisation); the LR $[2, 0, 2]$ state at $E = 8$ ($\mathcal{P} = -1$) has non-zero C_ℓ only at the even bridges $\ell \in \{2, 4, 6\}$, its odd bridges vanish. Not explained by $\mathcal{P}$ or $\mathcal{Q}$ (other LR states with the same eigenvalues are non-zero); both have integer energy (candidates for singular or exceptional Bethe states) [num, 10^{-13}].

Open: whether the physical ABJM structure constant should use a $\mathcal{P}$-symmetrised bra (both bridge placements, starting on A or F), which would change the LR question; the twisted version (2603 works at generic twist with $\omega = 1$).

10 Other three-point configurations and parity selection rules

[`abjm-3pt-configs.wb`] Two-block BPS bras: block 1 on ℓ sites with (a_1, f_1), block 2 on the rest with (a_2, f_2), $a_i \cdot f_i = 0$, *same polarisations for every L*. Up to $GL(4)$ and site rescalings the only invariants are whether the cross pairings $a_1 \cdot f_2$, $a_2 \cdot f_1$ vanish (plus linear dependences), giving six classes; with block $1 = (e_1, e_4)$, block 2: generic $(e_1 - e_4, e_4 + e_1)$ (the note's), $a_2 \cdot f_1 = 0$: $(e_2, e_1 + e_3)$, $a_1 \cdot f_2 = 0$: $(e_4 + e_2, e_3)$, both zero: (e_2, e_3), $a_2 = a_1$: (e_1, e_3), $f_2 = f_1$: (e_2, e_4); bridge starting on A or F; whole multiplets probed by random global rotations; 180 configurations at $L = 2, 3, 4$. (In §12 the two blocks are labelled 2 and 3 after the operators O_2, O_3, $(a_1, f_1, a_2, f_2) = (v_2, u_2, v_3, u_3)$, so the cross pairings are $a_1 \cdot f_2 = p_{23}$ and $a_2 \cdot f_1 = p_{32}$.)

- LR: no configuration couples only to LR states at $L = 3, 4$.

[7] The values $C_1 = 1/\sqrt{8}, 1/\sqrt{24}$ for the adjoints and $C_2 = -0.25, 1/\sqrt{15}$ for the singlets in `ABJM-3pt-transfer-matrices.md` §5.7 (the $\theta = 0$ table) and in the "Where it stands" paragraph of `NEXT.md` belong to the singular chain $\theta_A = \theta_F$ and are superseded by the table above (ABJM point).

[8] The note of 2026-10-12 (`abjm-transfer-matrices-symmetries.tex`, §7(ii)) and the summary cell of `abjm-LR-symmetry.wb` call the rule kinematic on the whole sector; §12 supersedes that reading.

- P: length-independent selection rules ($C = 0$ unless $Q_1(u) \propto Q_{123}(2h-u)$), verified at operator level on the cyclic sector, $\langle B|X_P(u)|r\rangle = 0$ for cyclic r, 10^{-16} at $L = 3, 4$ (vacuous at $L = 2$) [num, 10^{-16}]:

block 2	bridge ℓ	start
generic (note's W_{-1})	1 or $2L-1$	A or F
$a_2 \cdot f_1 = 0$	1	A
$a_1 \cdot f_2 = 0$	1	F
both cross pairings zero	2 or $2L-2$	A or F

  Control: the note's configuration at $\ell = 2$ fails ($10^{-2.4}$). The full operator identity (without the cyclic restriction) fails, as it must.

- LP: no uniform rule.

The P-rules in the table are statements about specific bridge lengths and configurations; they are not to be confused with the bridge-parity rule of §9(ii), which concerns $L + p$ and, as shown in §12, depends on the polarisation. By the criterion above (both cross pairings non-zero, no linear dependences) the note's polarisation and the example P^* of §12 lie in the same $GL(4)$ class; they differ by a transformation outside the Borel subgroup fixed by a highest-weight Ψ, so for a highest-weight Ψ they can give different overlaps.

11 Factorised bras annihilated by the symmetry differences

[`abjm-LR-factorized.wb`, `abjm-other-symmetries.wb`] $X_{\rm LR} = \bar{t}_1(u) - \bar{t}_3(u+h)$, $X_{\rm P} = \bar{t}_1(u) - \bar{t}_1(2h-u)$, $X_{\rm LP} = \bar{t}_1(u) - \bar{t}_3(3h-u)$.

- LR, $L = 2$, untwisted: left kernel 136/256 ($X_{\rm LR}$ linear in u); product states in it only the BPS orbit [exact]. Twisted, $\Lambda = \mathrm{diag}(z_1, z_2, 1/z_2, 1/z_1)$ ($z_1 z_4 = z_2 z_3 = 1$, so the t_3 twist $\det\Lambda\,\Lambda^{-1}$ is the Weyl image of Λ): 36 LR states, kernel 36, product states only $e_a \otimes e_{5-a}$; the note's W_a only at $a = 0$ [exact].

- P: every two-site state $(v \otimes w)^{\otimes L}$ is annihilated (10^{-16}, $L = 2, 3, 4$) – integrable two-site boundary states [num, 10^{-16}].

- LP: product solutions at $L = 2$ (SU(2)×SU(2)-sector states) fail at $L = 3, 4$.

(Per-length product searches are superseded by the length-independent requirement of §10.)

12 Two-block polarisation forms

[`twoblock-polarisation-forms.wb`, `data/abjm-twoblock-forms.json`, `checks/twoblock-polforms/`]

Status: the data and the structural statements come from python (numpy, sympy; [python]); the notebook redoes the computation in Wolfram, but only its sections 0–1 (set-up, highest-weight check of all 184 states) were run; every other cell is written in full with the python result pasted as markdown, and a rerun is pending. All coefficients are numerical (the states are accurate to 10^{-13}); exponents and vanishing patterns are exact in the sense of a gap of more than ten orders of magnitude between zero and non-zero coefficients.

The object. For the 184 HW cyclic states and $M = 0, \ldots, L$ define

$$F_\Psi(v_2, u_2; v_3, u_3; M) = \big\langle (v_2 \otimes u_2)^{\otimes(L-M)} \otimes (v_3 \otimes u_3)^{\otimes M} \big| \Psi \big\rangle, \qquad v_b \cdot u_b = 0 \ \text{(BPS pairs)}, \tag{58}$$

the first $L-M$ pairs carrying (v_2, u_2) and the last M pairs (v_3, u_3) (blocks labelled 2 and 3 after O_2, O_3; bilinear pairing, phase of Ψ as in §9). It is a polynomial of multidegree $(L-M, L-M, M, M)$ in the four vectors. The note's even bridge is $C_\ell = F_\Psi$ at $\ell = 2(L-M)$ and $(v_2, u_2; v_3, u_3) = (e_1, e_4; e_1 - e_4, e_4 + e_1)$.

Borel invariants. For a HW state F_Ψ is invariant under the simultaneous action of the upper unitriangular group, $v \to vn$, $u \to un^{-T}$ on all four vectors, i.e. it is annihilated by $D_{ij} = \sum_{b=2,3} \left(v_{b,i} \partial_{v_{b,j}} - u_{b,j} \partial_{u_{b,i}} \right)$, $i < j$, and it is torus covariant with the weight of Ψ (checked as polynomials for all six D_{ij}: max coefficient $4 \cdot 10^{-15}$ at $L = 2, 3$, $3 \cdot 10^{-14}$ at $L = 4$ [python]). The eight building blocks are

block	weight	meaning
$a_2 = (v_2)_1$, $a_3 = (v_3)_1$	ϵ_1	first components of the A vectors
$d_2 = (u_2)_4$, $d_3 = (u_3)_4$	$-\epsilon_4$	last components of the F vectors
$m_v = (v_2)_1 (v_3)_2 - (v_2)_2 (v_3)_1$	$\epsilon_1 + \epsilon_2$	leading 2×2 minor of the A vectors
$m_u = (u_2)_3 (u_3)_4 - (u_2)_4 (u_3)_3$	$-\epsilon_3 - \epsilon_4$	trailing 2×2 minor of the F vectors
$p_{23} = v_2 \cdot u_3$, $p_{32} = v_3 \cdot u_2$	0	cross pairings

(plus $p_{22} = v_2 \cdot u_2$, $p_{33} = v_3 \cdot u_3$ if the pairs are not BPS).

Structural observations [python]. (1) For all 184 states and all M, F_Ψ is a polynomial in $a_2, a_3, d_2, d_3, m_v, m_u, p_{23}, p_{32}$, as an identity of 16-variable polynomials (coefficient residual $\le 10^{-14}$ at $L = 2, 3$ and $\le 4 \cdot 10^{-14}$ at $L = 4$, with and without the BPS condition; least-squares fit residual at 60 random points $4 \cdot 10^{-16}$), and F_Ψ vanishes identically wherever no invariant monomial of the right weight and degree exists. (2) *Exponents.* With $n_2 = L - M$, $n_3 = M$ and weight w of Ψ, the monomials are

$$\begin{aligned} &a_2^{e_1} a_3^{e_2} d_2^{e_3} d_3^{e_4} \, m_v^{e_5} m_u^{e_6} \, p_{23}^{e_8} p_{32}^{e_9}, \\ &e_5 = w_2 = \frac{2q + r - p}{4}, \quad e_6 = -w_3 = \frac{p + 2q - r}{4}, \quad e_8 + e_9 = s := L - \frac{p + 2q + r}{2}, \end{aligned} \tag{59}$$

with $e_1 = n_2 - e_5 - e_8$, $e_2 = n_3 - e_5 - e_9$, $e_3 = n_2 - e_6 - e_9$, $e_4 = n_3 - e_6 - e_8$ (all ≥ 0); so $e_5 + e_6 = q$ and $|p - r| \le 2q$. (3) *Structure count.* The number of independent structures per (state, M) is

$$N(L, M, [p, q, r]) = \max\Big(0,\ \min(s,\ n_2 - e_5,\ n_3 - e_6) - \max(0,\ s - n_3 + e_5,\ s - n_2 + e_6) + 1\Big) \tag{60}$$

when e_5, e_6, s are non-negative integers and 0 otherwise, labelled by e_8 between the two bounds (0 mismatches against the monomial enumeration in all 882 (state, M) pairs; the notebook identifies N with the multiplicity of $[p, q, r]$ in $[n_2, 0, n_2] \otimes [n_3, 0, n_3]$). (4) *Minimal and complete.* For every multidegree $(L-M, L-M, M, M)$, $L = 2, 3, 4$, and every integer weight (7248 sectors, with and without the BPS condition), the dimension of the space of polynomials annihilated by D_{12}, D_{23}, D_{34} equals the number of monomials in the eight (ten) invariants (exact integer linear algebra, no mismatch); the eight BPS invariants are algebraically independent (Jacobian rank 8). Hence nothing can be removed or replaced.

Reduced form. With $F_{\text{vac}} = (a_2 d_2)^{L-M} (a_3 d_3)^M$ (the overlap of the vacuum state),

$$\frac{F}{F_{\text{vac}}} = \sum c \, x^{e_8} y^{e_9} \mu_v^{e_5} \mu_u^{e_6}, \qquad x = \frac{p_{23}}{a_2 d_3}, \quad y = \frac{p_{32}}{a_3 d_2}, \quad \mu_v = \frac{m_v}{a_2 a_3}, \quad \mu_u = \frac{m_u}{d_2 d_3}, \tag{61}$$

the coefficients c ("reduced structure constants") being polarisation independent. At the note's polarisation $a_2 = a_3 = d_2 = d_3 = 1$, $m_v = m_u = 0$, $p_{23} = 1$, $p_{32} = -1$, i.e. $(x, y, \mu_v, \mu_u) = (1, -1, 0, 0)$, so

$$C_M^{\text{note}} = \frac{F}{F_{\text{vac}}}\Big|_{(1,-1,0,0)} = \sum_{e_5=e_6=0} (-1)^{e_9} c, \tag{62}$$

reproduced against the saved C_ℓ for all 184 states and all M to $7 \cdot 10^{-15}$ (directly) and $2 \cdot 10^{-15}$ (from the invariant form) [python]. Two relations: *cyclicity*, which is exact, $F(v_2, u_2; v_3, u_3; M) = F(v_3, u_3; v_2, u_2; L - M)$ (at random points $2 \cdot 10^{-16}$), i.e. $c(M; e) = (-1)^{e_5+e_6} c(L - M; e$ swapped) with $a_2 \leftrightarrow a_3$, $d_2 \leftrightarrow d_3$, $p_{23} \leftrightarrow p_{32}$, $m_v \to -m_v$, $m_u \to -m_u$; and a $x \leftrightarrow y$ symmetry, which is empirical (not derived): for every state with $p = r$ the coefficients are symmetric, $c(e_8, e_9) = c(e_9, e_8)$ ($\mu_v \leftrightarrow \mu_u$ exchanged as well; $1.2 \cdot 10^{-15}$). It presumably is the parity operator $\mathcal{Q}$ of §8 acting on the bra. After cyclicity and this symmetry *one* constant remains per (state, M), except for $[2, 0, 2]$ at $L = 4$, $M = 2$, where $F/F_{\text{vac}} = c_0(x^2 + y^2) + c_1 xy$ has two and the note's value is $C = 2c_0 - c_1$ (the $E = 16$ pair has $c_0 = 0$).

Explicit forms. $L = 2$, BPS pairs (the only non-zero entries; $M = 0, 2$ vanish except for the vacuum), with the phase of Ψ fixed as above:

state	E_{MZ}	M	F_Ψ	F/F_{vac}
$[2, 0, 2]$	0	$0, 1, 2$	$a_2^2 d_2^2,\ a_2 a_3 d_2 d_3,\ a_3^2 d_3^2$	1
$[0, 2, 0]$	16	1	$-\frac{1}{2}\, m_v m_u$	$-\frac{1}{2}\mu_v \mu_u$
$[1, 0, 1]$ (two)	12	1	$(a_3 d_2 p_{23} + a_2 d_3 p_{32})/\sqrt{12}$	$(x + y)/\sqrt{12}$
singlet	20	1	$p_{23} p_{32}/\sqrt{40}$	$xy/\sqrt{40}$
singlet	4	1	$-p_{23} p_{32}/\sqrt{24}$	$-xy/\sqrt{24}$

(closed forms: $|c|^2$ recognised as rationals). $L = 3$ (26 states), BPS pairs; c denotes a state-dependent constant, listed where a closed form is known:

$[p, q, r]$	E_{MZ} (number)	$M = 1$ ($M = 2$ by cyclicity)	constant
$[3, 0, 3]$	0 (1)	1	—
$[1, 2, 1]$	12 (2)	$-\frac{1}{3}\mu_v \mu_u$	exact
$[2, 0, 2]$	8 (2)	$c\,(x + y)$	$c = (-1 \pm i)/\sqrt{24}$
$[2, 1, 0]$	12 (3)	$c\, y\, \mu_u$	$\lvert c\rvert^2 = \frac{1}{72}, \frac{1}{72}, \frac{1}{18}$
$[0, 1, 2]$	12 (3)	$c\, x\, \mu_v$	$\lvert c\rvert^2 = \frac{1}{72}, \frac{1}{72}, \frac{1}{18}$
$[1, 0, 1]$	$16 \mp 4\sqrt{2}$ (1 each)	$c\, xy$	$\lvert c\rvert^2 = (58 \pm 39\sqrt{2})/3864$ (PSLQ, 12 digits)
$[1, 0, 1]$	23.544, 6.456 (2 each)	$c\, xy$	numerical (data file)

(for $[2, 1, 0]$ and $[0, 1, 2]$ the $M = 2$ entry is $-c\, x\, \mu_u$, resp. $-c\, y\, \mu_v$; for $[1, 0, 1]$ the upper sign is $E = 16 - 4\sqrt{2}$). All other $L = 3$ states vanish: $[4, 0, 0]$, $[0, 0, 4]$, $[0, 2, 0]$ and the four singlets for every M and every polarisation (no invariant exists, $e_5 < 0$ or $e_6 < 0$ or $N = 0$), and the $[2, 0, 2]$ state at $E = 16$ by an accidental zero (see below). At $L = 4$ (152 states), BPS pairs, with $M = 3, 4$ by cyclicity:

$[p,q,r]$	states	$M=1$	$M=2$
$[4,0,4]$	1	1	1
$[2,2,2]$	3	$-\frac{1}{4}\mu_v\mu_u$ ($E=8$ pair), $+\frac{1}{4}\mu_v\mu_u$ ($E=16$)	$-\frac{1}{2}\mu_v\mu_u$ ($E=8$ pair), 0 ($E=16$, accidental)
$[3,0,3]$	4	$c(x+y)$	$c'(x+y)$
$[3,1,1]$, $[1,1,3]$	8, 8	$c\,y\,\mu_u$, $c\,x\,\mu_v$	$c'(x-y)\mu_u$, $c'(x-y)\mu_v$
$[2,0,2]$	20	$c\,xy$	$c_0(x^2+y^2)+c_1xy$
$[1,0,1]$	23	0	$c\,xy(x+y)$
$[1,2,1]$	13	0	$c\,(x+y)\mu_v\mu_u$
$[0,2,0]$	16	0	$c\,xy\,\mu_v\mu_u$
$[0,0,0]$	10	0	$c\,x^2y^2$
$[4,2,0]$, $[0,2,4]$	1, 1	0	$-\frac{1}{2\sqrt{3}}\mu_u^2$, $-\frac{1}{2\sqrt{3}}\mu_v^2$
$[0,4,0]$	2 ($E=24,8$)	0	$\frac{1}{12}\mu_v^2\mu_u^2$, $\frac{1}{4}\mu_v^2\mu_u^2$

($[2,1,0]$, $[0,1,2]$, $[2,3,0]$, $[0,3,2]$: no structure at $M=1$ and zero at $M=2$ by cyclicity; $[4,0,0]$, $[0,0,4]$, $[1,0,5]$, $[5,0,1]$: $e_5<0$ or $e_6<0$, zero for every polarisation and M.) The complete data, including the full (non-BPS) forms and the 16-variable polynomials at $L\le 3$, are in `data/abjm-twoblock-forms.json`.

Zero patterns. $F_\Psi=0$ for all polarisations (kinematic zero) whenever $e_5<0$, $e_6<0$, $s<0$ or $N=0$. Of the (state, M) pairs with allowed monomials, 39 vanish dynamically: 34 are forced by cyclicity (at $M=L/2$ the structure $e_8=e_9$ maps to itself with sign $(-1)^q$, so q odd gives $F=0$; $L=4$, $M=2$, classes $[2,1,0]$ (16 states), $[0,1,2]$ (16), $[2,3,0]$ (1), $[0,3,2]$ (1)), and 5 are *accidental* zeros: at $L=3$ the $[2,0,2]$ state at $E=16$ ($M=1,2$), and at $L=4$, $M=2$, the $[2,2,2]$ state at $E=16$, the $[1,2,1]$ state at $E=16$ and the $[1,0,1]$ state at $E=32$ (the all-bridge zero of §9(iv)). All accidental zeros sit at $E=16$ or 32. *The* $[2,0,2]$, $E=16$ *state at* $L=3$ *vanishes for every two-block polarisation.* Classes with $s=0$ ($p+2q+r=2L$: the vacuum, $[0,2,0]$ at $L=2$, $[1,2,1]$ at $L=3$, $[2,2,2]$, $[0,4,0]$, $[4,2,0]$, $[0,2,4]$ at $L=4$, …) do not depend on the pairings, so they are seen even when $p_{23}=p_{32}=0$ (doubly extremal); the singly extremal classes (one pairing allowed to vanish) occur only for $M\neq L/2$, since at $M=L/2$ the structures come in $x\leftrightarrow y$ pairs. F need not vanish when one pairing vanishes (e.g. the $L=2$ $[1,0,1]$ pair), so "both pairings needed" is a statement about the O_2–O_3 contraction, not about F.

What the note's polarisation does not see [python]. On the BPS locus the sublocus $\mu_v=\mu_u=0$ forces $x+y=0$; in the frame $v_2=e_1$, $u_2=e_4$ one finds

$$x+y=-\mu_v\,\frac{(u_3)_2}{(u_3)_4}+\mu_u\,\frac{(v_3)_3}{(v_3)_1} \tag{63}$$

(from $v_3\cdot u_3=0$; checked on random BPS points). The note's polarisation lies on this sublocus, so it probes only the single alternating combination $\sum(-1)^{e_9}c$ and *vanishes identically for every state whose* $F/F_{\rm vac}$ *is divisible by* $x+y$ *or contains* μ_v *or* μ_u. Consequently:

- it is blind to all states with s odd, i.e. $L+p$ odd for $q=0$ (a symmetric binary form of odd degree is divisible by $x+y$: $c(x+y)$ for $[2,0,2]$ at $L=3$ and $[3,0,3]$ at $L=4$, $c\,xy(x+y)$ for $[1,0,1]$ at $L=4$): the even-bridge parity rule of §9(ii) is a property of the polarisation, not of the states;
- it is blind to all $q>0$ states, which need μ_v or $\mu_u\neq 0$ (those with $s=0$ depend on the polarisation only through m_v,m_u and a,d).

Example P^*: $v_2=e_1$, $u_2=e_4$, $v_3=(1,1,0,1)$, $u_3=(1,1,0,-2)$ (BPS: $v_3\cdot u_3=0$) gives $(x,y,\mu_v,\mu_u)=(-\frac{1}{2},1,1,0)$. States invisible at the note's point but visible at P^* (number of states with $F/F_{\rm vac}^{\rm note}=0\neq F/F_{\rm vac}^{P^*}$, example value):

L	$[p,q,r]$	M	states	example $F/F_{\rm vac}(P^*)$
2	$[1,0,1]$	1	2	0.144338
3	$[0,1,2]$	1, 2	3, 3	$0.0295-0.0510i$, $0.0589-0.1021i$
3	$[2,0,2]$	1, 2	2, 2	$-0.1021+0.1021i$
4	$[0,2,4]$	2	1	-0.288675
4	$[1,0,1]$	2	22	$-0.00096-0.00166i$
4	$[1,1,3]$	1, 2, 3	8, 8, 8	$0.0360-0.0456i$, $0.1600-0.2025i$, $0.0720-0.0911i$
4	$[3,0,3]$	1, 2, 3	4, 4, 4	$0.1035+0.0752i$, $0.1674+0.1217i$, $0.1035+0.0752i$

The invariant form reproduces the direct overlap at P^* to $2.7\cdot10^{-15}$. The $L=2$ $[1,0,1]$ pair gives $F/F_{\rm vac}(P^*)=(x+y)/\sqrt{12}=0.144338$ although $C_2^{\rm mote}=0$. (Rows with $q>0$ and $\mu_u=0$ at P^* are seen through $\mu_v=1$; states needing $\mu_u\neq0$ require a different polarisation.)

13 Separation of variables for the alternating chain

Four sites A F A F ($L=2$ pairs; L is written only for formulas valid at any number of pairs): θ_1,θ_3 on the A sites, θ_2,θ_4 on the F sites, companion twist $G=G_c(z)$, generic θ,z and finite twist, except where the ABJM point is named (§13.6). F sites are built with t_1^n, A sites with t_a (naming of §2); $\tau_a(u)$ is the eigenvalue of $t_a(u)$ on the state under discussion.

13.1 Basis at four sites

[`sov-companion-L4.wb`, `sov-Q-dictionary-L4.wb`]

Words.

$$\langle a_1,k_2,a_3,k_4| = \langle 0|\, t_{a_1}(\theta_1-h)\, t_1(\theta_2)^{k_2}\, t_{a_3}(\theta_3-h)\, t_1(\theta_4)^{k_4}, \qquad a_i,k_i\in\{0,1,2,3\}, \tag{64}$$

rank 256 [exact]. Their expansion in the companion-frame product basis is exactly z-independent and block diagonal in the charge $|x|=a_1+k_2+a_3+k_4$ (equal to $\sum_A(i-1)+\sum_F(4-j)$ on the product vector with e_i on A sites, e_j on F sites); the 13 blocks have sizes 1, 4, 10, 20, 31, 40, 44, 40, 31, 20, 10, 4, 1 [exact]. For an eigenstate Ψ of the twisted transfer matrices, with $x=(a_1,k_2,a_3,k_4)$,

$$\langle x|\Psi\rangle = \tau_x(\Psi)\,\langle 0|\Psi\rangle, \qquad \tau_x=\tau_{a_1}(\theta_1-h)\,\tau_1(\theta_2)^{k_2}\,\tau_{a_3}(\theta_3-h)\,\tau_1(\theta_4)^{k_4}. \tag{65}$$

SoV vacuum. RV's $\langle 0|$ (RV2 (4.12)) is the lowest-weight Gelfand–Tsetlin (GT) vacuum: (3.19), $\langle 0|T_{j1}=\delta_{j1}\nu\langle 0|$, applied recursively through the embeddings, means that $\langle 0|$ is killed by every $T_{ij}(u)$ with $i>j$ of the untwisted monodromy and is an eigenbra of the T_{ii}. (3.19) alone leaves a 15-dimensional space; the full condition has a unique solution, our $\langle 0|=(e_1\otimes e_4)^{\otimes2}$ in the companion frame [exact] (generic and ABJM θ).

B-operator. RV2 (2.25), untwisted T (qmin: quantum minors of §2):

$$B(u)=\sum_{k_1\in\{1,2,3\}}\ \sum_{K_2\subset\{1,2,3\},\,|K_2|=2} T_{k_1,1}(u)\,\mathrm{qmin}_{K_2;(1,k_1+1)}(u+h) \times \mathrm{qmin}_{(1,2,3);(1,K_2+1)}(u+2h). \tag{66}$$

$[B(u),B(v)]=0$ and at generic θ all 256 words are B-eigenbras [exact]: a genuine SoV basis, equal to RV's up to normalisation (§13.2).

BPS expansions (generic θ; number of nonzero coefficients among the 256; unchanged for the shifted word bases used before the regularisation of §13.6) [exact]:

BPS state $v\otimes w\otimes v\otimes w$	$\#\{c_x\neq 0\}$	BPS state	$\#\{c_x\neq 0\}$
vacuum	1	A only: $\exp(\kappa E_{12})$	13
F only: $v=e_1$, $w\in e_1^{\perp}$ general	9	A only: $w=e_4$, $v\in e_4^{\perp}$ general	56
F only: $\exp(\kappa E_{34})$	4	$SU(2)\times SU(2)$ sector	37
F only: $\exp(\kappa E_{24})$	5	torus $\mathbb{C}[G_c^T]$	134–236
		generic	256

The 9 F-only words are $a_1=a_3=0$, $k\le 2$; the 13 A-only words have $a\le 2$, $k\le 1$. $\exp(\kappa E_{34})$ is the $\{2,4\}$ common rotation of §5 (not a product state: F–F exchange term), $\exp(\kappa E_{12})$ the $\{1,3\}$ one.

Localisation on sites 1,2 [exact]. Of the 16 site-1,2 words exactly $\{1,t_3(\theta_1-h)\}\times\{1,t_1(\theta_2)\}$ leave sites 3,4 at the vacuum, and the constants factorise,

$$\langle 0|\big(1+\alpha\,t_3(\theta_1-h)\big)\big(1+\beta\,t_1(\theta_2)\big)=(e_1+\alpha k_3 e_4)\otimes(e_4+\beta k_1 e_3)\otimes e_1\otimes e_4, \tag{67}$$

$k_1=-2263/14000$, $k_3=-21489/120736$ at the standard point. The localised span is five-dimensional; its fifth element (a combination of $t_1(\theta_2)^2$ and $t_1(\theta_1-h)t_1(\theta_2)$) is entangled and adds no product states.

Localisation on sites 1,2,3 [exact]. Localised single words $\{1,t_3(\theta_1-h)\}\times\{1,t_1(\theta_2)\}\times\{1,t_3(\theta_3-h)\}$, span 15-dimensional. Product states: (A) $(e_1+ae_4)\otimes(e_4+ce_3)\otimes e_1\otimes e_4$; (B) $(e_1+ae_4)\otimes e_3\otimes(e_1+be_4)\otimes e_4$ for arbitrary a,b (words $\langle 0|t_1(\theta_2)\times$ degree ≤ 1 in $t_3(\theta_1-h)$, $t_3(\theta_3-h)$, with an exchange correction). The generic three-site state $(e_1+ae_4)\otimes(e_4+ce_3)\otimes(e_1+be_4)$ is not localisable: there is no three-parameter extension.

13.2 Dictionary to Q-functions

[`sov-Q-dictionary-L4.wb`; `notes/rv-sov-dictionary.md`] [hp] unless stated.

Baxter equation and gauge. With $\tau_0=1$, $\tau_4=\det G\cdot\mathrm{qdet}$, the four solutions $Q_1,\dots,Q_4$ obey

$$\sum_{a=0}^{4}(-1)^a\,\tau_a(u)\,Q(u-ah)=0. \tag{68}$$

On the vacuum $Q_i=z_i^{u/h}\,\Phi(u)\,q_i(u)$ with $\Phi(u)=V_4(u)\Phi(u-h)$ (RV's Gamma gauge), $V_4=\prod_A(u-\theta_A)\prod_F(u-\theta_F-h)$; for a general eigenstate the q_i are polynomials (degrees in §7).[9] Casoratians: $Q_{123}(u)=\det_{i,j\le 3}Q_i(u-(j-1)h)$ (the polynomial Q_{123} of §7 is this divided by its vacuum value).

Wave functions of the word basis. The eigenvalues in (65) are Q-data, verified to 10^{-42} on all 256 four-site states for every j and every triple I (and exactly on the 16 states of one pair):

$$\text{F sites } (s=2,4):\quad \tau_1(\theta_s)=\frac{Q_j(\theta_s)}{Q_j(\theta_s-h)}\quad(\text{any } j), \tag{69}$$

$$\text{A sites } (s=1,3):\quad \tau_a(\theta_s-h)=\frac{D_a(\theta_s)}{D_0(\theta_s)},\quad D_a(\theta)=\det_{i\in I,\ j\in\{1,2,3,4\}\setminus\{a+1\}}Q_i(\theta-jh)\quad(\text{any triple } I). \tag{70}$$

[9] Eq. (68) is the Baxter equation of §7 for the full Q's (the polynomial-gauge equation there has the extra factors $\prod V_4$ and arguments $u+h-ah$). RV's Q^{RV} obeys (68) with $Q^{\mathrm{RV}}(u)=Q(u-h)$; `rv-sov-dictionary.md` and `sov-onepair-determinant.wb` use that labelling, in which every argument below is shifted by $+h$. All formulas here are in the labelling of (68).

They are (68) at special points: at $u = \theta_F$ the terms $a \geq 2$ drop since $\tau_{a\geq 2}(\theta_F) = 0$; at $u = \theta_A - h$ the term $a = 4$ drops since $\tau_4(\theta_A - h) = 0$, and Cramer's rule on three of the four Q_i gives $\tau_{1,2,3}$. Here $D_0 = Q_{123}(\theta - 2h)$ and $\tau_3(\theta_A - h) = Q_{123}(\theta_A - h)/Q_{123}(\theta_A - 2h)$ is a single dual-Q ratio, whereas τ_1, τ_2 are *gapped* Casoratians (columns $\theta_A - \{1,3,4\}h$ and $\theta_A - \{1,2,4\}h$ over D_0) with no single-Q form.

Separated variables, normalisation. The words equal RV2's unnormalised basis (4.12)=(5.18) and B acts diagonally with a degree-six polynomial per site, $\langle x|B(u) = \prod_s b_s(u)\langle x|$. The three dynamical roots of b_s (the others are frozen) are: θ_F ($3-k$ times) and $\theta_F + h$ (k times) at F sites; $\theta_A - bh$, $b \in \{0,1,2,3\} \setminus \{a\}$, at A sites. RV2's product of 3×3 minors (5.20) in these variables (on Q^{RV}) reproduces (65). In RV's normalisation the dictionary needs no x-dependent constant, $N_x = 1$, in the Gamma gauge (RV2 (5.9), (5.21)); in terms of the polynomials q_i it carries $N_x = \prod_F V_4(\theta_F)^k \prod_A \prod_{b\leq a} V_4(\theta_A - bh)$. $\langle 0|\Psi\rangle$ itself is fixed by RV's $\langle \Omega_\sigma|\Psi\rangle = 1$ and cannot be tested without a dual basis (next subsection).

13.3 Dual ket basis, Gram matrix and measure

[`sov-Q-dictionary-L4.wb`, `Wa-sov-expansion-L4.wb`, `abjm-sov-norm-L4.wb`]

Dual basis. On the ket vacuum $|\bar{0}\rangle = e_4 \otimes e_1 \otimes e_4 \otimes e_1$ the mirror construction uses $t_1(\theta_A)^m$ on A sites and $t_n(\theta_F + nh)$ on F sites,

$$|y\rangle = \prod_{s\in A} t_1(\theta_s)^{m_s} \prod_{s\in F} t_{n_s}(\theta_s + n_s h)|\bar{0}\rangle, \qquad m_s, n_s \in \{0,1,2,3\}, \tag{71}$$

and $\langle \Psi^{\mathrm{L}}|y\rangle = \tilde{\tau}_y(\Psi)\langle \Psi^{\mathrm{L}}|\bar{0}\rangle$, where $\langle \Psi^{\mathrm{L}}|$ is the left eigenvector of the transfer matrices and $\tilde{\tau}_y = \prod_A \tau_1(\theta_s)^{m_s} \prod_F \tau_{n_s}(\theta_s + n_s h)$. The ket wave functions mirror the bra ones: powers of one dual Q on A sites, a 3×3 determinant of three dual Q's on F sites [exact] (checked on twisted vacua, §13.7).

Gram matrix and measure. $\mathrm{Gram}_{xy} = \langle x|y\rangle$ is nonzero only if $|x| + |y| = 12$ (charge conservation), is triangular, and has 1456 nonzero entries [exact]. It is *not* diagonal: there is no diagonal measure for the mixed $\mathbf{4}/\bar{\mathbf{4}}$ chain (GLMR/GLMRV and 2603 have dual bases and measures only for symmetric/fundamental representations and the $\mathbf{6}$). With $M = \mathrm{Gram}^{-1}$ (again 1456 nonzero entries, $M_{yx} \neq 0$ only for $|x| + |y| = 12$) completeness reads $1 = \sum_{x,y} |y\rangle M_{yx}\langle x|$ [exact], and for every eigenstate, with $\langle 0|\Psi\rangle \neq 0$ on all 256 states,

$$\frac{1}{\mathcal{N}(\Psi)} = \frac{\langle \Psi^{\mathrm{L}}|\Psi\rangle}{\langle 0|\Psi\rangle\langle \Psi^{\mathrm{L}}|\bar{0}\rangle} = \sum_{x,y} \tilde{\tau}_y(\Psi)\, M_{yx}\, \tau_x(\Psi), \qquad \mathcal{N}(\mathrm{vac}) = (\lambda_1\lambda_4)^L. \tag{72}$$

This defines the SoV norm $\mathbb{B} = 1/\mathcal{N}$, the $\mathbb{B}$ of 2603's $C = \mathcal{N}\,\mathbb{W}\mathbb{A}/\sqrt{\mathbb{B}}$ with $\mathbb{W} \equiv 1$. The 12×12 GLMR-type determinant for $1/\mathcal{N}$ suggested by the dictionary (F sites: single Q on the bra side and a Slater determinant of dual Q's on the ket side; A sites: the mirror) does *not* fit (72); no determinant form is known.

Finite-twist norm statements. (The norm $\mathbb{B}$ is known to Paul; the following, from agent E, are cross-checks of the Q-assembly, not a derivation of it.) Let $\lambda_j = \prod_{m\neq j}(z_j - z_m)^{-1}$, $\lambda = (\lambda_1, \ldots, \lambda_4)$, $\mathbf{1} = (1,1,1,1)$, and let μ be the torus weight of Ψ (μ_j = number of A sites in state e_j minus number of F sites in state e_j). In the diagonal frame $\langle 0| = \lambda \otimes \mathbf{1} \otimes \lambda \otimes \mathbf{1}$, $|\bar{0}\rangle = \mathbf{1} \otimes \lambda \otimes \mathbf{1} \otimes \lambda$.

(E1) $\langle 0|\Psi\rangle = \lambda^\mu \langle \bar{0}^T|\Psi\rangle$, $\lambda^\mu = \prod_j \lambda_j^{\mu_j}$, for every state of weight μ (weight counting) [exact]; checked to 10^{-59} on all 256 states.

(E2) $\mathcal{N}(\Psi) = 1/\sum \tilde{\tau} M \tau$ as in (72), on all 256 states at generic θ, z [hp] (relative error $< 10^{-47}$).

(E3) There is *no* exact finite-twist Euclidean identity $|\langle 0|\Psi\rangle|^2 = \lambda^\mu \mathcal{N}\|\Psi\|^2$. In the diagonal frame with real Ψ, $|\langle 0|\Psi\rangle|^2 = H\lambda^\mu\mathcal{N}\|\Psi\|^2$ with $H = \langle\Psi^T|\bar{0}\rangle\langle\Psi^{\mathrm{L}}|\Psi\rangle/(\langle\Psi^{\mathrm{L}}|\bar{0}\rangle\,\Psi^T\Psi) \neq 1$: the right eigenvectors of t^Λ are not orthogonal and Ψ^{L} is proportional neither to Ψ^T nor to $\Psi^\dagger$ (cosines 0.37–1 at the standard point) [hp]. $H = 1 + O(\varepsilon)$ along the untwisting chain $z = 1 + \varepsilon\phi$ (pair-by-pair θ); for the two even-bridge singlets $H - 1 = \frac{E}{2}(\theta_{F2} - \theta_{F1}) + O(\varepsilon^2)$, $E = 20, 4$. Exactly, with Ψ^{L}_* normalised by $\langle\Psi^{\mathrm{L}}_*|\bar{0}\rangle = \langle\bar{0}^T|\Psi\rangle$: $\langle 0|\Psi\rangle^2/\langle\Psi^{\mathrm{L}}_*|\Psi\rangle = \lambda^\mu/\sum\tilde{\tau}M\tau$ [hp].

13.4 One pair exactly

[`sov-onepair-determinant.wb`, python `checks/session-2026-10-12/onepair/`] Two-site chain A F, all 16 states, words $\langle a,k| = \langle 0|t_a(\theta_1 - h)t_1(\theta_2)^k$, $a,k \in \{0,\ldots,3\}$, $\theta := \theta_{12}$.

Coefficients. For any $v \otimes u$, $\langle v \otimes u| = \sum_{a,k} c_{ak}(v,u)\langle a,k|$ with c_{ak} bilinear, z-independent, depending on θ, h only, and supported on the antidiagonal $i-j = a+k-3$ (grade of $e_i \otimes e_j$ is $i-j+3$; block sizes 1, 2, 3, 4, 3, 2, 1) [exact]:

$$c_{00} = v_1u_4, \quad c_{01} = \frac{(h-\theta)v_1u_3 + hv_2u_4}{h\theta(h-\theta)}, \quad c_{33} = -\frac{v_4u_1}{6h^6\theta^3(h-\theta)(2h-\theta)(4h-\theta)}. \tag{73}$$

Numbers of nonzero c_{ak} [exact]: generic BPS pair 16; one site generic, the other at its vacuum vector 7; A site twist-adapted of degree one ($v = e_1(\alpha + \beta G^T)$, $u = e_4$) 3; F site degree one ($v = e_1$, $u = e_4(\gamma + \delta G)$) 2; both 5.

Determinant formula [exact] (16 of 16 states, three random rational (v,u) each; [hp] 10^{-46}). With $x_b = \theta_1 - (b+1)h$, $D_0 = Q_{123}(\theta_1 - 2h)$,

$$\frac{\langle v\otimes u|\Psi\rangle}{\langle 0|\Psi\rangle} = \sum_{a,k} c_{ak}(v,u)\,\tau_a(\theta_1-h)\,\tau_1(\theta_2)^k = \frac{\det\left[\,w_b\,;\;Q_1(x_b)\,;\;Q_2(x_b)\,;\;Q_3(x_b)\,\right]_{b=0}^3}{D_0},$$

$$w_b = (-1)^b \sum_{k=0}^{3} c_{bk}(v,u)\,r^k, \tag{74}$$

with $r = \tau_1(\theta_2) = Q_j(\theta_2)/Q_j(\theta_2 - h)$ for all j; the 4×4 matrix has the weight row $(w_0,\ldots,w_3)$ first and then the three rows $Q_i(x_b)$. This is the Laplace expansion $\sum_b (-1)^b w_b D_b$ combined with $\tau_b(\theta_1 - h) = D_b/D_0$.[10]

Weight row, closed form [exact]. With $R = r/(h\theta)$, $w_b = \mathrm{Num}_b/N_b$, $N_b = b!\,h^b\prod_{m=1}^{b+1}(mh-\theta)$ (the Gamma-gauge factor of column b),

$$\mathrm{Num}_b = u(R)\Big[(bh-\theta)v_{b+1} + h\sum_{i>b} v_iR^{i-b-1}\Big] - hR^{3-b}\,\mathcal{S}(R), \tag{75}$$

$$u(R) = \sum_j u_jR^{4-j}, \qquad \mathcal{S}(R) = \sum_{i\geq j} v_iu_jR^{i-j}.$$

Equivalently, with $\mathcal{T}(R) = \sum_{i<j} v_iu_jR^{3+i-j}$, the overlap is $u(R)$ times an A-site 4×4 determinant plus $h\,\mathcal{T}(R)$ times the Casoratian of the four functions $R^{u/h}, \tilde{Q}_1, \tilde{Q}_2, \tilde{Q}_3$ ($\tilde{Q}_j = z_j^{u/h}q_j$), which vanishes exactly on the 12 product eigenstates (there: F-site polynomial $u(R)$ times A-site determinant). The weight row is cubic in r, while a determinant of Q-values with F columns at $\theta_2 - h, \theta_2$ is at most linear in r, so no single Q-only determinant reproduces a general bra.

[10] In the labelling of `sov-onepair-determinant.wb`: $x_b = \theta_1 - bh$, denominator $Q_{123}(\theta_1 - h)$, $r = Q_j(\theta_2 + h)/Q_j(\theta_2)$.

Torus [exact]. For $X \in \mathbb{C}[G^T]$ the global rotation $\rho(X) = X \otimes X^{-T}$ commutes with all t_a and, the 16 words being independent, $\rho(X) = \sum_{a,k} c_{ak}(e_1 X, e_4 X^{-T})\, t_a(\theta_1 - h)\, t_1(\theta_2)^k$ as operators. Hence the determinant is multiplied by the character, $\det[w(gX); Q]/D_0 = \chi_X \det[w(g); Q]/D_0$, $\chi_X = f(z_a)/f(z_b)$ on the product eigenstate labelled (a, b) ($X = f(G^T)$), $\chi_X = 1$ on the four zero-weight states.

Two pairs: no factorisation. The 16×16 matrix of coefficients $c_{(a_1,k_2),(a_3,k_4)}$ (pair-1 word labels against pair-2 word labels) has rank 16 for a generic product bra (3 if pair 2 is at the vacuum, 2 for an F-only degree-one bra); no normalisation factorises. For F sites $u_s = e_4(\gamma_s + \delta_s G)$ ($s = 2, 4$) and A sites at the vacuum [exact],

$$\frac{\langle e_1 \otimes u_2 \otimes e_1 \otimes u_4|\Psi\rangle}{\langle 0|\Psi\rangle} = (\gamma_2 + \delta_2\rho_2)(\gamma_4 + \delta_4\rho_4) - \frac{h\gamma_2\delta_4\rho_2}{\theta_{24}} - \frac{h\delta_2\delta_4\rho_2\rho_4}{\theta_{24} + h},$$
$$\rho_2 = \frac{\tau_1(\theta_2)}{h\theta_{12}\theta_{23}(\theta_{24} - h)}, \qquad \rho_4 = \frac{\tau_1(\theta_4)}{h\theta_{14}\theta_{24}\theta_{34}}, \quad (76)$$

the exchange terms coming from $\langle 0|t_1(\theta_4) = h\theta_{14}\theta_{34}\big[h\, e_1 \otimes e_3 \otimes e_1 \otimes e_4 + \theta_{24}\, e_1 \otimes e_4 \otimes e_1 \otimes e_3\big]$. The F–F exchange is *not* a Vandermonde measure: the cross term $\gamma_2\delta_4$ carries ρ_2, not ρ_4, and the determinant of the 2×2 bilinear form, $h\rho_2[\rho_2/\theta_{24} - \rho_4/(\theta_{24} + h)]$, is not separable; the same holds for A–A pairs. So no one-row-per-site determinant exists in the left-basis/τ language; a 2603-type determinant needs a right SoV basis and a non-diagonal measure for the mixed chain (open).

13.5 Two-block bras and the companion vacuum in the highest-weight orbit

[`sov-twoblock-expansion-L4.wb`, `even-bridge-sov-formula.wb`]

Cost of block bras [exact]. In the basis (64) a bra with a generic block on sites $\{1, 2\}$ and $\{3, 4\}$ at the vacuum needs 36 words ($a_3 = 0$, $k_4 \le 2$; pair-cut rank 3); the reverse needs 130 words (pair-cut rank 16); the common rotation $1 - \kappa G_c^T$ on pair 1 needs 20 words.

The companion SoV vacuum is in the orbit of the physical vacuum [exact]. With $\lambda_j = \prod_{m \ne j}(z_j - z_m)^{-1}$ and $r_j = \lambda_j/\lambda_1$, in the diagonal frame ($\rho(g) = g$ on A sites, g^{-T} on F sites), at every L,

$$\langle 0|_{\rm comp} = \lambda_1^L\, \langle {\rm vac}^{\otimes L}|\, \rho(n'D'), \qquad n' = \begin{pmatrix} 1 & 1 & 1 & r_4 \\ 0 & 1 & 0 & -r_2 \\ 0 & 0 & 1 & -r_3 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad D' = {\rm diag}(1, r_2, r_3, 1), \qquad (77)$$

finite as $z \to 1$. So the SoV vacuum is in the Borel (highest-weight) orbit of the physical vacuum ${\rm vac} = e_1 \otimes e_4$ per pair. Rotating the two blocks by $g_1 = 1 + b\,\mathbb{E}$, $g_2 = 1 + b'\mathbb{E}$ ($\mathbb{E} = E_{14}$ in the twist eigenbasis) gives the physical two-block bras; the sparsest choice is $b' = 0$.

One-pair rotation. For $b' = 0$ the bra in the companion frame is

$$B(b) = (e_1 + b\lambda_1 v_{z_4}) \otimes (e_4 - b\ell_1) \otimes e_1 \otimes e_4, \quad v_{z_4} = e_1 \prod_{m \ne 4}(G_c^T - z_m), \quad \ell_1 = \lambda_1\, e_4 \prod_{m \ne 1}(G_c - z_m), \quad (78)$$

36 SoV words (1, 17, 36 words at orders b^0, b^1, b^2) [exact]; the 36-term sum equals the direct overlap to 10^{-44} (exact c_x, 60-digit eigenvalues; double precision fails, cond $M \sim 10^{18}$–10^{25}) [hp]. Untwisting: for the two $L = 2$ singlets, $\langle B(b)|\Psi_\varepsilon\rangle/(\lambda_1^2 b^2)$ tends linearly in ε to the data value C_2 ($-0.15807 \to -1/\sqrt{40}$, $0.20408 \to 1/\sqrt{24}$) [num, 10^{-4}]. With $r_2 = \tau_1(\theta_2)$, $r_4 = \tau_1(\theta_4)$ and polynomials $P_\beta(r_2, r_4)$ collecting the F-site words,

$$K_b = \frac{\langle B(b)|\Psi\rangle}{\langle 0|\Psi\rangle} = \sum_{a=0}^{3} \tau_a(\theta_1 - h) P_a(r_2, r_4) = \frac{\det\big[\,(-1)^\beta P_\beta(r_2, r_4)\,;\; Q_i(x_\beta)\,\big]_{\beta=0}^{3}}{D_0}, \qquad (79)$$

one bordered 4×4 Casoratian (x_β as in (74)); no finer structure was found (no sparser word basis, tensor rank 7).

13.6 The ABJM point

[`sov-abjm-point-limit.wb`, `abjm-sov-norm-L4.wb`, python `checks/abjmlimit/`] Pair-by-pair $\theta=(\theta_{F1}+d,\theta_{F1},\theta_{F2}+d,\theta_{F2})$, $d=\theta_A-\theta_F\to 2h$ (§4), $u_0:=\theta_A-h=\theta_F+h$. RV's genericity condition (2.9) fails ($\theta_A^{\rm RV}-\theta_F^{\rm RV}=h$), the generic words degenerate, and the "degenerate B" of the first write-up (169 distinct eigenvalues, $169=13^2$) is resolved as follows.[11]

Degeneration of the generic basis [exact]. Pair-by-pair θ: the generic words have rank 256 for generic d and $81=9^2$ at $d=2h$ (the 16×16 one-pair word matrix $M(d)$ has rank 9 there, $\det M(d)\propto(d-2h)^7\cdots$). For fully pair-periodic θ (coincident inhomogeneities, e.g. $t_a(\theta_1-h)=t_a(\theta_3-h)$) the generic words have rank 70 for every d.

Exact identities and Jordan structure [exact]. At u_0, as operator identities (the exact identities of §4),

$$t_3(u_0)=0,\qquad t_2(u_0)=t_1(u_0)\,t_1(\theta_F),\tag{80}$$

hence $\tau_3(u_0)=0$ and $\tau_2(u_0)=\tau_1(u_0)\tau_F$ on states, $\tau_F:=\tau_1(\theta_F)$: the collisions of the B-spectrum. On the ket side $t_2(\theta_A)=t_1(\theta_A)t_1(u_0)$, $t_3(\theta_A+h)=0$. The LR map $\mathcal{P}$ ($\mathcal{P}\bar t_1(u)\mathcal{P}^{-1}=\bar t_3(u+h)$, §8) sends $\bar t_1(\theta_F)$ to $\bar t_3(\theta_A-h)$ exactly at $d=2h$, and the five-word pair rotation K_κ of §16 is built from exactly these regularised pair words ($X_1=t_1(\theta_F)$, $X_2=t_1(u_0)X_1$, $X_3\propto Z$ below). B has Jordan blocks: per resonant pair 10 simple eigenvalues and three J_2 blocks ($10+3\cdot2=16$), i.e. 13 distinct eigenvalues per pair. In Q-language the columns $Q_i(\theta_A-2h)$, $Q_i(\theta_A-3h)$ of the A-site Casoratian are proportional (ratio τ_F, by (69)), so $D_0=D_3=0$, $D_2=\tau_F D_1$, and (70) becomes $0/0$: its content moves to the u-derivatives of the Baxter equation.[12]

Regularised bra basis (the $d\to2h$ limit of RV's basis = Jordan basis of B). For every resonant A site replace the word operator by

a	$O(a,k)$	factor $\lambda(a,k)$ in the wave function
0	1	1
1	$t_1(u_0)$	$x=\tau_1(u_0)$
2	$W=t_2'(u_0)-t_1'(u_0)t_1(\theta_F)$ $(k\le2)$	$w=\tau_2'(u_0)-\tau_1'(u_0)\tau_F$ $(k\le2)$
	$t_2(u_0)$ $(k=3)$	$x\tau_F$ $(k=3)$
3	$Z=t_3'(u_0)=g_3'\,\widehat{t}_3(u_0)$	$y=\tau_3'(u_0)$

($'=\partial_u$; $\widehat{t}_3$ the reduced t_3). The words $\langle0|O_1\,t_1(\theta_2)^{k_2}\,O_3\,t_1(\theta_4)^{k_4}$, $O_1=O(a_1,k_2)$, $O_3=O(a_3,k_4)$, have rank 256 [exact], and $\langle x|\Psi\rangle=\langle0|\Psi\rangle\prod_{\rm pairs}\lambda(a,k)\,\tau_F^k$ [hp] ($3\cdot10^{-48}$). The data x,w,y follow from the u-derivative of (68) at u_0 (a 3×3 Cramer system, any three of the four Q_i),

$$Q_i'(u_0)-x\,D_i+w\,Q_i(u_0-2h)-y\,Q_i(u_0-3h)=0,\qquad D_i=Q_i'(u_0-h)-\tau_F Q_i'(u_0-2h),\tag{81}$$

with τ_F to $4.5\cdot10^{-49}$ and x,w,y to $3\cdot10^{-27}$ [hp]. The coefficients of a *fixed* bra diverge like $\delta^{-\#\text{resonant pairs}}$, $\delta=d-2h$, but the overlaps converge (verified to 10^{-45}).

[11] This replaces "the SoV basis survives at the ABJM point with shifted evaluation points" (the words at θ_A-3h or θ_A+2h are a basis, but only the 16 F-only words are B-eigen) and "the Ryan–Volin B is degenerate at the ABJM point".

[12] The source (session report of 12-10-2026, §9) writes the proportional columns at θ_A-h,θ_A-2h, i.e. in the labelling shifted by h (footnote to (68)); re-indexed here.

Regularised ket basis. The resonant ket sites are the F sites: levels $n = 0, 1, 2, 3$ use the operators 1, $t_1(u_0)$, $\widetilde{W} = t_2'(\theta_A) - t_1(\theta_A)t_1'(u_0)$, $\widetilde{Z} = t_3'(\theta_A + h)$ and A sites keep $t_1(\theta_A)^m$. The pair factor of the ket wave function is $\tau_1(\theta_A)^m \{1, x, \tilde{w}, \tilde{y}\}$, $\tilde{w} = \tau_2'(\theta_A) - \tau_1(\theta_A)\tau_1'(u_0)$, $\tilde{y} = \tau_3'(\theta_A + h)$ (rank 256 [exact]). The Gram matrix of the regularised bases has 3323 nonzero entries at every tested ε and is *no longer* triangular or charge graded. $1/\mathcal{N} = \sum \tilde{\tau} M \tau$ with $M = \mathrm{Gram}^{-1}$ reproduces the direct value of (72) on the 28 zero-weight states for $\varepsilon = 1/50, \ldots, 1/1600$ (relative error $< 10^{-29}$ at 100–120 digits, limited by the degeneration of Gram as $\theta_{F1} - \theta_{F2} = O(\varepsilon)$) [hp].

Caveats. (i) For fully pair-periodic θ the regularisation needs shift-derivatives up to order 7; the homogeneous limit is open, and pair-by-pair θ_F must be used. (ii) At $z_2 = z_3$ the companion frame is singular and the diagonal-twist chain has a $\mathfrak{gl}(2)$ symmetry, so there is no SoV basis there; the K_κ route of §16 (which needs $z_2 = z_3$) must be combined with SoV at $z_2 - z_3 = O(\varepsilon)$, which suffices for the limit (§16).

13.7 SoV expansion of $\langle W_a^L|$ and of twisted vacua

[`Wa-sov-expansion-L4.wb`; python `checks/session-2026-10-13/wasov/`] $L = 2$ pairs, generic θ, z.

Companion image [exact]. Per pair $W_a = (e_1 + a e_4) \otimes (e_4 - a e_1)$ (diagonal frame). With $v(x) = (x^3, x^2, x, 1)$, $\tilde{u}_j$ the j-th row of V^{-1} and $R^{-1} = V^T$ (A) $\otimes V^{-1}$ (F),

$$\langle W_a^L | R^{-1} = \big[(v(z_1) + a\, v(z_4)) \otimes (\tilde{u}_4 - a\, \tilde{u}_1)\big]^{\otimes L} \tag{82}$$

with coefficient exactly 1 (symbolic a). For $a = 0$ this is the companion Bethe vacuum $\langle 0_w|$ at $w = z$ (below), *not* the SoV vacuum $(\lambda \otimes \mathbf{1})^{\otimes 2} R^{-1} = e_1 \otimes e_4 \otimes e_1 \otimes e_4$; overlaps are $\langle W_a | \Psi_\Lambda \rangle = \langle W_a R^{-1} | \Psi_c \rangle$ with $|\Psi_c\rangle = R|\Psi_\Lambda\rangle$.

The expansion is dense. Write $\langle W_a^L | R^{-1} = \sum_x c_x(a) \langle x|$, $c_x(a) = (\langle W_a | R^{-1} Y)_x$, Y the inverse word matrix (block diagonal in $|x|$). Each $c_x(a)$ is a degree-4 polynomial in a containing all five powers, for all 256 words, and every component $\langle \mathrm{vac}^L | \rho(E_{14})^n / n!$ is dense too [exact] (physical frame sparse, SoV frame dense). The words are z-independent, the bra is not:

$$c_x(a; \kappa z) = \kappa^{6 - |x|}\, c_x(a; z) \qquad \text{[exact] (symbolic } a,\ \kappa = 2/3). \tag{83}$$

Ranks of the components $n = 0, \ldots, 4$ (each with 256 nonzero words and site ranks 4, 4, 4, 4) [exact]: pair cut (12)|(34): 11, 13, 16, 13, 11; cut (13)|(24): 4, 12, 16, 12, 4; rank of $m_{a_1 a_3}$ over $\mathbb{Q}(r_2, r_4)$ (below): 2, 2, 4, 2, 2; the generic bra ($a = 3/7$) has 16, 16, 4. The ranks are palindromic under $n \leftrightarrow 4 - n$ ($W_a = -a^2 \langle \mathrm{LW} | \rho(e^{E_{41}/a})$ per pair) but there is no simple map between the coefficient tensors. The $n = 0$ and $n = 4$ components are exact eigenbras of $t^{G_c(z)}$ [exact], $(v(z_1) \otimes \tilde{u}_4)^{\otimes 2}$ with the vacuum eigenvalue of the twisted-vacuum formula below and $(v(z_4) \otimes \tilde{u}_1)^{\otimes 2}$ with $(z_1, z_2, z_3, z_4) \to (z_4, z_2, z_3, z_1)$ in it; $n = 1, 2, 3$ are not eigenbras.

Sectors and exactness. $\langle \mathrm{vac}^L | \rho(E_{14})^n / n! \, |\Psi\rangle \neq 0$ only for weight $(2-n, 0, 0, n-2)$: sector sizes 1, 12, 28, 12, 1 (54 of the 256 eigenstates), and $\langle W_a | \Psi \rangle = a^n \langle W_1 | \Psi \rangle$. The SoV sum $\sum_x c_x^{(n)} \tau_x(\Psi) \langle 0 | \Psi \rangle$ equals the direct overlap on all 256 states, including the vanishing on the other 202 (maximal error $\leq 5 \cdot 10^{-41}$) [hp]. Since the 256 monomials τ_x are independent functions on the 256 states, $c_x(a)$ is *the* unique representation valid on all 256 states.

The A–A coefficient matrix; no determinant form. $\langle W^{(n)} | \Psi \rangle / \langle 0 | \Psi \rangle = \sum_{a_1, a_3} \tau_{a_1}(\theta_1 - h)\, m^{(n)}_{a_1 a_3}(r_2, r_4)\, \tau_{a_3}(\theta_3 - h)$, $r_s = \tau_1(\theta_s)$, $m_{a_1 a_3} = \sum_{k_2, k_4} c_{a_1 k_2 a_3 k_4} r_2^{k_2} r_4^{k_4}$ of bidegree $(3, 3)$. Rank 2 ($n = 0, 1, 3, 4$) means a sum of two products of bordered 4×4 Casoratians $B_k^{(1)} B_k^{(3)} / (D_0(\theta_1) D_0(\theta_3))$,

equivalently one 8×8 determinant with two border rows; rank 4 ($n=2$ and generic a) is no compression, so the singlet sector (weight 0, 28 states) is the structurally hardest. For $n=0,4$ the entries share a common factor of bidegree $(2,2)$ and the column space is cut out by two bilinear equations (the vacuum structure); for $n=1,3$ there is no common factor and the 2×2 minors are products of two irreducible $(3,3)$ polynomials [exact]. Candidates tested on the 256 eigenstates and excluded [hp]: a product over pairs of bordered Casoratians (the pair-cut rank would have to be 1, it is 11–16), a sum $\alpha P_1+\beta P_2+\gamma$ of one-pair overlaps, a single 8×8 bordered determinant for $n=2$ or generic a, a few pair-factorised terms (≥ 11 needed), and a GLMR/2603-type bracket determinant (it needs the non-diagonal measure of §13.3). So on all 256 states there is no determinant or pair-factorised form of $\langle W_a^L|\Psi\rangle$. Likewise the dual wave functions $y\mapsto\langle W^{(n)}|y\rangle$ do not factorise [exact]: tensor ranks 1, 1, 1, 1 (pair cut 1) for $n=0,4$; 4, 4, 4, 4 (pair cut 6) for $n=1,3$; 4, 4, 4, 4 (pair cut 15) for $n=2$; for $a=3/7$ site ranks 4, 3, 4, 3 and pair cut 9 ("$\langle W_a^L|$ has the Q-functions of $\langle$vac$|$ but is not a twisted vacuum").

Twisted vacua [exact]/[hp]. Words, kets and hence Gram, M are independent of the twist: those built with $t^{G_c(w)}$ equal those built with $t^{G_c(z)}$ (exact, 256×256). The twisted vacuum $\langle 0_w|=(v(w_1)\otimes\tilde u_w)^{\otimes 2}$ ($\tilde u_w=$ row 4 of $V(w)^{-1}$) is an exact eigenbra of $t_a^{G_c(w)}$, $a=1,2,3$, with the vacuum eigenvalues $\tau_a(u)=\sum_{|I|=a}\prod_{k=1}^a w_{i_k}V_{i_k}(u-(k-1)h)$ (V_i as in §2). Its dual wave functions *factorise*,

$$\langle 0_w|y\rangle=\langle 0_w|\bar 0\rangle\prod_s f_s(y_s),\qquad f_A(m)=\tau_1(\theta_A)^m,\quad f_F(n)=\tau_n(\theta_F+nh),\tag{84}$$

where τ are the vacuum eigenvalues of twist w, for all 256 y (relative error exactly 0, all tensor ranks 1; $\langle 0_w|\bar 0\rangle=651015225/2427364$ at the standard point). In terms of the dual Q's of the twist-w vacuum ($w_I(u)\propto Z_I^{u/h}q_I(u)$, $Z_I=\prod_{i\in I}w_i$, triples I), $\tilde\tau_y(\mathrm{vac}_w)$ is per A site the m-th power of a ratio of one dual Q at two points and per F site of level n a ratio of two 3×3 determinants of dual Q's, up to rational constants independent of twist and state [exact]. The bra-side dictionary (69)–(70) holds on the same vacua. Completeness gives $\langle 0_w|=\sum_x\big(\sum_y\langle 0_w|y\rangle M_{yx}\big)\langle x|$ (dense, 256 words), hence the finite Q-sum

$$\mathbb{W}_w(\Psi):=\frac{\langle 0_w|\Psi_z\rangle}{\langle 0_w|\bar 0\rangle\langle 0|\Psi_z\rangle}=\sum_{x,y}\tilde\tau_y(\mathrm{vac}_w)\,M_{yx}\,\tau_x(\Psi),\tag{85}$$

exact on all 256 eigenstates Ψ_z (maximal relative deviation $<10^{-44}$; $\min_n|\mathbb{W}_w|=2.9\cdot 10^{-8}$) [hp]. Biorthogonality is the case $w=z$: $\mathbb{W}_z(\Psi)=0$ on all 255 non-vacuum states ($<10^{-45}$ relative to $|\mathbb{W}_z(\mathrm{vac})|=1.3477\cdot 10^{-4}$) [hp]. No determinant form of (85) was attempted (it needs the mixed-chain measure).

Global Borel rotations of $\langle 0|$ [exact]. For $\langle 0|\rho(1+bE_{ij})$ on all four sites, the six generators $E_{12},E_{34},E_{14},E_{13},E_{24},E_{23}$ transported from the diagonal frame ($g_c=V^{-T}gV^T$ on A, $Vg^{-T}V^{-1}$ on F) all give dense bras, with $1,30,140,236,256$ words at orders $b^0,\dots,b^4$ (union 256). Acting directly in the companion frame the numbers of words are 13 for E_{12} (1, 4, 8 at b^0,b^1,b^2), 4 for E_{34}, 5 for E_{24}, 33 for E_{13}, 70 for E_{14} and 1 for E_{23}. So no physical-frame Borel rotation is few-word in the SoV basis (same reason as for $\langle W_a|$); the few-word global rotations are the $E_{12}/E_{34}/E_{24}$-type steps of the *companion* $\mathfrak{gl}(4)$ (§5).

Nevertheless a Q-form on the relevant sectors. The dense expansion is the unique representation on all 256 states and encodes the vanishing off the sectors; it does not preclude a short formula on the weight sectors $(\mu,0,0,-\mu)$ where $\langle W_a^L|$ has support. There $\langle W_a^L|\Psi\rangle$ equals, up to explicit factors, the overlap with the twisted vacuum $\langle 0_w|$, $w=(w_1,z_2,z_3,w_4)$, and the $w_1\to\infty$ limit of (85) isolates exactly 9 F-only words $\langle 0|t_1(\theta_2)^{k_2}t_1(\theta_4)^{k_4}$, $k_s\leq 2$: at $L=2$ a polynomial of bidegree $(2,2)$ in the two F-site single-Q ratios $\tau_1(\theta_2),\tau_1(\theta_4)$ (10^{-46} to 10^{-51} on the 54 states of

the sectors) [hp]. The derivation, the factorised L-pair form and its use for the even bridge are in §18.[13]

14 ABJM three-point functions from transfer matrices: setup

This section and the next two write the ABJM structure constants of the note `StructuringABJM.pdf` as overlaps $\langle B|\Psi\rangle$ of a transfer-matrix eigenstate Ψ with a bra B that is obtained from an integrable reference bra by a *word* in transfer matrices, following arXiv:2603.26617 ("2603"). Ψ is a cyclic highest-weight (HW) eigenstate of the ABJM chain (Secs. 4, 7); conventions are those of Sec. 2. Bras are row vectors, $\rho_{\rm pair}(g)$ is the group element g acting on an A–F pair ($v \mapsto vg$ on the A site, $u \mapsto ug^{-T}$ on the F site), $J = \sum_s \rho_s(E_{14})$ is the global raising operator, $\Lambda = \mathrm{diag}(z_1, \dots, z_4)$ is the twist in the physical (diagonal) frame, and $\tau_a(u)$ is the eigenvalue of $t_a(u)$ on Ψ. Evidence tags are those of Sec. 1.

14.1 Pair vectors and the bras of the note

On one pair put

$$\mathrm{vac} = e_1 \otimes e_4, \qquad W_a = (e_1 + a e_4) \otimes (e_4 - a e_1) = \mathrm{vac} \cdot \rho_{\rm pair}(e^{aE_{14}}), \qquad v \cdot u = 0. \tag{86}$$

$W_0 = \mathrm{vac}$ is an O_2 pair and $W_{-1} = (e_1 - e_4) \otimes (e_4 + e_1)$ an O_3 pair. The bra of the note (its (3.6)) is

$$\langle B_M| = \langle \mathrm{vac}^{\otimes(L-M)} \otimes W_{-1}^{\otimes M}|, \qquad C_M(\Psi) = \langle B_M|\Psi\rangle / \|\Psi\|, \tag{87}$$

which is the C_ℓ of Sec. 9 with $\ell = 2(L-M)$ O_2 sites (data normalisation: unit-norm Ψ, $\langle \mathrm{vac}|\mathrm{vac}\rangle = 1$). Even bridges are ℓ even (a whole number M of O_3 pairs); odd bridges have ℓ odd, i.e. one field of a pair is rotated, e.g. $\ell = 1$: $\langle e_1 \otimes (e_4 + e_1) \otimes W_{-1}^{\otimes(L-1)}|$. Because Ψ is cyclic (translation by one pair), the position of the block in B_M is immaterial.

Identification (2603 analogue) [python] [`abjm-bridges.wb`, `ABJM-3pt-transfer-matrices.md`]. A pair twist rotation $\rho_{\rm pair}(D)$ with $D = \mathrm{diag}(d_1, \dots, d_4)$ maps

$$W_a \, \rho_{\rm pair}(D) = \frac{d_1}{d_4} \, W_{a\, d_4/d_1}. \tag{88}$$

With $D_\kappa = 1 - \kappa\Lambda$ this is $W_a \to W_{a'}$, $a' = a(1-\kappa z_4)/(1-\kappa z_1)$, and $\kappa = 1/z_4$ gives $a' = 0$: it turns an O_3 pair into an O_2 pair. The F-site factor $1/d_4$ is singular at $\kappa = 1/z_4$; the regular (polynomial) normalisation is

$$T_\kappa(v \otimes u) = (vD_\kappa) \otimes (u \,\mathrm{adj}\, D_\kappa), \qquad T_\kappa(W_a) = (1-\kappa z_1)^2(1-\kappa z_2)(1-\kappa z_3)\, W_{a'}, \tag{89}$$

which at $\kappa = 1/z_4$ gives $(1 - z_1/z_4)^2(1 - z_2/z_4)(1 - z_3/z_4)\,\mathrm{vac}$. Hence the note's (3.6) is $\langle W_{-1}^{\otimes L}|$ with its first $L - M$ pairs rotated by D_{1/z_4}, exactly 2603's statement "$O_2 = O_3$ with the first ℓ sites rotated by $1 - \kappa\cdot$twist" on a block. The companion SoV vacuum is a global rotation of the physical vacuum with all components non-zero in the diagonal frame (Sec. 13), so, as in 2603, the three-point reference is taken directly as a diagonal-frame global rotation of $\mathrm{vac}^{\otimes L}$, namely $\langle W_a^{\otimes L}| = \langle \mathrm{vac}^{\otimes L}|\rho(e^{aJ})$.

[13] This supersedes the first-pass conclusion "no closed Q-form" (morning of 13-10-2026; sections 3 and 6 of the report of `Wa-sov-expansion-L4.wb`), which is withdrawn: what holds is that no determinant or pair-factorised form exists on all 256 states, while on the sectors of weight $(\mu, 0, 0, -\mu)$ the F-only Q-form exists.

What 2603 provides (reading of arXiv:2603.26617 App. K, `rv-sov-dictionary.md` §5). A site of 2603 is the $\mathbf{6} = \wedge^2\mathbf{4}$ with vacuum $Z = e_1 \wedge e_2$. Its κ-family is the one-point word $1 - \kappa t_+(\theta_1) + \kappa^2 t_0(\theta_1)$ (t_0 the $\wedge^2$ transfer matrix), which is the *group element* $\wedge^2(1-\kappa g)$ on the site. Localisation needs only $\langle Z|\mathcal{L}_{jk} = 0$ for $j < k$ plus two bonus zeros $\mathcal{L}_{21} = \mathcal{L}_{43} = 0$; it holds for any global rotation of the vacuum (no goodness condition), and the pairing of a rotated with an unrotated vector is quadratic in $D = 1 - \kappa g$, hence finite and non-extremal. The ABJM analogue needed is a *common pair rotation*: A by $D = 1 - \kappa\Lambda$, F by $D^{-1} \propto \operatorname{adj} D$ (cubic). It is automatically BPS; its pairings with the reference are $-\kappa\, u\Lambda v^T$ and $\sum_j v_j u_j/(1-\kappa z_j)$, which are $O(1)$ precisely for references with $u\Lambda v^T \neq 0$ ("non-good" references such as W_a, $a \neq 0$). This is what Sec. 16 constructs.

Pair localisation at generic θ [python] [`abjm-bridges.wb`, `L4-ffbar-free-rotations.wb`]. On $\langle W_{-1}|^{\otimes 2}$, of the 16 words $t_a(\theta_1 - h)t_1(\theta_2)^k$ exactly $\{1, t_3(\theta_1 - h)\} \times \{1, t_1(\theta_2)\}$ localise on pair 1; all 320 pair-1-localising words (up to two library factors per site) span only 4 dimensions and give product states only $\rho_1(\Lambda^i)\rho_2(\Lambda^j)$ with $(i,j) \in \{(0,0),(-1,0),(-1,1)\}$. The global torus is $t_1(\theta_1)^{-1}t_1(\theta_2)t_1(\theta_3)^{-1}t_1(\theta_4) = c\,\rho(\gamma)$. This four-dimensional algebra of generic θ is all one sees away from the ABJM point; it led to the earlier (superseded) conclusion that a Λ-rotation of an F site alone, hence the even bridge, is out of reach. It is not: the missing fifth word $t_2(\theta_F + h)$ localises only at $\theta_A - \theta_F = 2h$, and the even bridge is constructed in Sec. 16.

14.2 The highest-weight trick and why it fails in the untwisting limit

[python] [`abjm-bridges.wb`]. For a HW state $J\Psi = 0$, and e^{bJ} shifts every pair vector of a product bra, $W_a \to W_{a+b}$; hence

$$\langle \mathrm{vac} \otimes W_{-1}|\Psi\rangle = \langle W_b \otimes W_{b-1}|\Psi\rangle \qquad \text{for all } b \qquad (10^{-16}). \tag{90}$$

The end-growing chain of Sec. 15.2 gives, exactly at finite twist, $\langle W_{b-1}^{\otimes L}|t_1(\theta_1)^{-1}t_1(\theta_2) = c_{12}\langle W_b \otimes W_{b-1}^{\otimes(L-1)}|$ when $(b-1)z_1/z_4 = b$, i.e. $b = z_1/(z_1 - z_4)$; this is a one-word finite-twist formula for the right-hand side of (90). But $b \to \infty$ as $z \to 1$, and the resulting untwisting limit is wrong and state dependent. 2603 avoids this by a *continuous* κ-family on the block. For ABJM pairs that family is the pair κ-family of Sec. 16: there the free parameter is a finite a with $a' - a \to -1$, so that the HW shift needed is finite (Sec. 16.4). The old "F-site gap" framing of the even bridge as an open problem is thereby resolved (Sec. 16).

15 Odd bridges

Odd bridges (one field of a pair rotated, $\ell = 1, 3, \dots$) are reachable with the words of the end-growing chain of the reference $\langle W_a^{\otimes L}|$, exactly at finite twist, and have a regular untwisting limit. They see only the vacuum and the states with $n = 1$, i.e. $[L-1, 0, L-1]$ (Sec. 15.6).

15.1 The exact odd-bridge formula

[python] [`abjm-bridges.wb` §3; proved for all L in §15.2]. On any $\langle W| = \langle W_a^{\otimes L}|$,

$$\langle W|t_1(\theta_1)^{-1} = c_1\,\langle W|\rho_1(\Lambda^{-1}), \qquad \frac{1}{c_1} = h\prod_{F\,s}\theta_{1s}\prod_{A\,s\neq 1}(\theta_{1s}+h), \tag{91}$$

where $\rho_1(\Lambda^{-1})$ multiplies the vector on the A site 1 from the right by Λ^{-1} ($\theta_{ks} = \theta_k - \theta_s$). On the A vector $e_1 - e_4$ of W_{-1} one has $(e_1 - e_4)(1 - z_4\Lambda^{-1}) = (1 - z_4/z_1)e_1$, so $\langle W_{-1}^{\otimes L}|\big(1 - \frac{z_4}{c_1}t_1(\theta_1)^{-1}\big) = (1 - z_4/z_1)\,\langle e_1 \otimes (e_4 + e_1) \otimes W_{-1}^{\otimes(L-1)}|$, and

$$\langle e_1 \otimes (e_4 + e_1) \otimes W_{-1}^{\otimes(L-1)}|\Psi\rangle = \frac{1 - z_4/(c_1\tau_1(\theta_1))}{1 - z_4/z_1}\,\langle W_{-1}^{\otimes L}|\Psi\rangle, \tag{92}$$

exact at any twist, for every eigenstate Ψ of t_1 with eigenvalue τ_1.[14] Verified through the untwisting limit at the ABJM point with a generic inhomogeneous deformation [num, 10^{-11}] (down to 10^{-14} in places) [`abjm-bridges.wb` §3]; there $\langle W|\Psi\rangle \sim \epsilon$ and the bracket in (92) is $\sim 1/\epsilon$. The general-L forms below contain (92) as the case $k=1$, $\kappa = z_4$, $a=-1$ of the A-site family (exact at finite ϵ).

15.2 The end-growing chain: closed forms and proof

[exact] [`odd-bridge-general-L.wb` §1; exact at $L=2,3$ on three parameter sets including generic h and a generic BPS pair, 10^{-16} at $L=4$; fully reproduced in the rerun of 2026-10-13]. Let $\Omega = (v\otimes u)^{\otimes L}$ with $v\cdot u = 0$ and the *same* pair at every position, and let $\langle P_k|$ be $\langle\Omega|$ with the first k sites rotated by the global-torus element γ (A site: $v \mapsto v\Lambda^{-1}$; F site: $u \mapsto u\Lambda$), so $P_0 = \Omega$ and $P_{2L} = \rho(\gamma)\Omega$. The end-growing chain uses the inverse letter on A sites and the forward letter on F sites:

$$\langle P_{k-1}|\, t_1(\theta_k)^{-1} = d_k\, \langle P_k| \qquad (k \text{ odd, A site}), \tag{93}$$

$$\langle P_{k-1}|\, t_1(\theta_k) = d_k\, \langle P_k| \qquad (k \text{ even, F site}), \tag{94}$$

with

$$\frac{1}{d_k} = h \prod_{s\neq k} \vartheta_{ks} \quad (k \text{ odd}), \qquad \vartheta_{ks} = \begin{cases} \theta_{ks} & s \text{ F} \\ \theta_{ks}+h & s \text{ A}, \end{cases} \tag{95}$$

$$d_k = -h \prod_{s\neq k} \tilde\vartheta_{ks} \quad (k \text{ even}), \qquad \tilde\vartheta_{ks} = \begin{cases} \theta_{ks} & s \text{ A} \\ \theta_{ks}-h & s \text{ F}. \end{cases} \tag{96}$$

The cumulative constants $c_k = d_1\cdots d_k$ are those of the word $t_1(\theta_1)^{-1}t_1(\theta_2)t_1(\theta_3)^{-1}\cdots$ (k letters) acting on Ω: $\langle\Omega|$ (first k letters) $= c_k\langle P_k|$. In particular

$$c_1 = d_1 = \Big[h \prod_{F\, s} \theta_{1s} \prod_{A\, s\neq 1} (\theta_{1s}+h)\Big]^{-1} \quad (= (91), \text{ now proved}), \tag{97}$$

$$c_{12} := c_2 = d_1 d_2 = \prod_{A,\, s\geq 3} \frac{\theta_{2s}}{\theta_{1s}+h} \prod_{F,\, s\geq 4} \frac{\theta_{2s}-h}{\theta_{1s}}, \tag{98}$$

$$c_{2L} = \frac{\prod_{F<F'}(h^2-\theta_{FF'}^2)}{\prod_{A<A'}(h^2-\theta_{AA'}^2)} \qquad \text{(full chain, global torus)}. \tag{99}$$

Example, $L=2$ at $\theta = (2/7, -3/5, 8/11, -11/8)$, $h=\frac12$: $c_4 = -1108723/174000$ [exact].

Proof (all L). One-site lemma (BPS contraction). Let a site carry the bra vector w and contract a column c (or row r) of the auxiliary space into its Lax operator. For $\mathcal{L}_A = x + hK$: $\sum_j c_j\mathcal{L}_{ij} = x\, c_i\langle w| + h\, w_i\langle c|$ and $\sum_i r_i\mathcal{L}_{ij} = x\, r_j\langle w| + h\,(r\cdot w)\langle j|$; for $\mathcal{L}_F = x - hP$: $\sum_j c_j\mathcal{L}_{ij} = x\, c_i\langle w| - h\,(w\cdot c)\langle i|$ and $\sum_i r_i\mathcal{L}_{ij} = x\, r_j\langle w| - h\, w_j\langle r|$. So a site acts on the auxiliary column or row as the scalar x if the two vectors are BPS-orthogonal, and as $x+h$ (A, $w=c$) or $x-h$ (F, $w=r$) if they coincide. At $u=\theta_k$ the Lax operator of site k is hK (A) or $-hP$ (F), so $t_1(\theta_k)$ factorises through site k and every other site acts as a scalar θ_{ks}, $\theta_{ks}+h$ or $\theta_{ks}-h$ as in (95). The leftover contraction at site k returns the unrotated vector for A (k odd: one finds $\langle P_k|t_1(\theta_k) = d_k^{-1}\langle P_{k-1}|$, equivalent to (93)) and the rotated vector for F (k even, (94)). The singlet terms $(r\cdot w)$, $(w\cdot c)$ vanish exactly because Λ^{-1} on A and Λ on F preserve $v\cdot u = 0$: this is why the only fixed rotation is the global-torus γ. No condition on h, θ, the diagonal twist or a is used except $v\cdot u = 0$ and that all pairs carry the same (v,u); for independent pairs the A–A and F–F exchange terms do not collapse [`abjm-nongood-reference-limit.wb`].

[14] An early version of the old note omitted the factor $1/(1-z_4/z_1)$; corrected 11-10-2026.

15.3 The two one-field families, at every placement

[exact] [`abjm-nongood-reference-limit.wb` (two pairs), `odd-bridge-general-L.wb` §2 ($L=3$, all placements)]. Two free rotations of a single field exist on $\langle\Omega|$:

$$\text{(b) A site:}\quad \langle\Omega|\big(\alpha+\beta\,t_1(\theta_1)^{-1}\big)=\langle\Omega| \text{ with the A vector of pair 1 replaced by } v(\alpha+\beta c_1\Lambda^{-1}), \tag{100}$$

$$\text{(a) F site:}\quad \langle\Omega|\,t_1(\theta_1)^{-1}\big(\alpha+\beta\,t_1(\theta_2)\big)=\langle\Omega| \text{ with pair 1 replaced by } (v\Lambda^{-1})\otimes u(\alpha c_1+\beta c_{12}\Lambda). \tag{101}$$

At every placement $k=1,\dots,2L$ the same holds after the chain: $\langle\Omega|\,(\text{letters } 1,\dots,k-1)(\alpha+\beta\,t_1(\theta_k)^{\mp1})$ rotates only the field at site k (A site via the inverse letter, F site via the forward letter) on top of the fixed $\Lambda^{\mp1}$-twisting of sites $1,\dots,k-1$. Products of (a) and (b) are not product states. With the normalised letters $\hat t_k=d_k t_1(\theta_k)$ (k odd), $\hat t_k=t_1(\theta_k)/d_k$ (k even), (93)–(94) read $\langle P_{k-1}|\hat t_k^{(-1)^k}=\langle P_k|$ with no constant, and the words are

$$W_F(p)=\Big[\prod_{j<k}\hat t_j^{(-1)^j}\Big]\big(c\epsilon-1+\hat t_k\big)\ \ (k=2p),\qquad W_A(p)=\Big[\prod_{j<k}\hat t_j^{(-1)^j}\Big]\big(1-\kappa\,\hat t_k^{-1}\big)\ \ (k=2p-1). \tag{102}$$

15.4 The untwisting limit of the one-field families

Take $z=1+\epsilon\varphi$, $\theta=\theta_{\rm ABJM}+\epsilon\theta^0$ (pair-preserving ABJM inhomogeneities), $\epsilon=10^{-2},\dots,10^{-4}$. Family (a) with $\alpha':=\alpha c_1=c\epsilon-1$, $\beta':=\beta c_{12}=1$ (i.e. the word $c\epsilon-1+\hat t_2$; $\alpha'+\beta'=c\epsilon$), for which $\alpha'+\beta' z_j=\epsilon(c+\varphi_j)$ on the F vector, gives

$$\langle\text{bra}_\epsilon|=\frac{\epsilon(c+\varphi_4)}{z_1}\,\langle N_\epsilon|,\qquad N_\epsilon=\Big[(e_1+a\tfrac{z_1}{z_4}e_4)\otimes(e_4-a'e_1)\Big]\otimes W_a^{\otimes(L-1)},\quad a'=a\frac{c+\varphi_1}{c+\varphi_4}, \tag{103}$$

and a finite HW shift e^{-aJ} maps the limiting bra to $\text{vac}^{\otimes L}$ with one F site equal to e_1 on pair 1: $\langle N_0|e^{-aJ}=\langle\text{vac}^{\otimes L}|+(a-a')\langle e_1e_1\otimes\text{vac}^{\otimes(L-1)}|$. Hence, at rate $O(\epsilon)$,

$$\langle N_\epsilon|\Psi_\epsilon\rangle\ \longrightarrow\ V_0+(a-a')\,X_{11}\quad\text{(F family)},\qquad \langle N^{\rm A}_\epsilon|\Psi_\epsilon\rangle\ \longrightarrow\ V_0+(a''-a)\,X_{44}\quad\text{(A family)}, \tag{104}$$

$a''=a(c+\varphi_4)/(c+\varphi_1)$ (the A family is the word $1-\kappa\hat t_1^{-1}$ with $\kappa=1-c\epsilon$: A vector $\propto(e_1+a''e_4)$, F vector e_4-ae_1; the shift e^{-aJ} then gives $\text{vac}+(a''-a)e_4e_4$), with $V_0=\langle\text{vac}^{\otimes L}|\Psi_0\rangle$ ($=1$ on the vacuum, 0 otherwise), $X_{11}=\langle e_1e_1\otimes\text{vac}^{\otimes(L-1)}|\Psi_0\rangle$, $X_{44}=\langle e_4e_4\otimes\text{vac}^{\otimes(L-1)}|\Psi_0\rangle$ (pair 1, by cyclicity any pair). The limit is independent of φ, c and of the placement p (on cyclic states pairs 2, 3 give the same limit; at finite ϵ the placements differ by $O(\epsilon)$), and $X_{11}=X_{44}=C_1$ of the data table (HW shift plus cyclicity). Values: $L=2$, the two $[1,0,1]$ states: $C_1=(1\mp i\sqrt{3})/\sqrt{48}=0.144338\mp0.25i$ (Sec. 9); $L=3$, the three $[2,0,2]$ states ($E_{\rm MZ}=8,8,16$): $C_1=+i/\sqrt{24}$, $-i/\sqrt{24}$ (the two $E_{\rm MZ}=8$ states) and $1/\sqrt{24}$ ($E_{\rm MZ}=16$); the vacuum gives $V_0=1$ and all other saved states give 0 (the notebook tabulates $(T_\epsilon-V_0)/(a-a')$, resp. $(T_\epsilon-V_0)/(a''-a)$, against C_1 for several (φ,c) and both families). [num, $O(\epsilon)$] [`odd-bridge-general-L.wb` §2, rerun 2026-10-13]. The special member $\kappa=z_4$, $a=-1$ of the A family reproduces (92) at finite ϵ exactly. The finite-twist identity $\langle\text{bra}_\epsilon|\Psi_\epsilon\rangle=(\text{word eigenvalue})\langle W_a^{\otimes L}|\Psi_\epsilon\rangle$ with the closed-form d_k, c_k holds to 10^{-15} for every state at $L=2,3$ and chain positions $k=1,2$ [num, 10^{-15}].

15.5 Homogeneous ABJM-point limit and the normalised $\hat\tau$ form

[exact] / [num, $O(\epsilon)$] [`odd-bridge-general-L.wb` §4]. At the ABJM point every constant above is finite and non-zero in the fully homogeneous limit $\delta\to0$ (pair-periodic θ): d_k contains only θ_{ks} with s on the opposite sublattice ($=\pm2h$) or on the same sublattice shifted by $\pm h$ ($=\pm h$ after the shift). In closed form

$$c_1=(2h^2)^{-L},\qquad d_2=-h(-2h)^L(-h)^{L-1},\qquad c_{12}=c_1d_2=1, \tag{105}$$

($c_1 = 8$, $d_2 = 1/8$ at $L = 3$, $h = \frac{1}{2}$). The homogeneous limit is therefore *regular*; the singular locus is $\theta_A = \theta_F$ (the superseded "wrong chain", Sec. 4), where $d_k \to \infty$ (k odd) like $(\theta_A - \theta_F)^{-L}$ and $d_k \to 0$ (k even) like $(\theta_A - \theta_F)^{L}$, while the eigenvalues $\tau(\theta_k)$ go to zero at the same rate. Only the products $\hat{\tau}_k = d_k^{\pm 1}\tau_1(\theta_k)$ (the eigenvalues of $\hat{t}_k$) stay finite; on the twisted vacuum, whose eigenvalues are known exactly, $\hat{\tau}_j = z_1$ (j odd), $\hat{\tau}_j = z_4$ (j even) at every θ. In terms of them, exactly at finite twist,

$$\langle \mathrm{bra}_\epsilon | \Psi_\epsilon \rangle = \Big[\prod_{j<k} \hat{\tau}_j^{(-1)^j} \Big] \big(c\epsilon - 1 + \hat{\tau}_k \big) \langle W_a^{\otimes L} | \Psi_\epsilon \rangle \tag{106}$$

(F family, $k = 2p$; for the A family replace the last factor by $1 - \kappa/\hat{\tau}_k$). For the one-field F bridge on pair 1 (same form at every L)

$$V_0 + (a - a')C_1 = \lim_{\epsilon \to 0} \frac{c\epsilon - 1 + \hat{\tau}_2}{\hat{\tau}_1 \, \epsilon \, (c + \varphi_4)} \, \langle W_a^{\otimes L} | \Psi_\epsilon \rangle, \qquad a' = a \frac{c + \varphi_1}{c + \varphi_4}. \tag{107}$$

On the vacuum $\hat{\tau}_1 = z_1$ exactly and the right side is 1. On the states with $V_0 = 0$ (weight $(L-1)(e_1 - e_4)$, the $n = 1$ states of Sec. 15.6) one finds numerically $\hat{\tau}_1 = \hat{\tau}_2$ in the limit (a fourth root of unity: $+i, -i, -1$ for the three $[2,0,2]$ states at $L = 3$), so that

$$C_1 = \big(1 - 1/\hat{\tau}_1\big) K, \qquad K = \lim_{\epsilon \to 0} \frac{\langle W_a^{\otimes L} | \Psi_\epsilon \rangle}{\epsilon \, a \, (\varphi_4 - \varphi_1)}, \tag{108}$$

independent of φ, c and a (tested at $\delta = 0$) [num, $O(\epsilon)$]. The $\hat{\tau}$ are eigenvalues of t_1 at two inhomogeneities, hence Q-expressible (Sec. 18); (107) and (108) express the odd bridge through such data and the overlap $\langle W_a^{\otimes L} | \Psi \rangle$ of Sec. 18.

15.6 The odd one-field bridge as a T_{14} form factor; which states it sees

Let $T_{14}(u) = \mathrm{tr}(T\, E_{41})$ be the $(1,4)$ monodromy element and $G_a = \Lambda + a(z_1 - z_4)E_{41} = e^{aE_{41}} \Lambda e^{-aE_{41}}$. Then $\langle W_a^{\otimes L}|$ is the vacuum eigenbra of $t^{G_a}(u)$ with eigenvalue $\tau_{\mathrm{vac}}(u)$, $t_1^{G_a} = t_1^{\Lambda} + a(z_1 - z_4)T_{14}$ [exact] [`even-bridge-epsilon-expansion.wb`, $L = 2$], and for every eigenstate Ψ of t^{Λ} at finite twist

$$\big(\tau_{\mathrm{vac}}(u) - \tau_\Psi(u) \big) \langle W_a^{\otimes L} | \Psi \rangle = a(z_1 - z_4) \, \langle W_a^{\otimes L} | T_{14}(u) | \Psi \rangle \tag{109}$$

[`even-bridge-epsilon-expansion.wb`: exact 30-digit agreement, deviation $< 10^{-29}$ on all 41 eigenstates of the sectors $(2,0,0,-2)$, $(1,0,0,-1)$, $(0,0,0,0)$ at $L = 2$].

Orders in ϵ. For Ψ of weight $(\mu_1, 0, 0, -\mu_1)$ put $n = L - \mu_1$; then $\langle W_a^{\otimes L} | \Psi \rangle = (a^n/n!) \langle \mathrm{vac} | J^n | \Psi \rangle$ and $\langle W_a^{\otimes L} | \Psi_\epsilon \rangle = (a^n/n!)\, \epsilon^n \omega_n + \ldots$, while $\langle W_a^{\otimes L} | \Psi \rangle \equiv 0$ for every other weight. The bra of (104) carries an explicit factor ϵ, so a finite limit needs $\langle W | \Psi_\epsilon \rangle = O(\epsilon)$: *odd bridges see only the vacuum ($n = 0$) and the states with $n = 1$, i.e.* $[L-1, 0, L-1]$; all other Ψ give 0 (Sec. 15.4; at $L = 3$ exactly the vacuum and the three $[2,0,2]$ states). The even bridge $M = 1$ sees only $n = 2$, $[L-2, 0, L-2]$ (Sec. 16.4); hence for a fixed state the two are never both non-zero and there is *no even/odd ratio relation* [python] [`even-bridge-epsilon-expansion.wb`, `checks/session-2026-10-13/epsexp/`].

Form factor. Matching orders in (109) gives $\omega_1 = (\varphi_1 - \varphi_4)\langle \mathrm{vac} | T_{14}(u) | \Psi_0 \rangle / (\tau_{\mathrm{vac}}(u) - \tau_{\Psi_0}(u))$, independent of u. The odd-bridge chain gives $\omega_1(m) = (\varphi_4 - \varphi_1) C_1(m)/D_0(m)$ with $D_0 = (\hat{\tau}_2 - 1)/\hat{\tau}_1$ the $\epsilon \to 0$ value of the chain-word eigenvalue ($D_0 = 3/2 \pm (\sqrt{3}/2)i$ at $L = 2$ for the two $[1,0,1]$ states, same for $k = 1, 2$; $D_0 = 1 \pm i$, 2 at $L = 3$ for the $[2,0,2]$ states, i.e. $1 - 1/\hat{\tau}_1$ of (108)). Therefore, for the HW states m of $[L-1, 0, L-1]$,

$$\langle \mathrm{vac} | T_{14}(u) | m \rangle = - \big(\tau_{\mathrm{vac}}(u) - \tau_m(u) \big) \, \frac{C_1(m)}{D_0(m)}, \tag{110}$$

numerically to 10^{-4} ($O(\epsilon)$) for the $2 + 3$ states at $L = 2, 3$ [python] [`scan2.py`, `odd_check.py` in `checks/session-2026-10-13/epsexp/`]. So the odd one-field bridge is the T_{14} form factor $\langle \mathrm{vac} | T_{14} | m \rangle$ normalised by the $\hat{\tau}$ ratio D_0.

Relation to the even bridge. The only relation between odd and even data is first-order perturbative: with $\kappa_m = -C_1(m)/D_0(m)$ and $g_m = \langle\tilde{m}|T_{14}(u_1)|\Psi_0\rangle/(\tau_m(u_1) - \tau_{\Psi_0}(u_1))$,

$$C_{M=1}(\Psi_0) = \frac{K_1(\tau_{\Psi_0})}{\tau_{\rm vac}(u) - \tau_{\Psi_0}(u)} \sum_m \kappa_m \big(\tau_{\rm vac}(u) - \tau_m(u)\big) g_m, \tag{111}$$

the sum over the HW states m of $[L-1, 0, L-1]$ (4 at $L=2$, 9 at $L=3$ including non-cyclic ones), with K_1 the pair-word eigenvalue of Sec. 16.2. It reproduces the data $C_{M=1}$ to 6 digits [python]; the new unknown is the form factor g_m between two excited states, which is not obtainable from Q-brackets of Ψ alone.

16 The pair κ-family and even bridges

This section solves the problem that the earlier notes called the "F-site gap": a Λ-rotation of a whole A–F pair (A by $D = 1 - \kappa\Lambda$, F by $D^{-1} \propto \mathrm{adj}\, D$) by a local transfer-matrix word, with a free continuous parameter κ. It exists at the ABJM point of Sec. 4, for a twist with $z_2 = z_3$, on the reference $\langle W_a^{\otimes L}|$.

Setting. $h = \frac{1}{2}$; pairwise ABJM inhomogeneities $\theta_{A,n} = \theta_{F,n} + 2h$ for every pair n ($\theta_{F,n}$ arbitrary, distinct or equal); diagonal twist $\Lambda = \mathrm{diag}(z_1, y, y, z_4)$ with $z_2 = z_3 =: y$; reference $\langle W_a^{\otimes L}|$ of (86). For pair 1 (sites 1, 2) write $\theta_F = \theta_2$, $\theta_A = \theta_1 = \theta_F + 2h$ and $u_0 := \theta_F + h = \theta_A - h$. $\hat{t}_k$ is t_k with exactly vanishing scalar prefactors $(u - \theta - jh) = 0$ dropped. Evidence: exact in `pair-group-rotation.wb` (rerun 2026-10-13: sections 1–6, 8 reproduced, $L=2$ sections 7, 9 partial; $L=3$ rows of the untwisting sections skipped as too heavy) and in the python of `checks/session-2026-10-12/pairgroup/`, `checks/session-2026-10-12/lpairs/`.

16.1 Pair-vacuum pattern and the column-support lemma (2603 App. K style)

[exact] [`pair-group-rotation.wb` §1; python `pairvac_pattern.py`, `scan_d.py`]. With $\langle S| = \sum_m \langle e_m e_m|$ the A–F singlet and $(\mathcal{L}_A\mathcal{L}_F)_{jk}$ the (j,k) auxiliary component of the Lax product on one pair (exact, symbolic in u, θ_A, θ_F),

$$\langle e_1 e_4|(\mathcal{L}_A\mathcal{L}_F)_{jk}(u) = (u-\theta_A)(u-\theta_F)\delta_{jk}\langle e_1 e_4| - h(u-\theta_A)\delta_{k4}\langle e_1 e_j| + h(u-\theta_F)\delta_{j1}\langle e_k e_4| - h^2\delta_{j1}\delta_{k4}\langle S|. \tag{112}$$

Seven components vanish identically, $(j,k) \in \{(2,1),(3,1),(4,1),(3,2),(4,2),(4,3),(2,3)\}$, and there are four *bonus* zeros: $(2,4),(3,4)$ at $u = \theta_A$ and $(1,2),(1,3)$ at $u = \theta_F$. This is the pattern of $\langle Z|\mathcal{L}_{jk}$ in 2603 (six lower-triangular zeros plus bonus zeros), here on the fused pair.

Lemma (column support). Let $\mathcal{D}_k$ be the set of k-subsets $I \subset \{1,2,3,4\}$ with $1 \notin I$, $4 \in I$. The rows $I \in \mathcal{D}_k$ of the $\wedge^k$ auxiliary space are diagonal on *both* vacua $\langle e_1|$ (A) and $\langle e_4|$ (F), with I-independent eigenvalues, so pairs $2, \ldots, L$ in the vacuum are transparent on $\mathcal{D}_k$. If the Lax product of pair 1 at u_0 has column support in $\mathcal{D}_k$, then for *any* twist, hence for any global rotation of $\langle \mathrm{vac}^{\otimes L}|$ (which intertwines t^Λ with $t^{G'}$),

$$\langle \mathrm{vac}^{\otimes L}|t_k(u_0) = \lambda_k(u_0)\, \langle \mathrm{vac}_{\rm pair}|t_k^{(1\ \rm pair)}(u_0) \otimes \langle \mathrm{vac}^{\otimes(L-1)}|, \tag{113}$$

$$\lambda_k(u) = \prod_{n\geq 2} \Big[\prod_{j=0}^{k-1} (u - \theta_{A,n} - jh)\ (u - \theta_{F,n} - kh) \prod_{j=0}^{k-2} (u - \theta_{F,n} - jh) \Big] \tag{114}$$

(vanishing factors dropped). The F site at $u_0 = \theta_F + (k-1)h$ kills all columns without a 4; the A site at $u_0 = \theta_A - h$ kills all columns containing a 1. Hence

- $k=1$: $t_1(\theta_F)$ localises (F-site condition only; any geometry);
- $k=3$: $\hat{t}_3(\theta_A - h)$ localises (A-site condition only; any geometry);

- $k = 2$: $t_2(\theta_F + h)$ localises *only if* both conditions hold, i.e. *only at* $\theta_A - \theta_F = 2h$ (scan $d = (\theta_A - \theta_F)/h = -5, \ldots, 5$ on 4 sites with random twist and random BPS reference, words $t_k(\theta_F + mh)$, $k \le 3$, $m = -6, \ldots, 7$).

This is why all searches at generic θ failed (Sec. 16.6) and why the ABJM point, already singled out by $[H_{\rm MZ}, t_a] = 0$ (Sec. 4), is the place where the pair κ-family lives. Every single localising word equals $\lambda_k \times$(the same word on the one-pair chain), exactly [`pairword_check.py`, Fractions, random rational twist and BPS reference].[15]

Operator identities at the ABJM point [exact] [`even-bridge-lpairs.wb` §1a; exact mod $P = 268435399$, random product bras, $L = 2, 3$]:

$$t_2(u_0) = t_1(u_0)\, t_1(\theta_F), \qquad t_3(u_0) = 0 \qquad (u_0 = \theta_F + h = \theta_A - h), \tag{115}$$

the latter after removing the vanishing scalar prefactor ($\hat{t}_3(u_0)$ is the regular remainder).

The localised algebra on pair 1 [exact] [`pair-group-rotation.wb`; python `span_full.py`, `pair_sym.py`]. All individually localising words (products of length ≤ 4 of $t_k(\theta_F + mh)$, $k \le 3$, $m = -2, \ldots, 5$) span, on any reference, exactly

$$\{\, 1,\ X_1 = t_1(\theta_F),\ X_2 = t_2(\theta_F + h),\ X_3 = \hat{t}_3(\theta_F + h)\ (= \hat{t}_3(\theta_A - h)),\ X_3X_1 \,\} \tag{116}$$

which is 5-dimensional at the ABJM point and 4-dimensional at generic θ (no X_2). The same-point generating functions $\sum_k c_k t_k(u_0)$ localise nothing beyond the individually localising terms. On the pair vacuum with a general twist matrix G (pair-1 chain; S the singlet vector)

$$\langle \mathrm{vac}|X_1 = 2h^2\, e_1 \otimes e_4 G - h^2 G_{41}\, S, \qquad \langle \mathrm{vac}|X_3 = 12h^5\, (e_1 \,\mathrm{cof}\, G) \otimes e_4, \qquad X_3X_1 \propto \rho_{\rm pair}(G^{-T}), \tag{117}$$

and X_2 is given by the 2×2 minors of G with rows $\{2,4\}$, $\{3,4\}$ (no group form). For a diagonal twist the pair-level normalised words $\xi_1 = X_1/(2h^2)$, $\xi_2 = -X_2/(2h^4)$, $\xi_3 = X_3/(12h^5)$ have eigenvalues on the charged weight states $e_i \otimes e_j$ ($i \ne j$; A index i, F index j)

$$\xi_1 \to z_j, \qquad \xi_2 \to z_j\,(e_1(z) - z_i - z_j), \qquad \xi_3 \to \det\Lambda/z_i, \qquad \xi_3\xi_1 \to \det\Lambda\, z_j/z_i, \tag{118}$$

whereas $\rho_{\rm pair}(1 - \kappa\Lambda)$ needs $(1 - \kappa z_i)/(1 - \kappa z_j)$ on these states and 1 on the zero-weight states. For a twist with generic spectrum the rank-one (product) elements of the five-word span are only $v(a + bG^{-T}) \otimes u$ and $vG^{-T} \otimes u(a + bG)$: there is *no common group rotation of a pair* on any reference with $u\Lambda v^T \ne 0$. At $z_2 = z_3$ on W_a the span does reach it (next subsection).

16.2 The exact five-word K_κ

[exact] [`pair-group-rotation.wb`; python `pair_solve.py`, `word_verify.py`; pair-level solution symbolic in $z_1, y, z_4, h, a, \kappa$]. With $s = 1 - \kappa(z_1 + z_4)$ and $\lambda_1 = \lambda_1(\theta_F)$, $\lambda_2 = \lambda_2(\theta_F + h)$, $\lambda_3 = \lambda_3(\theta_F + h)$ (the products over the other pairs of the lemma),

$$K_\kappa = s^2 + \kappa s\Big[\frac{X_1}{h^2\lambda_1} + \frac{X_2}{2h^4 y\,\lambda_2} + \frac{X_3}{6h^5y^2\lambda_3}\Big] + \kappa^2 \frac{X_3X_1}{24h^7y^2\lambda_1\lambda_3}, \tag{119}$$

and

$$\langle W_a^{\otimes L}|K_\kappa = (1 - \kappa z_1)^2\, \langle W_{a'} \otimes W_a^{\otimes(L-1)}|, \qquad a' = a\,\frac{1 - \kappa z_4}{1 - \kappa z_1}. \tag{120}$$

[15] Rerun 2026-10-13 (`pair-group-rotation.wb`, cell 17): the python check fitted only the constant for *products*. Four of the nine two-factor words tested, $t_1(\theta_F)t_1(\theta_F + h)$, $t_1(\theta_F)\hat{t}_3(\theta_F + 3h)$, $t_1(\theta_F + 2h)\hat{t}_3(\theta_F + h)$ and $\hat{t}_3(\theta_F + h)\hat{t}_3(\theta_F + 2h)$, do localise and are proportional to the one-pair word, but with constants $\frac{51}{31}$ resp. $\frac{71}{31}$ (ratios 51/31, 71/31, 71/31, 51/31) times $\prod\lambda_k$, not $\prod\lambda_k$. K_κ below is unaffected: its only product word X_3X_1 does have constant $\lambda_1\lambda_3$.

Equivalently $K_\kappa = (1-\kappa z_1)(1-\kappa z_4)\rho_{\rm pair}(1-\kappa\Lambda)$ on W-pairs, cf. (88), (89). Exact at $L=2,3$ for $a\in\{\frac{1}{3},-1\}$ and $\kappa\in\{\frac{3}{10},-\frac{7}{5},1/z_4\}$ (the last has $a'=0$). In pair-level normalisation (all $\lambda_k=1$)

$$K = s^2 + \kappa s\big(2\xi_1 - \xi_2/y + 2\xi_3/y^2\big) + \kappa^2\xi_3\xi_1/y^2, \tag{121}$$

with coefficients $(s^2,\ \kappa s/h^2,\ \kappa s/(2h^4y),\ \kappa s/(6h^5y^2),\ \kappa^2/(24h^7y^2))$ on $(1, X_1, X_2, X_3, X_3X_1)$; at $\kappa=3/10$, $z=(3/2,5/4,5/4,4/7)$ these are (2809/19600, 159/350, 636/875, 1696/4375, 192/625), identical to the Wolfram calibration cell. The word is *unique* (the linear system for the five coefficients has a unique solution), independent of a, and local (the pair-level coefficients depend on L only through λ_k).

Structure of K_κ [exact] [`even-bridge-epsilon-expansion.wb`, $L=2$; python for $L=3$]. Write, with the eigenvalues $\tau_F=\tau_1(\theta_F)$, $\tau_2(u_0)=x\tau_F$, $\hat\tau_3(u_0)$ of the words,

$$\mathsf{A}=\frac{\tau_F}{h^2\lambda_1},\qquad \mathsf{B}=\frac{\tau_2(u_0)}{2h^4y\,\lambda_2},\qquad \mathsf{C}=\frac{\hat\tau_3(u_0)}{6h^5y^2\lambda_3},\qquad K_\kappa(\tau)=s^2+\kappa s(\mathsf{A}+\mathsf{B}+\mathsf{C})+\tfrac14\kappa^2\,\mathsf{AC}. \tag{122}$$

At $\kappa_\star=1/(z_1+z_4)$ ($s=0$) $K_\kappa\propto X_3X_1$, the fixed twist rotation $W_a\to W_{ar}$, $r=z_1/z_4$. At the untwisted point ($\kappa=1$, $y=1$, $s=-1$) one finds $\mathsf{AC}=4$ on all 6 ($L=2$) and 26 ($L=3$, python) HW states ($\mathsf{A}=2e^{i\vartheta}$, $\mathsf{C}=2e^{-i\vartheta}$). Explanation (exact at $L=2$): $\hat X_3X_1=24h^7\lambda_1\lambda_3\,S=\frac{9}{256}\,S$ with S the one-pair translation (the swap of the two pairs at $L=2$), so $\mathsf{AC}=4$ on translation-invariant states; this is the shift test $U\propto S$ of Sec. 4. Therefore

$$K_1(\tau) = 2-\mathsf{A}-\mathsf{B}-\mathsf{C} = 2-4\cos\vartheta-\mathsf{B}. \tag{123}$$

K_1 vanishes on the vacuum ($\mathsf{A}=\mathsf{C}=2$, $\mathsf{B}=-2$) and on $[L-1,0,L-1]$ ($\mathsf{A}+\mathsf{B}+\mathsf{C}=2$), and is non-zero on other irreps: $L=2$: 10 and 2 on the two singlets, 8 on the zero-weight states of $[0,2,0]$. The characteristic polynomials of the untwisted K_1 operator at $L=2$ in the sectors $(2,0,0,-2)$, $(1,0,0,-1)$, $(0,0,0,0)$ are $-x$, $x^{11}(x-8)$, $x^{24}(x-2)(x-8)^2(x-10)$ [exact].[16]

16.3 Why $z_2=z_3$ is needed

[exact] [`pair-group-rotation.wb` §4; python `negcontrols.py`, `combo_search.py`, `combo_L3.py`; exact]. The test on $W_a^{\otimes 2}$:

point	twist	rank of 5-word span	span + target
ABJM	$z_2\neq z_3$	5	6 (fails)
generic θ	$z_2=z_3$	4	5 (fails)
ABJM	$z_2=z_3=y$	5	5 (**works**)

Mechanism. At $z_2\neq z_3$ the obstruction lies in the zero-weight directions: X_1 has a zero-weight component $\propto E_{22}+E_{33}$ and X_2 one $\propto z_3E_{22}+z_2E_{33}$, so the leftover is spanned by these two (6 directions needed against 5 operators). The cyclic space of W_a on one pair is 6-dimensional (2 weight states plus 4 zero-weight eigenstates) while the algebra is 5-dimensional; at $z_2=z_3$ the zero-weight eigenstate odd under $2\leftrightarrow 3$ decouples, the cyclic space becomes 5-dimensional and equals the algebra. On a generic BPS reference at $z_2=z_3$ the same word gives a pair state of rank 2 (entangled, traceless), 3.5% away from the rotation $d_1d_4\rho(D)$. Non-localising combinations: at $L=2$ universal combinations of 253 words reach $W_{a'}\otimes W_a$ even at $z_2\neq z_3$ and generic θ (the two-pair algebra is too rich: meaningless); at $L=3$, ABJM with $z_2\neq z_3$: rank 5, target residual $2.3\cdot10^{-2}$ (fails); generic θ with $z_2\neq z_3$: rank 4, fails; ABJM with $z_2=z_3$ reaches; generic θ with $z_2=z_3$ reaches but the L^1-sparsest combination has 80 words with coefficients $10^{-3}\ldots10^{7}$ (no structure; L-universality untested) [python].

[16] The statement "$K_{\kappa=1}(\tau)=0$ for all non-vacuum states" made earlier on 2026-10-13 is false: K_1 vanishes exactly on the vacuum tower and $[L-1,0,L-1]$ (i.e. $n\le 1$, with $n=L-\mu_1$ as in Sec. 15.6); non-cyclic states can have accidental zeros ($L=2$: $[2,1,0]$, $[0,1,2]$).

Finite twist versus the limit [python] [`task3b.py`, `task3c.py`]. At finite generic twist with $d = z_2 - z_3 \neq 0$ the relative residual of the best word is linear in d: $5.25\cdot10^{-3}$, $5.25\cdot10^{-4}$, $5.25\cdot10^{-5}$, $5.25\cdot10^{-6}$ for $d = 10^{-1}$, 10^{-2}, 10^{-3}, 10^{-4} ($L = 3$, $\kappa = \frac{3}{10}$), equal to the least-squares optimum over all five coefficients (the symmetric choice $y \to (z_2+z_3)/2$, $y^2 \to z_2 z_3$ in the X_2 term is the optimum; $y = z_2$ is six times worse): $z_2 = z_3$ is a genuine restriction at finite twist. In the untwisting limit, however, only $z_2 - z_3 = O(\epsilon)$ is needed: for $z_2 - z_3 = \frac{1}{2}\epsilon^q$ with $q = 1, 2, 3$ (so $q = 1$ has $\varphi_2 \neq \varphi_3$) the finite-twist identity is violated by $|\mathrm{RHS} - \mathrm{LHS}| \approx 0.1\,(z_2 - z_3)^2/\epsilon$ (e.g. state #4, $[2, 0, 2]$: $2.6\cdot10^{-4}$, $2.7\cdot10^{-5}$, $2.7\cdot10^{-6}$ for $q = 1$), and the *limits are unchanged* (identical to the $z_2 = z_3$ runs). A separation $z_2 - z_3 \gg \sqrt{\epsilon}$ would spoil it. *Caveat* (`even-bridge-sov-formula.wb`): at exactly $z_2 = z_3$ the companion frame is singular and the diagonal-twist chain has a $\mathfrak{gl}(2)$ symmetry, so there is no SoV basis; the K_κ route has to be combined with SoV at $z_2 - z_3 = O(\epsilon)$, which suffices for the limit (Sec. 18).

16.4 The one-pair even bridge and its untwisting limit

[exact] [`pair-group-rotation.wb`, `even-bridge-lpairs.wb`; operator identity; at $L = 3$ on all 26 HW states [num, 10^{-6}], limited by cancellation]. Since (120) is an operator identity, for any eigenstate Ψ of t^Λ with $z_2 = z_3$

$$\langle W_{a'} \otimes W_a^{\otimes(L-1)} | \Psi \rangle = \frac{K_\kappa(\tau_\Psi)}{(1-\kappa z_1)^2}\, \langle W_a^{\otimes L} | \Psi \rangle, \qquad K_\kappa(\tau) = (122). \tag{124}$$

At $\kappa = 1/z_4$ (vacuum pair, $a' = 0$), with $\Sigma = \mathsf{A} + \mathsf{B} + \mathsf{C}$ and $\Pi = \frac{1}{4}\mathsf{AC}$ the eigenvalues of $2\xi_1 - \xi_2/y + 2\xi_3/y^2$ and $\xi_3\xi_1/y^2$,

$$\langle \mathrm{vac} \otimes W_a^{\otimes(L-1)} | \Psi \rangle = \frac{z_1^2 - z_1\Sigma + \Pi}{(z_1 - z_4)^2}\, \langle W_a^{\otimes L} | \Psi \rangle. \tag{125}$$

Untwisting recipe [num, $O(\epsilon)$] [`even-bridge-lpairs.wb` §2, rerun 2026-10-13; python `untwist.py`, `task2.py`]. $z = 1 + \epsilon\varphi$ with $\varphi_2 = \varphi_3$ (standard $\varphi = (0.9, 0.35, 0.35, -1.1)$), $\theta_{F,n} = \epsilon\theta^0_n$ ($\theta^0 = (0, 0.37, -0.52)$), $\theta_A = \theta_F + 2h$; Ψ_ϵ is the twisted eigenvector in the weight sector connected to the untwisted HW state Ψ_0 (all non-degenerate, overlap $\to 1$). The right-hand side of (124) is a $0 \cdot \infty$ product: $K_\kappa = O(1)$ or smaller, $\langle W_a^{\otimes L} | \Psi_\epsilon \rangle = O(\epsilon^n)$ and $(1 - \kappa z_1)^2 = O(\epsilon^2)$ (details below).

(i) $\kappa = 1/z_4$, $a = -1$: $\to \langle \mathrm{vac} \otimes W_{-1}^{\otimes(L-1)} | \Psi_0 \rangle$ ($M = L - 1$). $L = 2$: singlets $-1/\sqrt{40}$, $1/\sqrt{24}$ (errors $1.6, 2.0\cdot10^{-3}$ at $\epsilon = 10^{-2}$, decreasing linearly), $[1, 0, 1]$: 0, vacuum: 1.

(ii) $\kappa \to 1$ with $a' - a \to -1$: $\kappa = 1/(z_1 - a(z_1 - z_4))$ with $a = \frac{1}{3}$ (then $a' - a = -1$ exactly), or the linear prescription $\kappa = 1 - c\epsilon$, $a = (\varphi_1 - c)/(\varphi_1 - \varphi_4)$, for which $a' - a = a\kappa(z_1 - z_4)/(1 - \kappa z_1) \to a(\varphi_1 - \varphi_4)/(c - \varphi_1) = -1$. The finite HW shift e^{-aJ} gives $\langle W_{a'} \otimes W_a^{\otimes(L-1)} | \Psi_0 \rangle = \langle W_{-1} \otimes \mathrm{vac}^{\otimes(L-1)} | \Psi_0 \rangle = \langle \mathrm{vac}^{\otimes(L-1)} \otimes W_{-1} | \Psi_0 \rangle$ (cyclicity), i.e. $M = 1$.

In (ii) the shift parameter a stays *finite*, which is exactly what the highest-weight trick of Sec. 14.2 lacked ($b \to \infty$). The limits are independent of direction and of c: for $\varphi = (.9, .35, .35, -1.1)$, $(-.7, 1.3, 1.3, .4)$ and $(1.5, -.6, -.6, .2)$ with $c = 0.3$ and $c = 2.0$, states $0, 4, 16, 17, 18, 20, 22, 25$, all six runs converge as $O(\epsilon)$ to the same values (e.g. state 16, target -0.171126: errors $3.4, 19, 3.4, 11, 1.8, 32\cdot10^{-5}$ at $\epsilon = 10^{-4}$). $\kappa = 1/z_4$ is the member $c = \varphi_4$ ($a = 1$, $a' = 0$).

Families at $L = 3$. [num, $O(\epsilon)$] [`even-bridge-lpairs.wb` §2]. All 26 HW states reproduced:

family	κ	a	$a'-a$	limit (after HW shift)
F1	$1/(z_1-a(z_1-z_4))$	$\frac{1}{3}$	-1	$\langle W_{-1}\otimes \mathrm{vac}^{\otimes 2}\vert\Psi_0\rangle = C_4$ $(M=1)$
F2	$1/(z_1+a(z_1-z_4))$	$\frac{1}{3}$	$+1$ (shift by $-a'$)	$\langle \mathrm{vac}\otimes W_{-1}^{\otimes 2}\vert\Psi_0\rangle = C_2$ $(M=2)$
F3	$1/z_4$	-1	$+1$	C_2 $(M=2)$
F4	$1/z_4$	$+1$ (shift by -1)	-1	C_4 $(M=1)$
G	global $\rho(1-\kappa\Lambda)$ on all pairs	—	—	C_0 $(M=3)$; weight factor $\left[\frac{1-\kappa z_4}{1-\kappa z_1}\right]^3\prod_i(1-\kappa z_i)^{\mu_i}$

(here C_ℓ is the data of Sec. 9, $\ell = 2(L-M)$ O_2 sites). All converge as $O(\epsilon)$ (on state 16 the errors at $\epsilon = 10^{-2}, 10^{-3}, 10^{-4}$ are $(1.5\text{–}6.0)\cdot 10^{-3}$, $(1.5\text{–}6.0)\cdot 10^{-4}$, $(1.5\text{–}6.1)\cdot 10^{-5}$ for F1–F4, and $3.3\cdot 10^{-5}, 3.3\cdot 10^{-7}, 3.3\cdot 10^{-9}$ for G, whose target is 0).[17] The non-zero targets at $L=3$ are the vacuum (1) and the six $[1,0,1]$ states, $C_2 = C_4 = -0.171126$, -0.027138, $-0.035872 \mp 0.028637i$, $0.129924 \mp 0.081713i$; the other 19 states have target 0, reproduced to $O(\epsilon)$.[18]

Coverage. For HW Ψ_0, $C_M = (-1)^{L+\mu_4} C_{L-M}$ (HW shift by $+1$ followed by $\rho(\mathrm{diag}(1,1,1,-1))$). The one-pair word therefore gives $M=1$ and $M=L-1$ at every L, and the global rotation $M=L$; at $L=3$ these are all even bridges. **The first case not covered is $L=4$, $M=2$.** It needs a second integrable family or a twist-dressed reference (Sec. 19).

Orders in ϵ (which states contribute) [num, 10^{-6}] [`even-bridge-epsilon-expansion.wb`; python `scan_eps.py`]. With $n = L-\mu_1$ for weight $(\mu_1,0,0,-\mu_1)$ one has $w := \langle W_a^{\otimes L}\vert\Psi_\epsilon\rangle = O(\epsilon^n)$ and $w\equiv 0$ for every other weight, $K = O(\epsilon^2)$ for $n\le 1$ (forced by finiteness of the left side; $K_1 = 0$) and $K = O(1)$ for $n\ge 2$. Hence the limit is 1 for the vacuum, 0 for $n=1$ and for $n=3$ ($w = O(\epsilon^3)$ against $(1-\kappa z_1)^2 = O(\epsilon^2)$), and for $n=2$ (the singlets and $[1,0,1]$ at $L=2,3$, i.e. $[L-2,0,L-2]$)

$$C_{M=1}(\Psi_0) = K_1(\tau_{\Psi_0})\,\frac{\omega_2}{2(\varphi_1-\varphi_4)^2}, \qquad \omega_2 = \lim \epsilon^{-2}\langle \mathrm{vac}\vert J^2\,\Psi_\epsilon\rangle \tag{126}$$

(Richardson to 10^{-7} on the two $L=2$ singlets and six $L=3$ $[1,0,1]$ states; ω_2 is bridge independent). Using the exact handle (109) with $[J,T_{14}]=0$ and $J\Psi_1 = (\varphi_1-\varphi_4)(M_0-\lambda_0)^{-1}T_{14}(u_1)\Psi_0$ (first-order perturbation theory),

$$\omega_2 = 2(\varphi_1-\varphi_4)^2\,\frac{\langle \mathrm{vac}\vert T_{14}(u)\left(t_1(u_1)-\tau_{\Psi_0}(u_1)\right)^{-1}T_{14}(u_1)\vert\Psi_0\rangle}{\tau_{\mathrm{vac}}(u)-\tau_{\Psi_0}(u)} \tag{127}$$

(untwisted operators, any u, u_1; the resolvent acts on the weight sector of $J\Psi_0$, dimension 12 at $L=2$, 24 at $L=3$). [exact] [`even-bridge-epsilon-expansion.wb`, cells (4), (5)]: at $L=2$ this gives the singlets exactly, $C_{M=1} = -1/\sqrt{40}$, $1/\sqrt{24}$, equal to the direct overlap $\langle W_{-1}\otimes \mathrm{vac}\vert\Psi_0\rangle$ and independent of (u,u_1); at $L=3$ the six $[1,0,1]$ states agree with the data to $8\cdot 10^{-15}$ (machine precision). Only the HW states of $[L-1,0,L-1]$ enter the resolvent (4 at $L=2$, 9 at $L=3$), and K_1 vanishes on all of them. The member $\kappa = 1/z_4$ gives the same product up to the sign of the symmetry above, so it is no independent handle on ω_2.

16.5 The ℓ-pair generalisation is pretwisted

[exact] (exact mod P) [`even-bridge-lpairs.wb` §1b–1e; python `task1b.py`–`task1k.py`]. At $L = 2,3,4$ and every pair p:

[17] In the table of section 2 of `even-bridge-lpairs.wb` the columns labelled F1 and F2 are swapped relative to the family definitions of its text (python `task2.py` uses the opposite sign of κ for "F1"); the notebook's F1/F2 columns equal the python F2/F1 columns. Both families converge to the same limit because $C_2 = C_4$ on these states, so no limit changes (rerun 2026-10-13).

[18] The earlier consolidated report (`pair-group-rotation` python, 2026-10-12) lists -0.153484 ($\times 2$) in place of the last pair; $\vert 0.129924 \mp 0.081713i\vert = 0.153484$, so this is the phase convention of the degenerate pair.

- The pair-p word $K^{(p)}$ (the X's at pair p, λ's over $n \neq p$) does *not* localise on pair p for $p \geq 2$: every pair-p word changes pairs $1, \ldots, p$ (block localisation, as 2603's `localized`). Naive products $\prod_{p \leq \ell} K^{(p)}$ fail ($\ell = 2$ at $L = 2, 3$; $\ell = 3$ at $L = 3$).

- All *multilinear* monomials $\prod_p w_p$, $w_p \in \{1, X_1, X_2, X_3, X_3X_1\}^{(p)}$, block-localise, $\langle W_a^{\otimes 3}| m = \lambda_m(\langle W_a^{\otimes 2}| m$ on the 2-pair chain$) \otimes \langle W_a|$; same-pair squares $(X_1^2, X_1X_2, X_2^2, X_2X_3, X_3^2)$ do not.

- *The target is not reachable:* $\langle W_{a'}^{\otimes \ell} \otimes W_a^{\otimes (L-\ell)}|$ is not in the span of the 5^ℓ multilinear monomials (rank $25 \to 26$ at $(L, \ell) = (2,2), (3,2), (4,2)$; $125 \to 126$ at $(3,3)$; $\kappa = \frac{3}{10}$ and $1/z_4$), nor of the monomials of per-pair degree ≤ 2 (100, rank 100); per-pair degree 3 (400 monomials) spans the whole 285-dimensional cyclic module (trivially, non-locally).

- The W-type product states in the two-block span form the 5-dimensional space $\mathrm{span}\{u_i\} \otimes W_a + W_{ar} \otimes \mathrm{span}\{u_j\}$, $r = z_1/z_4$ ($u_0 = e_1e_4$, $u_1 = e_4e_4 - e_1e_1$, $u_2 = -e_4e_1$, $W_b = u_0 + bu_1 + b^2u_2$): pair 1 rotated freely with pair 2 untouched, *or* pair 2 rotated freely with pair 1 twisted by the *fixed* Λ^{-1} ($a \to ar$). No $W_b \otimes W_c$ with both $b, c \neq a, ar$.

The structure is *pretwisted*:

$$\langle W_{ar}^{\otimes (p-1)} \otimes W_a^{\otimes (L-p+1)}| K_\kappa^{(p)} = (1 - \kappa z_1)^2 \, \langle W_{ar}^{\otimes (p-1)} \otimes W_{a'} \otimes W_a^{\otimes (L-p)}| \qquad (L = 3, 4, \text{ all } p), \tag{128}$$

and $(X_3X_1)^{(p)} = K^{(p)}$ at $\kappa_\star = 1/(z_1 + z_4)$ ($s = 0$):

$$\langle \cdots |(X_3X_1)^{(p)} = 24h^7y^2\lambda_1\lambda_3 z_4^2 \, \langle \cdots W_{ar} \text{ at pair } p \ \cdots |. \tag{129}$$

The chain word $(X_3X_1)^{(1)} \cdots (X_3X_1)^{(p-1)} K^{(p)}$ on $\langle W_a^{\otimes L}|$ equals $c\,(1 - \kappa z_1)^2 \langle W_{ar}^{\otimes (p-1)} \otimes W_{a'} \otimes W_a^{\otimes (L-p)}|$. Consequently $\prod_p K_{\kappa_p}^{(p)}$ rotates correctly iff $\kappa_1 = \cdots = \kappa_{\ell-1} = \kappa_\star$: the alternating-chain analogue of 2603's block localisation is the *end-growing chain* (fixed Λ^{-1} on the pairs left of the free one), *not* a common-κ block. Right-end words $Y_1 = t_1(\theta_{A,L})$, $Y_2 = t_2(\theta_{A,L})$, $Y_3 = \hat{t}_3(\theta_{A,L} + h)$ localise on pair L and $\{1, Y_1, Y_2, Y_3, Y_1Y_3\}$ rotates it freely; left $\times$ right do not combine (first and last pair cannot be rotated together). For $\ell = L$ the global $\rho(1 - \kappa\Lambda)$ is available, with eigenvalue $[(1 - \kappa z_4)/(1 - \kappa z_1)]^L \prod_i (1 - \kappa z_i)^{\mu_i}$. At $L = 4$, $\ell = 2$ the target $\langle W_{a'}^{\otimes 2} \otimes W_a^{\otimes 2}|$ is not in the 25 multilinear monomials (rank 25).

16.6 Locality brute force at generic θ and z

[python] (exact modulo P and with Fractions) [`pair-rotation-locality-L3.wb` (not rerun), `checks/session-2026-10-12/locality/`]. Diagonal frame, reference $W_a^{\otimes L}$, generic θ and generic z; words on pair 1 built from the 48 non-vanishing factors $t_a(\theta_s + mh)$, $s \in \{1, 2\}$, $m = -4, \ldots, 4$, products of ≤ 4 factors. The localised subspace has dimension 6 at $L = 2$ and 3. Targets $T_1 = W_{a'} \otimes W_a^{\otimes (L-1)}$ and T_2 (the common rotation) lie in the span with ≤ 2 factors at $L = 2$ but only with ≥ 4 factors at $L = 3$ (rank 563 of 588), densely (no sparse solution: OMP, L^1 and exhaustive ≤ 5-word searches all fail at $L = 2$); the needed word length grows $\sim 2(L - 1)$. *Obstruction:* $T_1, T_2 \in \mathrm{span}\{\psi_0, \psi_0 t_3(\theta_1 - h), \psi_0 t_3(\theta_1 - h) t_1(\theta_2), \psi_0 t_1(\theta_2)\} \oplus \mathbb{C}\,(S \otimes W_a^{\otimes (L-1)})$, where $\psi_0 = \langle W_a^{\otimes L}|$ and the isolated pair singlet $S \otimes w_0$ is not in the word span; the coefficients are identical at $L = 2, 3$: ψ_0-coefficient $9(8a' - 7)^2/169$, singlet coefficient $35(3a' - 1)(8a' - 7)/806$. Local universal identities confirmed: c_1, c_{12} of Sec. 15.2, c_3 for $t_3(\theta_1 - h) \to \rho_1(\Lambda^{-1})$,

$$c_3 = -\tfrac{3}{4} \det \Lambda \prod_F (\theta_{1s} - h)(\theta_{1s} - 2h)(\theta_{1s} - 4h) \prod_{A,\, s \geq 3} (\theta_{1s} - h)(\theta_{1s} - 2h)(\theta_{1s} - 3h), \tag{130}$$

and $\psi_0 t_1(\theta_2) = \alpha[\psi_0 \rho_2(\Lambda) + (\sigma/\alpha) S \otimes w_0]$ with $\sigma/\alpha = ha(z_1 - z_4)/\theta_{12}$ (L-independent). Calibration (i) holds only for $\psi_0 = W_a^{\otimes L}$. So the even bridge is *not* locally reachable at generic θ and z, consistent with Secs. 16.1–16.3: the few-word solution needs the ABJM point *and* $z_2 = z_3$.

16.7 Q-forms of τ_F, x and $\hat{\tau}_3(u_0)$

[num, 10^{-8}] [`even-bridge-lpairs.wb` (Q-form cell, rerun 2026-10-13; 0/0 guard added for three exactly vanishing rows); python `qf.py`, `qdict.py`, `task4a.py`–`task4d.py`]. Conventions of the SoV dictionary (Sec. 13): Baxter equation $\sum_{a=0}^{4}(-1)^a\tau_a(u)Q(u-ah)=0$, $Q_i=z_i^{u/h}\Phi(u)q_i(u)$ with q_i monic polynomials, $\Phi(u)=V_4(u)\Phi(u-h)$, $V_4(u)=\prod_A(u-\theta_A)\prod_F(u-\theta_F-h)$; $u_0=\theta_F+h=\theta_A-h$ (pair 1); "rest" denotes the product over the *other* pairs only; $F_3(u)=\prod_F(u-\theta_F)(u-\theta_F-h)$; for a triple I of Q's, $P_I(u)=\det[z_i^{2-s}q_i(u-sh)]_{i\in I,\,s=0,1,2}$, $q_I=P_I/F_3$ (reduced Casoratian = dual Q, a polynomial), $Z_I=\prod_{i\in I}z_i$. Then

$$\tau_F=\tau_1(\theta_F)=\frac{Q_j(\theta_F)}{Q_j(\theta_F-h)}=z_jV_4(\theta_F)\frac{q_j(\theta_F)}{q_j(\theta_F-h)}\qquad(\text{any } j),\tag{131}$$

$$x=\tau_1(u_0)=\frac{V_4^{\rm rest}(u_0)}{F_3^{\rm rest}(\theta_F)}\,Z_I^4\,\frac{\det\left[z_i^{-1}q_i(\theta_F+h),\,z_i^{-3}q_i(\theta_F-h),\,z_i^{-4}q_i(\theta_F-2h)\right]_{i\in I}}{q_I(\theta_F)},\tag{132}$$

$$\hat{\tau}_3(u_0)=h\,V_4^{\rm rest}(u_0)\,V_4(\theta_F)V_4(\theta_F-h)\,Z_I\,\frac{F_3^{\rm rest}(u_0)}{F_3^{\rm rest}(\theta_F)}\,\frac{q_I(\theta_F+h)}{q_I(\theta_F)}\qquad(\text{any triple } I),\tag{133}$$

and $\tau_2(u_0)=x\,\tau_F$ (the operator identity above). So X_1 is a single-Q ratio (F site), X_3 a single *dual*-Q ratio $q_I(\theta_F+h)/q_I(\theta_F)$ (A site), and $X_2=xX_1$ with x a gapped Casoratian divided by a reduced Casoratian. Derived as the $\delta\to0$ limit ($\theta_A=\theta_F+2h+\delta$) of the generic A-site Cramer formulas $\tau_a(\theta_A-h)=D_a/D_0$, the zeros of D_0 and of $V_4(\theta_A-h)$ being vacuum factors that cancel explicitly. Checks: $L=2$, all 256 states $\times$ 4 triples, maximal relative deviation $5.6\cdot10^{-12}$ (τ_F), $3.3\cdot10^{-8}$ ($x,\tau_2,\hat{\tau}_3$); $L=3$, sector $(2,0,0,-2)$ (24 states) 10^{-6}, sector $(2,1,-1,-2)$ (9 states) $2\cdot10^{-8}$ (float64, Baxter-conditioning limited); continuity: at $\delta=10^{-2},10^{-3},10^{-4}$, $\tau_1(\theta_A-h)\to x$ and $\tau_3(\theta_A-h)/\delta\to\hat{\tau}_3(u_0)$ linearly in δ. Finally $K(\kappa)=1-\kappa(2S-\Sigma)+\kappa^2(S^2-S\Sigma+\Pi)$, $S=z_1+z_4$, does *not* factor into rational linear factors in κ: at $L=2$, $z_2=z_3$, 53 of the 256 states with real eigenvalues have complex roots.

17 Which references admit a pair κ-family

[exact] [`reference-admissibility-Kkappa.wb`, all sections run in the Wolfram kernel; exact python fallback in `checks/session-2026-10-13/refmap/` agrees on every output (rows, ranks, coefficients, residuals, $L=3$ mod p, covariance, Lie algebra, generic-z patterns); `agent-2026-10-13-reference-map.md`]. The five-word K_κ of Sec. 16.2 was derived on W_a. Here every product reference $(v\otimes u)^{\otimes L}$, $v\cdot u=0$ (same pair everywhere), at the ABJM point is tested. *Admissible* means: for all κ tested ($\frac{3}{10}$, $-\frac{7}{5}$, $1/z_4$) the polynomially normalised rotation $T_\kappa=[v(1-\kappa\Lambda)]\otimes[u\,\mathrm{adj}(1-\kappa\Lambda)]$ of (89) lies in the span of the five localised words acting on the reference. On the chain every localising word equals $\lambda_w\times$ the same word on the one-pair chain (re-verified at $L=2$ and at $L=3$ (6 sites, 4096-dim) mod p for every reference that passes and for the negative controls), so admissibility is a pair-level statement; for W_a the pair-level coefficients are those of K_κ ($K_\kappa=T_\kappa/(1-\kappa y)^2$ there).

17.1 The admissible set at $z_2=z_3$

The centraliser of Λ is $GL(1)\times GL(2)\times GL(1)$ (acting $v\to vk$, $u\to uk^{-T}$; it commutes with all t_a and with $1-\kappa\Lambda$), so every v is Levi-equivalent to one of seven representatives, with u kept symbolic. Write $p=(v_2,v_3)$, $q=(u_2,u_3)$. Quantifier elimination (symbolic `Reduce`, Wolfram cell 13; $u_2'=0$ is the Levi-coordinate form of $p\cdot q=0$) gives:

Levi class of v	admissible iff
$(1,1,0,1)$ $(v_1,p,v_4\neq 0)$	$u_2'=0$ and $u_4=0$ ($\Rightarrow u_1=0$): $u=(0,p^\perp,0)$
$(1,1,0,0)$ $(v_4=0)$	$u_2'=0$ ($\Leftrightarrow p\cdot q=0$, then $u_1=0$): $u=(0,q,u_4)$, $q\perp p$
$(0,1,0,1)$	$u_4=0$ ($\Rightarrow p\cdot q=0$)
$(1,0,0,1)$ $(p=0)$	$q=0$ **or** $u_4=0$
$(1,0,0,0)$, $(0,1,0,0)$, $(0,0,0,1)$	always ($u_1=0$, resp. $p\cdot q=0$, resp. $u_4=0$ forced)

Collecting, $\mathcal{A}=\mathcal{A}_{\rm block}\cup\mathcal{A}_W$ with

$$\mathcal{A}_{\rm block}=\{v_1u_1=0,\ v_4u_4=0,\ p\cdot q=0\},\qquad \mathcal{A}_W=\{p=0=q\},\tag{134}$$

i.e. either v,u are BPS in every spectral block of Λ ($v|_C\cdot u|_C=0$ for $C=\{1\},\{2,3\},\{4\}$; equivalently $uf(\Lambda)v^T=0$ for *every* function f), or both are supported on the indices $\{1,4\}$ (the torus images of W_a, where $v_1u_1=-v_4u_4\neq 0$ is allowed). Inside the Borel orbit ($v_1u_4\neq 0$): $\mathcal{A}\cap\text{Borel}=(C)\cup(B)$ with $(C)=\{v_4=0,\ u_1=0,\ p\cdot q=0\}$, i.e. $v=(1,p,0)$, $u=(0,q,u_4)$, and $(B)=\{p=q=0\}$, i.e. W_a up to torus. (C) contains $(A)=\{v=e_1,\ u=(0,q,u_4)\}$ and the non-frozen references $v=(1,p,0)$: the vacuum rotated by $g=\exp(p\cdot E_{1,(23)})\exp(c\cdot E_{(23),4})$ with $p\cdot c=0$ (the "E_{12}/E_{34}-type" good rotations of the README); they are *good* ($u\Lambda v^T=0$), whereas W_a is not ($u\Lambda v^T=(z_1-z_4)v_1u_1\neq 0$). Block-BPS $\subset$ good, the converse fails. Named subfamilies [Wolfram §5c and python, identical]: $v=(1,0,0,v_4)$ with generic u: only $q=0$ (W); v generic, $u=(-v_4,0,0,1)$: fails; the $\mathfrak{gl}(2)$-symmetric $(1,v_2,v_2,v_4)\otimes(u_1,u_2,u_2,1)$ fails unless $v_4=u_1=0$ and $v_2u_2=0$; only $v_2,v_3\neq 0$ (any u with $p\cdot q=0$) and only $u_2,u_3\neq 0$ are always admissible. Exhaustive and random confirmation [python] [`smallint.py`, `scan.py`]: 8946 pairs $(v,u)\in\{-1,0,1,2\}^8$ with $v\cdot u=0$ (2690 admissible, 0 mismatches with (134)), plus 39 blockwise, 20 W-type and 137 random-zero-pattern samples.

Charged-sector structure [exact]. The words commute with the Cartan torus; on the 12 charged states $e_a\otimes e_b$ ($a\neq b$) their eigenvalues are (118). On a reference with charged support $S=\{(a,b):a\neq b,\ v_au_b\neq 0\}$ the five words act through the $|S|\times 5$ evaluation matrix, and a κ-family needs every κ-component to lie in the span, i.e. rank $|S|$ together with a compatible zero-weight sector. At generic z a generic reference has $|S|=12$: impossible.

Formulation through G' [exact] [`covariance.py`]. $\rho(g)\,t_k^\Lambda\,\rho(g)^{-1}=t_k^{G'}$ $(k=1,2,3)$ with

$$G'=g^{-T}\Lambda g^T=(g\Lambda g^{-1})^T\tag{135}$$

(the other three candidates fail), hence $\langle\text{vac}|\rho(g)X[t^\Lambda]=\langle\text{vac}|X[t^{G'}]\rho(g)$ for every word, checked for the whole K_κ and seven g. For upper-triangular g, G' is lower triangular with spectrum z, and $\langle\text{vac}^{\otimes L}|$ is its vacuum eigenbra (for W_a: $G'=\Lambda+cE_{41}$, $c\propto a(z_1-z_4)$, as in Sec. 15.6). With $v=e_1g$, $u=e_4g^{-T}$ one has $uf(\Lambda)v^T=[f(G')]_{41}$ (the column $[G'^n]_{41}$ vanishes exactly on the block-admissible rows), so

$$\text{admissible}\iff\ \text{(i) }[G'^n]_{41}=0\ \ (n=1,2,3)\quad\text{or}\quad\text{(ii) }G'\text{ is (Levi/torus-conjugate to) }\Lambda+cE_{41},\tag{136}$$

(i) being $e_4f(G')e_1^T=0$ for all polynomials f (for lower-triangular G': $G'_{41}=0$, $G'_{42}G'_{21}+G'_{43}G'_{31}=0$, …), (ii) exactly one non-zero off-diagonal entry, at $(4,1)$. Rows tested: R5 ($v=e_1$, $u=(0,2,3,5)$; G' off-diagonal at $(4,2),(4,3)$ only) and R2 ($v=(1,2,3,0)$, $u=(0,3,-2,5)$; G' at $(2,1),(3,1),(4,2),(4,3)$ with $G'_{42}G'_{21}+G'_{43}G'_{31}=0$) satisfy (i); W_a and torus$\times W_a$ satisfy (ii); a broken R2 with $(G'^2)_{41}=-4845/784\neq 0$, a generic Borel element and $W_a\times$R2 fail both.

The operators are class dependent [exact]. The five-word K_κ with the W_a coefficients acts as the rotation only on $\mathcal{A}_W$; it fails on $\langle 0|_{\rm comp}$, on R5, R6, R2 and at generic z. This refines the earlier statement "K_κ fails on $\langle 0|_{\rm comp}$": the *five-word* K_κ is specific to the W class, while block-BPS

references do admit *other* class-dependent words (next subsections) but have vanishing cross pairings (Sec. 17.4). Symbolic solutions of $T_\kappa \in$ span: for the A-frozen class $(0,1,-1,0)\otimes(u_1,u_2,u_2,u_4)$ the coefficients $(c_1,\dots,c_5)$ of $(1,X_1,X_2,X_3,X_3X_1)$ are $c_3=8\kappa^2$ (fixed), $c_1=1-\kappa e_1'+\kappa^2 e_2'-\frac{3}{8}c_4\,y z_1 z_4$, $c_2=2\kappa-\frac{3}{8}c_5\,y z_1 z_4$, with c_4,c_5 free and the result u-independent; for the class ($v=e_1$, $u=(0,u_2,u_3,u_4)$) there are three free coefficients; for $(1,1,0,0)\otimes(0,0,u_3,u_4)$ one free coefficient, the others *rational* in κ with a pole at $\kappa=1/y$ (unlike the polynomial K_κ).

17.2 At $z_2\neq z_3$: no family

[exact] [`reference-admissibility-Kkappa.wb`; python `genericz.py`, `obstruction.py`, `twloci_generic.py`]. At generic z enumerating all zero patterns with generic values, the admissible patterns are exactly those with *disjoint supports of v and u* (50 of 117); in the small-integer scan all 1890 blockwise candidates ($v_j u_j=0\ \forall j$) are admissible and no non-blockwise reference is. W_a and all W-type references fail.

No family on the twisted vacuum $\langle 0_w|$ at $z_2\neq z_3$. The twisted vacuum $\langle 0_w|$ (the vacuum eigenbra of $t^{G_c(w)}$, Sec. 18), tested consistently with words and reference at the same generic $z=(3/2,5/4,2/3,4/7)$, $w=(7/5,9/8,3/7,5/9)$: the five-word span has rank 5, the 13 κ-components D, B_{ab} have rank 13, and $\dim(\text{span}\cap\text{span}\{\kappa\text{-components}\})=2$, so *11 of the 13 needed directions are missing*; the charged support is $|S|=12$ (a 12×5 evaluation matrix of rank 5) and $\text{rank}(\text{span}+T_\kappa)=6$ at all three κ. A family exists iff $w_1\in\{z_1,\dots,z_4\}$ (then $v=e_a$ is frozen and $\pi_j=v_ju_j=0$ for all j); generic w, and $w_2,w_3\in\{z\}$, do not. A sixth localised operator cannot help: the missing space has dimension 11 and the localised algebra on pair 1 at the ABJM point is 5-dimensional (words of length ≤4, Sec. 16.1). Near $z_2=z_3$ ($z_2-z_3=\epsilon\Delta\varphi$, $z=1+\epsilon\varphi$, $w=1+\epsilon\omega$, $\kappa=1-c\epsilon$, $\varphi=(\frac{9}{10},\frac{7}{20},\frac{7}{20}+\Delta\varphi,-\frac{11}{10})$, $\omega=(\frac{7}{10},-\frac{3}{10},\frac{1}{5},\frac{1}{2})$, $c=\frac{4}{5}$) the relative best-fit residual of T_κ in the span is, at $\epsilon=10^{-4}$ (exact least squares printed as floats, equal to the 80-digit Wolfram result where run):

$\Delta\varphi$	twisted vac., all comps	twisted vac., charged comps	W_a, all	W_a, charged
0.01	0.0574	0.0271	0.0115	0
0.1	0.343	0.161	0.114	0
0.5	0.451	0.307	0.494	0

(for $\Delta\varphi\le0.1$ the values at $\epsilon=10^{-2},10^{-3}$ agree within 1%). Both residuals are *linear in $\Delta\varphi$ at small $\Delta\varphi$ and independent of ϵ*: there is no healing in the untwisting scaling. For W_a the leftover is purely in the zero-weight sector (E_{22},E_{33} directions, harmless for the physical overlaps as observed in Sec. 16.3); for the twisted vacuum it contains *charged* components (relative ~0.16 at $\Delta\varphi=0.1$), and there is no reason for these to be suppressed (the physical test of the charged leftover was not done). Quantitatively the block-BPS violation is $v_1u_1/(v_2u_2)$, $v_4u_4/(v_2u_2)=O(\Delta\varphi)$ because $\hat v_2=-\hat v_3\sim1/\Delta\varphi$. Hence changing the reference to $\langle 0_w|$ (the 2603 route) does not reproduce the 2603 structure at the ABJM point: at $z_2=z_3$ it collapses to an A-frozen reference, at $z_2\neq z_3$ the five words cannot rotate it. This route is superseded by the overlap identity of Sec. 18, where $\langle 0_w|$ enters only through the overlaps $\langle 0_w|\Psi\rangle=(v_1u_4)^L\langle W_a^{\otimes L}|\Psi\rangle$.

17.3 The A-frozen twisted-vacuum limit

[exact] [`reference-admissibility-Kkappa.wb`; chain $L=2,3$ mod p, python `r6_l3.py`]. At $z_2=z_3$ the diagonal-frame image of $\langle 0_w|$ degenerates (it exists only for $z_2\neq z_3$; at $z_2=z_3$ it is the rescaled limit $v=(c/\delta)(0,-1,1,0)+O(1)$, whose $O(1)$ part is lost): $v\to(0,1,-1,0)$, a *frozen* A site on the $\mathfrak{gl}(2)$-odd direction, projectively independent of w, with $u=u_w$ generic. On this limit reference a common pair rotation exists, but it is an F-site rotation only: the A site is an eigenvector of Λ, so "rotation" means $u\to u(1-\kappa\Lambda)^{-1}$, $v\to v\cdot\text{const}$. Choosing $c_4=c_5=0$ above,

with $e_1' = z_1 + y + z_4$, $e_2' = yz_1 + yz_4 + z_1z_4$,

$$K_\kappa^F = \left[1 - \kappa e_1' + \kappa^2 e_2'\right] + 2\kappa X_1 + 8\kappa^2 X_2 \qquad (h = \tfrac{1}{2};\ \text{general } h:\ \tfrac{\kappa}{2h^2}X_1 + \tfrac{\kappa^2}{2h^4}X_2) \tag{137}$$

at pair level (chain level: $c_1 + (2\kappa/\lambda_1)X_1 + (8\kappa^2/\lambda_2)X_2$; X_3, X_3X_1 are not needed). It is the one-site analogue of 2603's word $1 - \kappa t_+ + \kappa^2 t_0$. On an F index b it acts as $\prod_{j\in\{1,y,4\}}(1-\kappa z_j)/(1-\kappa z_b) = 1 - \kappa(e_1' - z_b) + \kappa^2(e_2' - z_b(e_1' - z_b))$; its eigenvalue on a state is $P_\kappa^F(\tau) = 1 - \kappa(e_1' - \xi_1) + \kappa^2(e_2' - \xi_2)$ with $\xi_1 = \tau_{X_1}/2h^2$, $\xi_2 = -\tau_{X_2}/2h^4$, a Q-form via x, τ_F of Sec. 16.7 ($c_3 = c_4 = 0$ in $P_K = c_0 + c_1\tau_F + c_2x\tau_F + c_3y + c_4y\tau_F$ of `even-bridge-sov-formula.wb`). Verified on the chain at $L = 2, 3 \pmod p$, on the twisted-vacuum limit, the companion-vacuum limit and a generic (u_1, u_2, u_2, u_4) for $\kappa \in \{\frac{3}{10}, -\frac{7}{5}, \frac{7}{4}, \frac{2}{9}\}$. The twisted vacuum is also admissible at its loci $w_1 = z_1$, $w_1 = z_4$, $w_2 = w_3$, $w_4 = z_4$ and combinations (rank of the span $1, 2$ or 3); every "yes" among the twisted-vacuum rows has a frozen A site.

17.4 Cross pairings: why W_a is the only useful reference

[exact] [`reference-admissibility-Kkappa.wb`; two-block Borel invariants of Sec. 12]. The cross-block pairings that make a two-block polarisation sensitive to the $s > 0$ states are, for the blocks (v, u) and (vD, uD^{-1}) with $D = \mathrm{diag}(c - \varphi_j)$ and $\pi_C = v|_C \cdot u|_C$,

$$p_{23} = \sum_C \frac{\pi_C}{c - \varphi_C}, \qquad p_{32} = \sum_C \pi_C\,(c - \varphi_C). \tag{138}$$

Every admissible reference except the W-type has all $\pi_C = 0$, hence $p_{23} \equiv p_{32} \equiv 0$; W_a is admissible exactly because $\pi_1 = -\pi_4 \neq 0$ is allowed at $z_2 = z_3$. So "cross pairing $\neq 0$" and "K-admissible" are compatible only for the W-type family. For the A-frozen polarisation $a_2 = a_3 = 0$, $m_v = 0$, $p_{23} = p_{32} = 0$ and only d_2, d_3, m_u survive; with the structure of Sec. 12 (F polynomial in $a_2, a_3, d_2, d_3, m_v, m_u, p_{23}, p_{32}$ with $e_8 + e_9 = s$) it can only see the $e_8 = e_9 = 0$ monomials: *it is not the generic two-block polarisation* the 2603-type construction was meant to produce (inference from the polarisation forms, not recomputed). For the non-frozen admissible family (C): $a_2 = 1$, $d_2 = u_4$, $m_v = p_2(\varphi_1 - \varphi_2)$, $m_u \neq 0$, but again $p_{23} = p_{32} = 0$. If an A-frozen polarisation is acceptable, the usable word is K_κ^F of (137). Open: the Q-form of $\langle 0_w|\Psi\rangle$ is not connected to any admissible K; a $z_2 \neq z_3$ generalisation of K_κ needs at least 11 additional localised operators (none known).

17.5 Lie algebra of pair-1 rotations inside the span

[exact] [`covariance.py` §9]. $\mathfrak{h}_r = \{X \in \mathfrak{gl}_4 :\ r\rho(X) \in \text{span of the five words}\}$; the stabiliser has dimension 10 and the orbit tangent dimension is 6 for every reference below.

reference	rank of span	$\dim \mathfrak{h}_r/\mathrm{stab}$	representatives mod stabiliser
W_a $(a = \frac{1}{3})$	5	2	E_{11}, E_{14} (torus and the a-shift)
R2: $v = (1, 2, 3, 0)$, $u = (0, 3, -2, 5)$	4	3	E_{11}, $E_{12} - \frac{3}{5}E_{14} + \frac{1}{2}E_{33} + \frac{1}{5}E_{34}$, $E_{22} + E_{33}$
R5: $v = e_1$, $u = (0, 2, 3, 5)$	2	2	E_{11}, $E_{22} + \frac{3}{5}E_{34}$
twisted-vac. limit $(0, 1, -1, 0) \otimes u_w$	3	3	E_{11}, $E_{21} + E_{31}$ (F-site mixing $u_1 \to e_2 + e_3$), $E_{22} + E_{33}$
R6: $(0, 2, 3, 0) \otimes (1, 3, -2, 4)$	3	3	E_{11}, $E_{21} - \frac{2}{3}E_{31}$, $E_{22} + E_{33}$
random BPS (not admissible)	5	1	one mixed element

For *every* admissible reference the whole torus orbit $\{[vf(\Lambda)] \otimes [uf(\Lambda)^{-1}]\}$ for $f = \Lambda, \Lambda^{-1}, 1 - \kappa\Lambda^{-1}, \Lambda^2$ and generic diagonal ($f_2 = f_3$) lies in the span; for the non-admissible random BPS only $f = \Lambda^{-1}$ does (the single word $X_3X_1 \sim \rho(\Lambda^{-1})$). On W_a the 5-dimensional span is "r, ∂_t(torus), ∂_a (E_{14}) plus two non-group directions"; on the A-frozen class it is the full torus-orbit span plus the mixing $E_{21} + E_{31}$.

17.6 Untwisting of a rotated pair: $D = \mathrm{diag}(c - \varphi_j)$

[exact] [python `untw.py`; exact]. With $z = 1 + \epsilon\varphi$, $w = 1 + \epsilon\omega$, $\kappa = 1 - c\epsilon$ one has $1 - \kappa z_j = \epsilon(c - \varphi_j) + O(\epsilon^2)$, so the rotation in the limit is $D = \mathrm{diag}(c - \varphi_j)$, *not* $c + \varphi_j$ (the latter corresponds to $\kappa = 1 + c\epsilon$ or $\varphi \to -\varphi$). For $\varphi = (\frac{9}{10}, \frac{7}{20}, \frac{7}{10}, -\frac{11}{10})$ ($\varphi_2 \neq \varphi_3$), $\omega = (\frac{7}{10}, -\frac{3}{10}, \frac{1}{5}, \frac{1}{2})$, $c = \frac{4}{5}$ the deviation of the limit of T_κ from $(\hat{v}D) \otimes (\hat{u}D^{-1})$, with $\hat{v}_j = \prod_{m \neq j}(\omega_1 - \varphi_m)/(\varphi_j - \varphi_m)$ and $\hat{u}_j = \prod_{m \leq 3}(\varphi_j - \omega_m)$, is

ϵ	10^{-2}	10^{-3}	10^{-4}	10^{-5}
$D = c - \varphi$	$3.0 \cdot 10^{-2}$	$3.0 \cdot 10^{-3}$	$2.9 \cdot 10^{-4}$	$2.9 \cdot 10^{-5}$ ($\propto \epsilon$)
$D = c + \varphi$	1.0	0.98	0.98	0.98 (wrong)

At $\varphi_2 = \varphi_3$ ($= \frac{7}{20}$) with the limit reference $(0, 1, -1, 0) \otimes u_w$ the same formula collapses to $(0, 1, -1, 0) \otimes (\hat{u}_j/(c - \varphi_j))$ (deviation $3.0 \cdot 10^{-2}, \ldots, 3.0 \cdot 10^{-5}$): the A site is frozen and $p_{23} = p_{32} = m_v = a_2 = a_3 = 0$. The normalised $\hat{v}$ as $\varphi_3 \to \varphi_2$ ($d = 0.1, 0.01, 0.001$) tends to $(0, -1, 1, 0)/\sqrt{2}$.

18 The even bridge in Q-functions

[`even-bridge-Qform.wb` (consolidated), `Wa-sov-expansion-L4.wb` cells 30–48, `Wa-Fonly-L6.wb`, `even-bridge-epsilon-expansion.wb`, `abjm-sov-norm-L4.wb`; notes `notes/w-a-reference-status.md`, `notes/agent-2026-10-13-*.md`.]
Throughout, Ψ is an eigenstate of the diagonal-twist transfer matrices $t_a^\Lambda(u)$ (§2.3), and companion-frame objects are transported with the R of the twist z of Ψ. Notation:

$$\lambda_j = \prod_{m \neq j} \frac{1}{z_j - z_m}, \qquad f(x) = (x - z_2)(x - z_3), \qquad u_4 = f(z_4), \tag{139}$$

$$\kappa_0 = \frac{\lambda_4}{\lambda_1} = -\frac{f(z_1)}{f(z_4)} = -\frac{(z_1 - z_2)(z_1 - z_3)}{(z_4 - z_2)(z_4 - z_3)}. \tag{140}$$

A state of weight $(\mu_1, 0, 0, -\mu_1)$ has $n := L - \mu_1$; for every other weight $\langle W_a^L|\Psi\rangle = 0$.

18.1 The problem

The pair κ-family (§16) gives, exactly at finite twist with $z_2 = z_3$,

$$\langle W_{a'} \otimes W_a^{L-1}|\Psi\rangle = \frac{P_\kappa(\tau_\Psi)}{(1 - \kappa z_1)^2}\,\langle W_a^L|\Psi\rangle, \qquad a' = a\,\frac{1 - \kappa z_4}{1 - \kappa z_1}, \tag{141}$$

with P_κ a polynomial in the one-pair eigenvalues $\tau_1(\theta_F)$, $\tau_2(u_0)$, $\hat{\tau}_3(u_0)$, all of which are known Q-expressions (§16). The missing piece is $\langle W_a^L|\Psi\rangle$.

$\langle W_a^L| = \langle \mathrm{vac}^L|\rho(e^{aE_{14}})$ is itself not a twisted vacuum. It is the vacuum eigenbra of the triangular twist $G_a = \Lambda + a(z_1 - z_4)E_{41}$, and of $\Lambda(w) + a(w_1 - w_4)E_{41}$ for every w, but never of a companion twist [`twisted-vacuum-companion-check.wb`, [exact] at $L = 2$]. A two-line argument rules out any twist sharing the companion SoV basis. The eigenvalue of $\langle W_a^L|$ is τ_{vac}, so its Q-functions are those of $\langle \mathrm{vac}^L|$. A factorised dual wave function would therefore coincide with that of $\langle \mathrm{vac}^L|$, and completeness would force $\langle W_a^L| \propto \langle \mathrm{vac}^L|$. Consistently, its SoV expansion is dense: all 256 words occur, in every power of a (§13). The resolution is that only the overlaps on the weights $(\mu, 0, 0, -\mu)$ matter, and there $\langle W_a^L|$ is indistinguishable from a twisted vacuum.

18.2 Overlaps of W_a are twisted-vacuum overlaps

The companion twisted vacuum $\langle 0_w| = (v(w_1) \otimes u_w)^{\otimes L}$ (§6) has the diagonal-frame image, per pair, [exact]

$$v_j = \lambda_j \prod_{m \neq j}(w_1 - z_m), \qquad u_j = \frac{\prod_{m \leq 3}(z_j - w_m)}{\prod_{m \leq 3}(w_4 - w_m)}, \qquad v \cdot u = 0. \tag{142}$$

Proposition 1. Let $w = (w_1, z_2, z_3, w_4)$. For every Ψ of weight $(\mu, 0, 0, -\mu)$,

$$\langle 0_w|\Psi\rangle = (v_1 u_4)^L \, \langle W_a^L|\Psi\rangle, \qquad a(w_1) = \frac{v_4}{v_1} = -\frac{u_1}{u_4} = \kappa_0 \, \frac{w_1 - z_1}{w_1 - z_4}. \tag{143}$$

Proof. With $w_2 = z_2$ and $w_3 = z_3$, $u_2 = u_3 = 0$, so every F site carries only e_1 and e_4. An A site in e_2 or e_3 then gives the bra a positive 2- or 3-weight that no F site cancels. So on the weights $(\mu, 0, 0, -\mu)$ only v_1 and v_4 contribute, each pair acts as $(v_1 e_1 + v_4 e_4) \otimes (u_1 e_1 + u_4 e_4) = v_1 u_4 \, W_{v_4/v_1}$, and $v \cdot u = v_1 u_1 + v_4 u_4 = 0$ gives $v_4/v_1 = -u_1/u_4$. □
Evidence: difference 0 to 40 digits on all eigenstates of the five sectors $(2 - n, 0, 0, n - 2)$, $n = 0..4$, at $L = 2$ (1, 12, 28, 12, 1 states). Both overlaps vanish in $(1, 1, -1, -1)$. In $(0, 1, 0, -1)$ only $\langle 0_w|\Psi\rangle \neq 0$, as the proof predicts. The Möbius map is symbolic [exact]: $a(z_1) = 0$ (vacuum), $a(z_4) = \infty$ (lowest-weight vacuum $e_4 \otimes e_1$), $a(\infty) = \kappa_0$. The parameter w_4 only normalises.

Q-sum. Since $\langle 0_w|$ shares the companion SoV basis with Ψ, its dual wave functions factorise, with the vacuum dual Q-functions of twist w, and

$$\mathbb{W}_w(\Psi) := \frac{\langle 0_w|\Psi\rangle}{\langle 0_w|\bar{0}\rangle\langle 0|\Psi\rangle} = \sum_{x,y} \tilde{\tau}_y(\mathrm{vac}_w) \, M_{yx} \, \tau_x(\Psi) \tag{144}$$

holds on all 256 states to 10^{-44} [hp] ($L = 2$, generic θ, z; M = inverse Gram matrix, §13). With (143) this expresses $\langle W_a^L|\Psi\rangle$ in Q-functions, checked to 10^{-46} on the 54 sector states for three values of w_1. At $w_2 = z_2$, $w_3 = z_3$ the $\tilde{\tau}_y(\mathrm{vac}_w)$ do not depend on w_4 and are polynomials in w_1. $\mathbb{W}_w$ is a polynomial of degree 8 in w_1, and on the sectors [hp]

$$\mathbb{W}_w(\Psi) = \big[\lambda_1 u_4 (w_1 - z_2)(w_1 - z_3)(w_1 - z_4)^2\big]^2 \, a(w_1)^n \, \Gamma_\Psi, \qquad \Gamma_\Psi = \frac{\langle W_1^L|\Psi\rangle}{\langle 0|\Psi\rangle} \qquad (L = 2). \tag{145}$$

All w_1-dependence is therefore explicit, and one number Γ_Ψ per state remains. It is computed most simply at $w_1 = \infty$.

18.3 The $w_1 \to \infty$ form: an F-site rotation of the SoV vacuum

As $w_1 \to \infty$, $v(w_1) \to w_1^3 e_1$ and $u_w \to \tilde{u}/((w_4 - z_2)(w_4 - z_3))$, with

$$\tilde{u} = e_4^T f(G_c) = (0,\ 1,\ -(z_2 + z_3),\ z_2 z_3) \qquad \text{(companion frame)}. \tag{146}$$

So the twisted vacuum becomes $(e_1 \otimes \tilde{u})^{\otimes L}$: the A sites are exactly those of the SoV vacuum $\langle 0|$, and every F site is rotated by the degree-2 twist polynomial $f(G_c)$. In the diagonal frame $f(\Lambda)$ kills e_2 and e_3, which is Proposition 1 in one line: the image is $(\lambda \otimes (f(z_1), 0, 0, f(z_4)))^{\otimes L}$. Hence, on the sectors,

$$\langle W_a^L|\Psi\rangle = \Big(\frac{a}{\kappa_0}\Big)^n (\lambda_1 u_4)^{-L} \, \big\langle (e_1 \otimes \tilde{u})^{\otimes L} \big| \Psi_c \big\rangle. \tag{147}$$

The top power w_1^8 of $\mathbb{W}_w$ is supported on exactly the 9 F-only words and equals the SoV expansion of $e_1 \otimes \tilde{u} \otimes e_1 \otimes \tilde{u}$, with ratio exactly 1 [exact].

18.4 The factorised closed form

F-only expansion. $(e_1 \otimes \tilde{u})^{\otimes L}$ lies in the span of the 3^L F-only SoV words $\langle 0| \prod_s t_1(\theta_s)^{k_s}$, $k_s \leq 2$, s running over the F sites [exact]. This holds at $L = 2$ (9 words) and at $L = 3$ (27 words, all 4096 components). No A-site words and no $k_s = 3$ are needed. The system is graded by the level $\sum_s k_s$ (blocks $1, 3, 6, 7, 6, 3, 1$ at $L = 3$).

Cherednik operators. On bras whose F sites have no e_1 component the A–F exchanges are scalars, and $\langle 0|\, t_1(\theta_s) = \kappa_s \langle 0|\hat{Y}_s$ with

$$\kappa_s = h \prod_{r<s} (\theta_r - \theta_s) \prod_{r>s} (\theta_s - \theta_r) \qquad \text{(product over all sites } r \neq s\text{)}, \tag{148}$$

$$\hat{Y}_s = R_{s,s+1} \cdots R_{s,L}\, N_s\, R_{s,1} \cdots R_{s,s-1}. \tag{149}$$

Here the indices of $\hat{Y}$ run over F sites, $R_{sf} = 1 - \frac{h}{\theta_s - \theta_f} P_{sf}$, and N is the twist raising $e_4 \to e_3 \to e_2$ (levels $0, 1, 2$). The $\hat{Y}_s$ commute. Checks [exact]: κ_s read off the exact chain words at $L = 3$ equals the formula; at $L = 2$, $\kappa_2 = h\theta_{12}\theta_{23}\theta_{24}$ and $\kappa_4 = h\theta_{14}\theta_{24}\theta_{34}$. The model words agree with the chain words on all 27×27 components.

Proposition 2 (closed form). With $\nu_{st} = \theta_s - \theta_t$ over F sites,

$$\big\langle (e_1 \otimes \tilde{u})^{\otimes L} \big| = \langle 0|\, B_{z_2}(\hat{Y})\, B_{z_3}(\hat{Y}), \tag{150}$$

$$B_z(\hat{Y}) = \sum_{S \subseteq \{\text{F sites}\}} (-z)^{L-|S|}\, g_S \prod_{s \in S} \hat{Y}_s, \qquad g_S = \prod_{s<t \in S} \frac{\nu_{st}^2}{\nu_{st}^2 - h^2}.$$

Evidence: exact on the full chain at $L = 2$ and $L = 3$, at three points (generic; $h = 2/3$ with other θ, z; the ABJM ϵ-chain at $\epsilon = 1/100$) [exact]. Exact in the F-chain model at $L = 4$, at one point [python]. Not proved for general L. B_z is the root operator $\prod_s (G - z)$ on the F sites written in the $\hat{Y}$'s, as a cluster expansion with pair weights only. For $h \to 0$ it is $\prod_s (\hat{Y}_s - z)$. At $L = 2$, $B_z = z^2 - z(\hat{Y}_2 + \hat{Y}_4) + g\,\hat{Y}_2\hat{Y}_4$ with $g = \nu^2/(\nu^2 - h^2)$ and $\nu = \theta_2 - \theta_4$. This is agent B's 3×3 matrix $\tilde{u}P\tilde{u}$, which collapses to $\prod_{z \in \{z_2, z_3\}} [(\hat{Y}_2 - z)(\hat{Y}_4 - z) + (g-1)\hat{Y}_2\hat{Y}_4]$.

Corollary (Q-form). On eigenstates $\langle 0|\hat{Y}^k|\Psi\rangle = \langle 0|\Psi\rangle \prod_s y_s^{k_s}$, so for Ψ of weight $(\mu, 0, 0, -\mu)$

$$\boxed{\frac{\langle W_a^L|\Psi\rangle}{\langle 0|\Psi\rangle} = \Big(\frac{a}{\kappa_0}\Big)^n (\lambda_1 u_4)^{-L}\, B_{z_2}(y)\, B_{z_3}(y), \qquad y_s = \frac{\tau_1(\theta_s)}{\kappa_s} = \frac{1}{\kappa_s} \frac{Q_j(\theta_s)}{Q_j(\theta_s - h)}}} \tag{151}$$

Here j is any index, in the Q-conventions of §13. The overlap is a polynomial of degree ≤ 2 in each F-site single-Q ratio, and it factorises over the two middle twist eigenvalues. Evidence [hp]: $L = 2$, all 108 sector states, generic and ABJM pair-by-pair θ, $< 10^{-41}$; $L = 3$, nine states across the sectors $n = 0..3$ (dimensions 1, 24, 141, 256), $< 2.5 \cdot 10^{-51}$. The coefficients involve only θ-differences with $\nu \neq 0, \pm h$, so the formula is regular at the pair-by-pair ABJM point. No regularised SoV basis is needed: only F-only words enter.

18.5 Assembly and comparison with the data

Combining (141) and (151):

$$C_{M=1} = \lim_{\epsilon \to 0} \frac{P_\kappa(\tau)}{(1 - \kappa z_1)^2} \Big(\frac{a}{\kappa_0}\Big)^n (\lambda_1 u_4)^{-L}\, B_{z_2}(y) B_{z_3}(y)\, \frac{\langle 0|\Psi_\epsilon\rangle}{\|\Psi_\epsilon\|}, \tag{152}$$

on the ABJM ϵ-chain:

- $z = 1 + \epsilon\varphi$ with $\varphi = (9/10, 7/20, 3/5, -11/10)$, so $z_2 - z_3 = O(\epsilon)$ suffices, and $y = (z_2 + z_3)/2$ in P_κ;
- pair-by-pair θ with $\theta_{F,n} = \epsilon\theta_n^0$;
- $\kappa = 1 - c\epsilon$ with $c = 3/10$, and $a = (\varphi_1 - c)/(\varphi_1 - \varphi_4)$, so $a' - a \to -1$;
- the finite highest-weight shift $e^{-aE_{14}}$ maps the bra to $\langle W_{-1} \otimes \mathrm{vac}^{L-1}|$ (§16).

Unit-norm Ψ_ϵ and 70–110-digit eigenvectors are needed, because of the ϵ^{-3} cancellation in $\langle 0|\Psi\rangle$.

L	state	extrapolated $\lvert C_{M=1}\rvert$ from (152)	data
2	singlet $E=20$	0.15811388301	$1/\sqrt{40}$ (diff $2\cdot10^{-12}$)
2	singlet $E=4$	0.20412414524	$1/\sqrt{24}$ (diff $9\cdot10^{-12}$)
3	$[1,0,1]$ #17	0.17112641	0.17112641
3	$[1,0,1]$ #18	0.027137747	0.027137747
3	$[1,0,1]$ #19, #20	0.045900846	$\lvert-0.035872\mp0.028637i\rvert$
3	$[1,0,1]$ #21, #22	0.15348408	$\lvert0.129924\mp0.081713i\rvert$

The consolidated notebook `even-bridge-Qform.wb` reaches 10^{-15} for the two singlets with a degree-4 fit. At every ϵ the Q-form equals the direct value $P_\kappa\langle W_a^L|\Psi_\epsilon\rangle/(1-\kappa z_1)^2$: to 10^{-66} at $L=2$ and to 10^{-41}–10^{-44} at $L=3$ [hp]. The formula gives $|C|$; the sign is the phase of Ψ. With the data phase, the signs agree $(-1/\sqrt{40},+1/\sqrt{24})$. By $C_M=(-1)^{L+\mu_4}C_{L-M}$ the same covers $M=L-1$.

18.6 Structure of the untwisting limit

[`even-bridge-epsilon-expansion.wb`, ported [exact]/[hp] at $L=2$, 10^{-14} at $L=3$; ϵ-scans [python].]

Orders. $\langle W_a^L|\Psi_\epsilon\rangle=\frac{a^n}{n!}\epsilon^n\omega_n+\dots$. $P_\kappa=O(\epsilon^2)$ for $n\le1$ and $O(1)$ for $n\ge2$. The $M=1$ even bridge sees only $n=2$ states $[L-2,0,L-2]$ (the $L=2$ singlets, the $L=3$ $[1,0,1]$). The vacuum gives 1, and $n=1$ and $n=3$ give 0. Odd one-field bridges see only $n=1$ (§15), so no even/odd ratio relation at fixed state exists.

Factorisation.

$$C_{M=1}(\Psi_0)=K_1(\tau_{\Psi_0})\cdot\Omega(\Psi_0),\qquad \Omega=\lim_{\epsilon\to0}\frac{\langle\mathrm{vac}|\rho(E_{14})^2|\Psi_\epsilon\rangle}{2\epsilon^2(\varphi_1-\varphi_4)^2}. \tag{153}$$

$K_1=P_{\kappa=1}$ is evaluated at the untwisted ABJM point. Ω is independent of the bridge and of the twist direction: only $(\varphi_1-\varphi_4)^2$ enters. This is exact at $L=2$, reproducing $-1/\sqrt{40}$ and $1/\sqrt{24}$, and holds to 10^{-14} at $L=3$. An exact finite-twist handle follows from $t^{G_a}=t^\Lambda+a(z_1-z_4)T_{14}$:

$$(\tau_{\mathrm{vac}}(u)-\tau_\Psi(u))\langle W_a^L|\Psi\rangle=a(z_1-z_4)\langle W_a^L|T_{14}(u)|\Psi\rangle, \tag{154}$$

$$\Omega=\frac{\langle\mathrm{vac}|T_{14}(u)\,(t_1(u_1)-\tau_{\Psi_0}(u_1))^{-1}\,T_{14}(u_1)|\Psi_0\rangle}{\tau_{\mathrm{vac}}(u)-\tau_{\Psi_0}(u)}. \tag{155}$$

The second formula uses untwisted operators, holds for any u,u_1, and its intermediate states are the HW states of $[L-1,0,L-1]$. It is a second-order T_{14} form factor. Proposition 1 now gives Ω as the ϵ^2 coefficient of the Q-form (151).

The pair factor. Write $A=\tau_1(\theta_F)/(h^2\lambda_1)$, $B=\tau_2(u_0)/(2h^4y\lambda_2)$ and $C=\hat\tau_3(u_0)/(6h^5y^2\lambda_3)$. Then $P_\kappa=s^2+\kappa s(A+B+C)+\kappa^2AC/4$ with $s=1-\kappa(z_1+z_4)$, and

$$K_1=2-A-B-C,\qquad AC=4. \tag{156}$$

$AC=4$ holds on cyclic states because $\hat X_3X_1=24h^7\lambda_1\lambda_3\,S$, with S the one-pair translation [exact] at $L=2$; this is the shift-operator statement $U\propto S$ of §4. K_1 vanishes on $n\le1$ and is nonzero on $n=2$. At $L=2$ the characteristic polynomial of the K_1 operator on the zero-weight sector is $x^{24}(x-2)(x-8)^2(x-10)$ [exact]. On the singlets $(A,B,C)=(2,-12,2)$ and $(-2,4,-2)$, so $K_1=10$ and 2. At $L=3$ the six $[1,0,1]$ states give 3.65685, -7.65685, 14.658 (×2), 1.842 (×2). "$K_1=0$ on all non-vacuum states" is false.

Analytic limit at $L = 2$ [`even-bridge-Qform.wb` Part 3]. Let $\tau = \tau_1(\theta_F)$, $\tau' = \partial_u \tau_1(\theta_F)$ and $x = \tau_1(\theta_F + h)$, all at the untwisted ABJM point, with $h = \frac{1}{2}$.

- *Pair factor.* At $\theta_{F1} = \theta_{F2}$ the prefactor of t_3 has a double zero at u_0, so $t_3'(u_0) = 0$ and the regularised $\hat{t}_3(u_0) = \frac{1}{2} t_3''(u_0)$. On translation-invariant states

$$K_1 = 2 - 8\tau - \frac{1}{2\tau} + 64 x \tau \qquad \text{[exact] on the singlets.} \tag{157}$$

- *Overlap factor.* The normalised y_s diverge, $\nu y_{2,4} \to \mp \tau/(4h^3)$, but the combinations $y_2 + y_4 \to -(\tau + h\tau')/(4h^4)$ and $g\, y_2 y_4 \to \tau^2/(16h^8)$ are finite. So $B_{z_2}, B_{z_3} \to B_1$ and $B_{z_2} B_{z_3} \to B_1^2$, with

$$B_1 = 1 + \frac{\tau + h\tau'}{4h^4} + \frac{\tau^2}{16h^8} = 1 + 4\tau + 2\tau' + 16\tau^2. \tag{158}$$

  This holds on all 28 zero-weight states, with $O(\epsilon)$ convergence [hp]. τ' is genuine extra data: a parity shortcut without it works only on the singlets.

- *Result.* With $\mathbb{B}_0$ the limit of the SoV norm,

$$\boxed{|C_{M=1}| = \frac{K_1\, B_1^2}{\sqrt{\mathbb{B}_0}}} \tag{159}$$

singlet	τ	τ'	x	K_1	B_1	$\mathbb{B}_0$	$K_1 B_1^2/\sqrt{\mathbb{B}_0}$
$E = 20$	$\frac{1}{4}$	1	$\frac{3}{4}$	10	5	$2.5 \cdot 10^6$	$1/\sqrt{40}$
$E = 4$	$-\frac{1}{4}$	1	$\frac{1}{4}$	2	3	7776	$1/\sqrt{24}$

- *Higher L.* At $L = 3$ the factors are not separately finite: $\mathbb{W} \sim \epsilon^2$, $\langle 0|\Psi\rangle/\|\Psi\| \sim \epsilon^{-3}$ and the prefactor $\sim \epsilon$. Only the product is finite, so the limit at $L \geq 3$ needs the combined scaling.

18.7 The norm

The SoV norm $\mathbb{B}$, i.e. $\langle 0|\Psi\rangle/\|\Psi\|$, is known in Q-functions (P. Ryan) and is not rederived here; in the numerics it is taken from the eigenvectors. As a cross-check [`abjm-sov-norm-L4.wb`], it was assembled from eigenvalue data alone with the regularised bra and ket bases at the ABJM point. The result $\mathbb{B} = \sum \tilde{\tau}_y M_{yx} \tau_x$ equals the direct pairing to 10^{-29}–10^{-48} on all 28 zero-weight states, and gives $\mathbb{B} \to 2.5 \cdot 10^6$ and 7776 on the singlets, hence $1/\sqrt{40}$ and $1/\sqrt{24}$ to 10^{-16}. At finite twist the Euclidean norm differs from the SoV pairing by $H = 1 + O(\epsilon)$, because twisted eigenvectors are not orthogonal; the limit is unaffected (§13).

18.8 Comparison with 2603

2603 has $C = \mathcal{N}\, \mathbb{W}_\omega \mathbb{A}_{\ell,\kappa}/\sqrt{\mathbb{B}}$. Here:

- $\mathbb{A} \leftrightarrow P_\kappa$: a one-pair word in Q-data at one pair, with limit K_1;
- $\mathbb{W}_\omega \leftrightarrow B_{z_2} B_{z_3}$, with 2603's ω played by w_1; the $w_1 \to \infty$ member is the simplest;
- $\sqrt{\mathbb{B}} \leftrightarrow$ the SoV norm.

The differences from 2603:

1. The κ-family rotates one pair only. The ℓ-pair structure is pretwisted, so the middle bridges $2 \leq M \leq L - 2$ are not covered (§16).
2. $z_2 = z_3$ is needed at finite twist; only $z_2 - z_3 = O(\epsilon)$ is needed in the limit.
3. $\mathbb{W}$ is a cluster expansion with F–F exchange weights g_S, not yet a determinant. The mixed $\mathbf{4}/\bar{\mathbf{4}}$ measure that would give determinants is unknown (§13).
4. The reference W_a is not itself the twisted vacuum. It shares the twisted vacuum's overlaps on the relevant sectors, while being the only admissible reference for the κ-family (§17).

19 Open problems

Structure constants.

1. *Middle even bridges* $2 \le M \le L-2$, first $L=4$, $M=2$. These are NOT covered: the pair κ-family gives $M \in \{1, L-1, L\}$ at every L, hence all even bridges up to $L=3$ only. The ℓ-pair words are pretwisted (end-growing), and no common-κ block exists (§16). Old candidates: a second integrable family on the twisted pairs; right-end words combined with another reference; twist-dressed references.

   New route (untested idea, 13 Oct, after Proposition 1). The proof of Proposition 1 works pair by pair. So any companion bra whose F sites are $e_4^T f(G_c) q_p(G_c)$ on pair p, with A sites chosen to make each pair BPS, has the overlaps of a multi-block W-bra on the sectors $(\mu, 0, 0, -\mu)$. In particular, rotating the first M pairs of $(e_1 \otimes \tilde{u})^{\otimes L}$ by $g = 1 - \kappa G_c$ gives

$$\left\langle \left(e_1^T(1-\kappa G_c^T) \otimes \tilde{u}(1-\kappa G_c)^{-1}\right)^{\otimes M} \otimes (e_1 \otimes \tilde{u})^{\otimes(L-M)} \middle| \Psi_c \right\rangle \;\propto\; \langle W_{a'}^M \otimes W_{\kappa_0}^{L-M} | \Psi \rangle,$$

$$a' = \kappa_0 \frac{1-\kappa z_4}{1-\kappa z_1}.$$

   The diagonal images are $v_j = \lambda_j(1-\kappa z_j)$ and $u_j = f(z_j)/(1-\kappa z_j)$, so $u_2 = u_3 = 0$ and $v \cdot u = 0$. This needs neither the κ-family nor $z_2 = z_3$: on the bare SoV vacuum every site localises (§5). The open question is the closed form of its SoV coefficients, which involve A-site words on the first M pairs, F-site words everywhere, and exchange and singlet-channel terms; the analogue of Proposition 2 is needed. Test first at $L=2$, $M=1$ against the κ-family result, then at $L=4$, $M=2$.

2. *Proof of Proposition 2* (the B_z cluster expansion) for general L. This is an identity for the commuting operators $\hat{Y}_s$, of degenerate-affine-Hecke type. Also needed: the full-chain check at $L=4$, and a determinant or Pfaffian form of $B_z(y)$ and hence of $\mathbb{W}$.

3. *The untwisted limit at* $L \ge 3$. The factors are not separately finite there; find the combined scaling and the analogue of $|C| = K_1 B_1^2/\sqrt{\mathbb{B}_0}$. Also express $\tau_1'(\theta_F)$ through Q-functions.

4. $z_2 \ne z_3$ at finite twist: a sixth localised pair operator, or an argument that $z_2 - z_3 = O(\epsilon)$ is the natural scheme.

5. *The polarisation gap.* The note's W_{-1} polarisation is blind to all states with $L+p$ odd and to all $q > 0$ states (§12). The Q-form should be redone for polarisations with nonzero minors. This needs a reference other than W_a, and §17 shows that W_a is the only admissible one for the present κ-family.

6. Odd bridges with three fields; mechanism of the parity selection rules; the integrability of $\langle W_a^L|$ (the note's (4.8)).

Separation of variables.

1. The measure of the mixed $\mathbf{4}/\bar{\mathbf{4}}$ chain, which would give determinant forms of overlaps and norms.

2. The homogeneous (fully pair-periodic) ABJM limit of the SoV basis, which needs derivatives up to order 7.

3. The factor $\frac{1}{3}e^{\mp i\pi/3}$ in the $\langle 0|$-based limits of the A-family (§13): observed, not derived.

Housekeeping.

1. Rerun the notebooks still unrun or partly run (Appendix A, `notes/rerun-2026-10-13.md`).

2. Investigate the rerun discrepancy in the pair-word localisation table, where constants of 51/31 and 71/31 appear (§16).

A Notebook and data index

All paths are relative to `Paul/Mathematica/su4-alternating/`. The date is the day the notebook was created (DD-MM-2026); later reruns are noted in the content column. Status as of 13-10-2026, counted from the notebook files: run means native outputs are stored on all code cells (or all but one set-up cell); part means only partly rerun or ported, as stated; **unrun** means *no* stored outputs: results of those notebooks are in markdown cells pasted from python, and the notebook must be rerun in VS Code (`notes/rerun-2026-10-13.md` and the NEXT item 0 list the reruns done so far; the rerun was stopped after four notebooks). Notebooks built as JSON by agents carry no native outputs unless they have been rerun.

A.1 Notebooks

file	date	status	content
Companion frame, qKZ form, localisation (11-10)			
`L2-ffbar-SoV-localisation.wb`	11-10-2026	**unrun**	$L = 2$ (A F; names opposite to Paul's): qKZ form, spectrum, $\mathfrak{gl}(4)$ B-operator, SoV basis $\langle a, k|$, one-site rotations; $L = 4$ preview. Spec `L2-ffbar-SoV-localisation-spec.md`. (§5)
`L4-ffbar-free-rotations.wb`	11-10-2026	**unrun**	$L = 4$: localisation inventory on $\langle 0|$ and on rotated references, one free κ per site, $\{1, 3\}$ common rotation, torus, fused-T scan, reach. Its four `Export`s write to `$TemporaryDirectory`. Spec `L4-ffbar-free-rotations-spec.md`. (§5)
Transfer matrices and the ABJM chain (12-10)			
`dual-lax-t3.wb`	12-10-2026	run	One-pass t_3 (dual Lax) and t_2 ($\wedge^2$ Lax) from the quantum minors, exact checks at 2 and 4 sites, vector form, $L = 6$ commutation test. (§3)
`abjm-chain.wb`	12-10-2026	run	H_{MZ} and sanity checks; $[H_{\mathrm{MZ}}, \bar{t}_a(u)] = 0$ iff $\theta_A - \theta_F = 2h$ (symbolic u and $\theta_A - \theta_F$); $H_{\mathrm{MZ}} = U(0)^{-1}U'(0) + \mathrm{const}$; $L = 3$ check. (§4)
`abjm-bridges.wb`	12-10-2026	run	Port of `checks/abjm_bridges*.py`: bridge overlaps; §1–2 at $\theta = 0$ superseded (singular point); shift test $U \propto S$; C_ℓ table and odd-bridge untwisting test at the ABJM point; `Lp` sets the number of pairs. (§14, §4)
States, Q-functions, symmetries, configurations (12-10)			
`abjm-LR-symmetry.wb`	12-10-2026	run	$\mathcal{P}$ (LR) and $\mathcal{Q}$ (parity); all highest-weight cyclic eigenstates at $L = 2, 3, 4$ $(6, 26, 152)$; C_ℓ for all bridges and vanishing rules; writes `data/abjm-hw-states.wxf`. (§7, §8, §9)
`abjm-Qfunctions.wb`	12-10-2026	run	Baxter equation, Q_1, Q_{12}, Q_{123} of all 184 states, degrees = magnon numbers, LR in Q-language; writes `data/abjm-Qfunctions.wxf` and the τ files. (§7)
`abjm-other-symmetries.wb`	12-10-2026	run	P and LP symmetries in eigenvalue and Q-form on all saved states; factorised P/LP bras. (§8, §11)
`abjm-LR-factorized.wb`	12-10-2026	run	Factorised bras annihilated by $\bar{t}_1(u) - \bar{t}_3(u + h)$, $L = 2$, untwisted and twisted. (§11)
`abjm-3pt-configs.wb`	12-10-2026	run	Two-block BPS configurations (six $GL(4)$ classes, bridge length, A/F start, $L = 2, 3, 4$): no LR rule, length-independent parity rules. (§10)
SoV at four sites and at the ABJM point (12-10)			

continued on the next page

file	date	status	content
`sov-companion-L4.wb`	12-10-2026	run	SoV word basis at 4 sites (rank 256), B-operator and vacuum checks, BPS expansions, localisation on sites $1,2$ and $1,2,3$, ABJM-point degeneracy. (§13)
`sov-Q-dictionary-L4.wb`	12-10-2026	run	Q-wave-function dictionary at 4 sites (F: single-Q ratios; A: 3×3 Casoratians), dual ket basis, Gram matrix and measure, $1/\mathcal{N}=\sum\tilde{\tau}M\tau$. Rerun and saved 13-10 (not logged in the rerun note). (§13)
`sov-onepair-determinant.wb`	12-10-2026	**unrun**	One pair exact: graded word basis, determinant formula $\langle v\otimes u\|\Psi\rangle=\langle 0\|\Psi\rangle\det[\dots]/Q_{123}$, two pairs never factorise, F–F exchange. (§13)
`sov-twoblock-expansion-L4.wb`	12-10-2026	**unrun**	Two-block bras in the SoV basis (operator-Schmidt rank 169 of M^{-1} across $\{1,2\}\|\{3,4\}$, 36-word sums), $\langle 0\|_{\text{comp}}$ in the Borel orbit of $\langle \text{vac}^L\|$, bordered Casoratian for the $M=1$ bridge. (§13)
`sov-abjm-point-limit.wb`	12-10-2026	**unrun**	ABJM-point limit of the SoV basis: rank collapse, $t_3(u_0)=0$, $t_2(u_0)=t_1(u_0)t_1(\theta_F)$, Jordan blocks of B, regularised basis and Q-forms of x,w,y. Was mid-run on 13-10 (cells 27 onwards cancelled; earlier cells may hold unsaved outputs). (§4.4)
Twisted vacua as references (12-10)			
`twisted-vacuum-localisation-L2.wb`	12-10-2026	**unrun**	$L=2$: $\langle 0_w\|$, obstruction $q(w,z)$, exact t_1 actions, inverses localise. (§6)
`twisted-vacuum-localisation-L4.wb`	12-10-2026	run	$L=4$: generic w (only end sites and end-growing chains localise), locus $w_1\in\{z\}$ (A frozen, F free), common F rotations. (§6)
`twisted-vacuum-blocks-L3.wb`	12-10-2026	run	$L=3$ pairs: block structure, common F rotations on the first ℓ pairs with closed-form coefficients. (§6)
`twisted-vacuum-wordspan-L2.wb`	12-10-2026	run	Word span on pair 1 at generic w (dimension 37; dimension 4 on sites $1,2$). (§6)
`twisted-vacuum-untwisting.wb`	12-10-2026	run	Diagonal-frame map, untwisting limit and overlaps with the saved states; neither limit gives the note's W_{-1}. (§6)
`twisted-vacuum-companion-check.wb`	13-10-2026	**unrun**	$L=2$, exact: the t_1-twist space of $\langle W_a\|$ is 11-dimensional and contains no companion $G_c(w)$ in either frame for $a\neq 0$; frame map $R=V^{-T}\otimes V$ verified. (§2.4)
Odd bridges, the pair κ-family, even bridges (12-10)			
`abjm-nongood-reference-limit.wb`	12-10-2026	run	On the non-good $\langle W_a^L\|$: the two exact end-pair families, untwisting of the F-site family, control against the odd bridge of `abjm-bridges.wb`. (§15)
`pair-rotation-wordsearch.wb`	12-10-2026	run	Search for a common pair rotation (A by $1-\kappa\Lambda$, F by the inverse) in the $t_{1,2,3}$ word span at 2 pairs; nearly trivial there. (§16)
`odd-bridge-general-L.wb`	12-10-2026	run	Proved chain constants d_k,c_k, one-field bridges at every placement, $L=3$ untwisting tables, regular homogeneous limit. Rerun 13-10, all numpy numbers reproduced; uses `hw_L3.json`. (§15)
`pair-group-rotation.wb`	12-10-2026	part	The pair κ-family K_κ at the ABJM point with $z_2=z_3$: column-support lemma, pair-vacuum pattern, localisation scan, locality at $L=2,3$, untwisting. Rerun 13-10: sections 1–6 and 8 reproduced; $L=3$ rows of sections 7 and 9 skipped as too heavy; one discrepancy (cell 17, constants 51/31, 71/31). (§16, §4.2)

continued on the next page

file	date	status	content
`pair-rotation-locality-L3.wb`	12-10-2026	**unrun**	Brute-force locality at 3 pairs, generic θ, z: the even bridge is not locally reachable; obstruction by the isolated pair singlet. (§16)
`even-bridge-lpairs.wb`	12-10-2026	run	ℓ-pair generalisation of K_κ (pretwisted end-growing structure), all even bridges at $L = 3$, Q-forms of $x, \hat{y}$. Rerun 13-10 (the F1/F2 labels of its section 2 table are swapped relative to the text; limits unchanged; the $L = 3$ Q-form checks remain python). (§16)
`even-bridge-sov-formula.wb`	12-10-2026	**unrun**	K_κ on the SoV vacuum $\langle 0\|_{\text{comp}}$ (not a common rotation; singlets invisible), SoV-normalised formula $\|C_K\| = \|P_K(\tau_\Psi)/P_K(\tau_{\text{vac}})\| \cdots$, regularised ket basis, Q-forms. (§17, §18)
`twoblock-polarisation-forms.wb`	12-10-2026	**unrun**	Two-block overlaps of all 184 states as polynomials in 8 Borel invariants; exponents from $[p, q, r]$; which polarisations are blind; writes `data/abjm-twoblock-forms.json`. (§12)
Reference states and the Q-form of the even bridge (13-10)			
`reference-admissibility-Kkappa.wb`	13-10-2026	run	Which references admit a pair κ-family: admissible set at $z_2 = z_3$, none on $\langle 0_w\|$ at $z_2 \neq z_3$, K^F for the A-frozen class, $GL(4)$ covariance, Lie algebra of pair rotations. Notes `agent-2026-10-13-reference-map.md`. (§17, §2.5)
`Wa-sov-expansion-L4.wb`	13-10-2026	run	Companion image and dense SoV expansion of $\langle W_a^L\|$ (no determinant form); twisted-vacuum Q-sum $\mathbb{W}_w$; follow-up cells 30–48: overlap identity $\langle 0_w\|\Psi\rangle = (v_1 u_4)^L \langle W_a^L\|\Psi\rangle$ and the F-only form at $L = 2$, $L = 2$ even bridge in Q's. Notes `agent-2026-10-13-Wa-sov.md`. (§18)
`even-bridge-epsilon-expansion.wb`	13-10-2026	part	ε-orders, $C_{M=1} = K_1\Omega$, $K_1 = 2 - A - B - C$, T_{14} identity, step 0a. Ported to the notebook: step 0a, T_{14} identity, K_1 spectra at $L = 2$, $C_{M=1}$ exact at $L = 2$ and to 10^{-14} at $L = 3$, $\hat{X}_3 X_1 = \frac{9}{256} S$; the ε-scans, Richardson limits and Hellmann–Feynman remain python. Notes `agent-2026-10-13-epsilon-expansion.md`. (§2.5, §18)
`Wa-Fonly-L6.wb`	13-10-2026	run	$L = 3$: $\langle (e_1 \otimes \tilde{u})^{\otimes 3}\|$ as 27 F-only words (exact), closed form $B_{z_2}(y) B_{z_3}(y)$, Q-form on states, $L = 3$ even bridge equals the data. Notes `agent-2026-10-13-Fonly-L6.md`. (§18)
`abjm-sov-norm-L4.wb`	13-10-2026	run	The SoV norm in Q's at the $L = 2$ ABJM point (regularised bra and ket bases, $\mathbb{B} = \sum \tilde{\tau} M \tau$); the $L = 2$ even bridge fully Q-based, $1/\sqrt{40}$, $1/\sqrt{24}$ at $4 \cdot 10^{-16}$. Notes `agent-2026-10-13-sov-norm.md`. (§18)
`even-bridge-Qform.wb`	13-10-2026	run	Consolidated: the three finite-twist ingredients, the assembled $M = 1$ bridge at $L = 2, 3$ against the data, and the $L = 2$ analytic limit $\|C_{M=1}\| = K_1 B_1^2/\sqrt{\mathbb{B}_0}$. Notes `agent-2026-10-13-Qform.md`. (§18)

The ten notebooks with no stored outputs on 13-10-2026 are `sov-abjm-point-limit`, `even-bridge-sov-formula`, `sov-onepair-determinant`, `sov-twoblock-expansion-L4`, `twoblock-polarisation-forms`, `twisted-vacuum-localisation-L2`, `twisted-vacuum-companion-check`, `pair-rotation-locality-L3`, `L2-ffbar-SoV-localisation` and `L4-ffbar-free-rotations`. Of these, `sov-abjm-point-limit` was interrupted mid-run on 13-10; `even-bridge-sov-formula`, `sov-onepair-determinant`, `sov-twoblock-expansion-L4`, `twoblock-polarisation-forms`, `twisted-vacuum-localisation-L2` and `twisted-vacuum-companion-check` are the "not

started" list of NEXT item 0; `pair-rotation-locality-L3` was not reached by the rerun; the two notebooks of 11-10 were built as JSON and never rerun with native outputs. For the notebooks marked `run` the python-only parts are stated in the content column and in the sections that use them.

A.2 Data files

file	content
`data/abjm-hw-states.wxf`	All highest-weight cyclic joint eigenstates of $\bar{t}_1, \bar{t}_3$ at the ABJM point, $L = 2, 3, 4$ $(6, 26, 152)$, with H_{MZ} energies, LR data and C_ℓ; keys `L2`, `L3`, `L4`, and a `"description"` key. Written by `abjm-LR-symmetry.wb`. *Never import the $L = 4$ key* (about 3 GB decompressed).
`data/abjm-tau.wxf`	Eigenvalue polynomials $\bar{t}_1, \bar{t}_2, \bar{t}_3(u)$ of all saved states, $L = 2, 3, 4$.
`data/abjm-tau-L4.wxf`	Eigenvalue data of the $L = 4$ states, saved alongside the Q-functions by `abjm-Qfunctions.wb`.
`data/abjm-Qfunctions.wxf`	Q_1, Q_{12}, Q_{123}, roots and degrees of all 184 states (Baxter gauge $d = V_4$); written by `abjm-Qfunctions.wb`.
`data/abjm-twoblock-forms.json`	Polynomial forms $F_\Psi(a_2, a_3, d_2, d_3, m_v, m_u, p_{23}, p_{32}; M)$ in the 8 Borel invariants for all 184 states, all sectors (7248); written by `twoblock-polarisation-forms.wb` via `checks/twoblock-polforms/gen_json.py`.
`hw_L2.json`, `hw_L3.json`	In `checks/session-2026-10-13/epsexp/`: the $L = 2$ and $L = 3$ highest-weight states exported from the `.wxf` file as raw JSON (with `.npz` copies for python); used by the reruns of `even-bridge-lpairs.wb` and `odd-bridge-general-L.wb` to avoid importing the 3 GB key.

A.3 Python check folders

All new computations go into notebooks. The python below is legacy or fallback (written while the kernel was unavailable) and is the cross-check for the notebooks; port a script to a notebook instead of extending it.

folder	content
`checks/` (top level)	Legacy numpy scripts of 11-10 (run `python checks/<name>.py`; scripts `exec` siblings): `ffbar*.py`, `check5.py`, `chk7.py`, `chk13.py` (qKZ fits, end sites, interior sites, $\{1, 3\}$ exchange); `check6.py` (SoV basis z-independence); `g2test.py`, `g2inv.py` ((G2)-type localisation inventory); `chk_common13.py`, `chk_rotref.py` with `rotbase.py`, `chk_span13_words.py`, `scan_*.py` (common rotations, good references, $3 \otimes 3$ word span, Ω_2 scan); `chk_frame.py` (frame map); `abjm_*.py` (ABJM three-point localisation, untwisting, bridge overlaps; script index in `ABJM-3pt-transfer-matrices.md` §7).
`checks/session-2026-10-12/` (`README.md`)	Verbatim python of the 12-10 agents (licence expired mid-session): `pairgroup/` (`pgbase.py`, `RESULTS.md`: K_κ), `lpairs/` (`eng.py`, `RESULTS.md`, `hw_L3.npz`), `sovq/` (`sov_l4.py`: Q-dictionary, dual basis), `sovformula/` (`RESULTS.md`: SoV-normalised formula, regularised bases), `twoblock/` (`RESULTS.md`), `onepair/`, `oddbridge/` (`ob.py`, tables), `locality/` (`RESULTS.md`).
`checks/abjmlimit/`	Python for `sov-abjm-point-limit.wb`: `chain.py`, `words.py`, `t1.py`–`t3.py`, exact and numerical $L = 2$ and $L = 4$ bases (`L2*.py`, `buildL4exact.py`, `numL4*.py`), Jordan structure (`bjordan.py`), regularised bases (`regL4.py`, `L2reg*.py`), d-scans (`dline*.py`).
`checks/twoblock-polforms/`	Python for `twoblock-polarisation-forms.wb`: `core.py`, `formula.py`, `gen_json.py` (writes `data/abjm-twoblock-forms.json`), `inv*.py` (Borel invariants), `symm.py`, `polyverify.py`, `run_all.py`.

folder	content
checks/session-2026-10-13/epsexp/	even-bridge-epsilon-expansion.wb: step0a.py (exact), scan_eps.py, scan2.py, an3.py, hf.py, pt.py, pt2.py, kspec.py, odd_check.py, base modules ue_base.py, ue2.py, outputs out_*.txt, tables_C.txt, and the state files hw_L2.json, hw_L3.json.
checks/session-2026-10-13/refmap/	reference-admissibility-Kkappa.wb: covariance.py, scan.py, obstruction.py, genericz.py, smallint.py, l3check.py, untw.py, symR2.py, symR6.py, twloci_generic.py, with *_out.txt.
checks/session-2026-10-13/wasov/	Wa-sov-expansion-L4.wb: wa_sov_core.py, run1.py–run13.py, logs, Wa_sov_coefficients.json.
checks/session-2026-10-13/fonly/	Wa-Fonly-L6.wb: the F-chain (Cherednik-type) model checks model_cherednik.py, model_L3.py, check_conj.py, check_conj_L4.py (run model_L3.py first).
checks/session-2026-10-13/norm/	abjm-sov-norm-L4.wb: gram_abjm.py, abjm_eval.py, abjm_eval2.py, logs.

A.4 Other files

README.md (conventions, results, file map), NEXT.md in the repository root (resume point), ABJM-3pt-transfer-matrices.md (transfer-matrix construction of the three-point functions, numpy era), the two *-spec.md files of the 11-10 notebooks, ref/*.txt (cell-source dumps of two gaudin-cg notebooks cited by the specs), and in notes/: w-a-reference-status.md (the four reference bras and the status of $\langle W_a^L|$), rv-sov-dictionary.md (Ryan–Volin, GLMR, 2603 formulas in our conventions), session-2026-10-12-agent-reports.md, plan-2026-10-13.md, rerun-2026-10-13.md, the six agent-2026-10-13-*.md reports, and the earlier note abjm-transfer-matrices-symmetries.tex (12-10), superseded by these complete notes. The folder img/ holds only wolfbook log files (btl.log), no figures.

B Superseded or corrected statements

- *Singular point.* Bridge tables, degeneracy analyses and the twisted-shift fix $\hat{S}$ at $\theta_A = \theta_F$ are artefacts of the singular point. They are replaced by the ABJM point (§4).
- "The SoV basis survives at the ABJM point with shifted evaluation points": the shifted words are a basis but not B-eigen. "The Ryan–Volin B is degenerate at the ABJM point" is resolved by the regularised ($d \to 2h$) basis (§13).
- Per-length product-state searches for P, LP and LR are superseded: the three-point bras must be length independent (§10).
- *Odd-bridge formula:* the factor $1/(1 - z_4/z_1)$ was missing in an early version (corrected 11-10-2026).
- *Even bridges:* "the even bridge is not reached by localising words; the F-site gap" (notes of 12 Oct, §6) is solved by the pair κ-family at the ABJM point with $z_2 = z_3$ (§16). The generic-θ locality searches failed because $t_2(\theta_F + h)$ localises only at the ABJM point.
- *Paul's formula:* "ask Paul for his twisted-vacuum overlap formula" (plan of 13 Oct, §0) was based on an unclear object. Then, on the morning of 13 Oct, "$\langle W_a^L|\Psi\rangle$ has no Q-form" was concluded from a bra-level argument. That conclusion is withdrawn: on the relevant sectors the overlap equals a twisted-vacuum overlap (Proposition 1, §18). The bra-level statement (no twist sharing the SoV basis has $\langle W_a^L|$ as eigenbra) remains true.
- *Reference states:* "K_κ fails on $\langle 0|_{\rm comp}$" is refined. The five-word coefficients are specific to the W class; block-BPS references admit class-dependent words but have vanishing cross pairings

(§17). "Changing the reference to $\langle 0_w|$" (route A of 13 Oct) fails, since no κ-family exists on it at $z_2 \neq z_3$, and it is not needed.

- *Pair factor:* "$K_{\kappa=1}(\tau) = 0$ for all non-vacuum states" is false; K_1 vanishes only for $n \leq 1$. The claimed even/odd ratio relation does not exist (§18.6).
- *Untwisting sign:* the untwisting of a rotated pair uses $D = \mathrm{diag}(c - \varphi_j)$, not $c + \varphi_j$ (§17).
- *Rerun of 13 Oct:* in `pair-group-rotation.wb` (cell 17), four two-factor pair words localise with constants (51/31) or (71/31) $\times \prod \lambda_k$, not $\prod \lambda_k$; the python only fitted the constant. K_κ is unaffected. In `even-bridge-lpairs.wb` §2 the F1 and F2 labels of the untwisting table are swapped relative to the text; the limits are the same.
- *Norm:* the derivation of the SoV norm $\mathbb{B}$ in Q-functions (agent E, 13 Oct) was unnecessary, since $\mathbb{B}$ is known (P. Ryan). It is kept as a cross-check (§18.7).

References. B. Bargheer, C. Bercini, B. Lefundes, P. Ryan, arXiv:2603.26617; J. A. Minahan, K. Zarembo, arXiv:0806.3951; D. Bak, S.-J. Rey, arXiv:0807.2063; P. Ryan, D. Volin, arXiv:1810.10996, arXiv:2002.12341; N. Gromov, F. Levkovich-Maslyuk, P. Ryan, D. Volin, arXiv:1910.13442; N. Gromov, F. Levkovich-Maslyuk, P. Ryan, arXiv:2011.08229; the ABJM note `StructuringABJM.pdf`.